

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J43 MLB SCHEMATIC DVT  
REV 6.5.0  
4/09/13

| REV   | ECN   | DESCRIPTION OF REVISION |
|-------|-------|-------------------------|
| <REV> | <ECN> | <ECO_DESCRIPTION>       |



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| 2    | BOM Configuration                   | J41_MLB  | 04/09/2013 |
| 3    | BOM Variants                        | K21_MLB  | 11/16/2010 |
| 4    | PD PARTS                            | MASTER   | MASTER     |
| 5    | CPU GFX/NCTF/RSVD                   | J41_MLB  | 02/06/2013 |
| 6    | CPU Misc/JTAG/CFG/RSVD              | J41_MLB  | 04/02/2013 |
| 7    | CPU DDR3/LPDDR3 Interfaces          | J41_MLB  | 02/06/2013 |
| 8    | CPU/PCH POWER                       | J41_MLB  | 04/09/2013 |
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| 10   | CPU Decoupling                      | WILL_J43 | 01/08/2013 |
| 11   | PCH Decoupling                      | J41_MLB  | 02/07/2013 |
| 12   | PCH Audio/JTAG/SATA/CLK             | J41_MLB  | 02/06/2013 |
| 13   | PCH PM/PCI/GFX                      | J41_MLB  | 02/06/2013 |
| 14   | PCH PCIe/USB/LPC/SPI/SMBus          | J41_MLB  | 02/06/2013 |
| 15   | PCH GPIO/MISC/LPIO                  | J41_MLB  | 04/02/2013 |
| 16   | CPU/PCH Merged XDP                  | J41_MLB  | 02/06/2013 |
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| 18   | Project Chipset Support             | J41_MLB  | 02/15/2013 |
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| 25   | Thunderbolt Host (1 of 2)           | J41_MLB  | 02/06/2013 |
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| 28   | Thunderbolt Connector A             | J41_MLB  | 02/07/2013 |
| 29   | Wireless Connector                  | J41_MLB  | 02/06/2013 |
| 30   | SSD Connector                       | J41_MLB  | 04/09/2013 |
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| 33   | SD READER CONNECTOR                 | MASTER   | 07/01/2011 |
| 34   | SD CONTROLLER (GL3219)              | MASTER   | 10/11/2010 |
| 35   | External A USB3 Connector           | J41_MLB  | 02/07/2013 |
| 36   | IPD Connector                       | J41_MLB  | 02/12/2013 |
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| 38   | SMC Shared Support                  | J41_MLB  | 02/06/2013 |
| 39   | SMC Project Support                 | J41_MLB  | 02/06/2013 |
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| 41   | High Side Current Sensing           | J41_MLB  | 03/28/2013 |
| 42   | Voltage & Load Side Current Sensing | J41_MLB  | 03/28/2013 |
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| 69   | 48 | Battery Connector               | MASTER      | MASTER     |
| 70   | 49 | DC-In & G3H Supply              | J41_MLB     | 02/06/2013 |
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| 111  | 67 | CPU Constraints                 | CONSTRAINTS | 09/25/2012 |
| 112  | 68 | PCH Constraints 1               | CLEAN_343   | 11/13/2012 |
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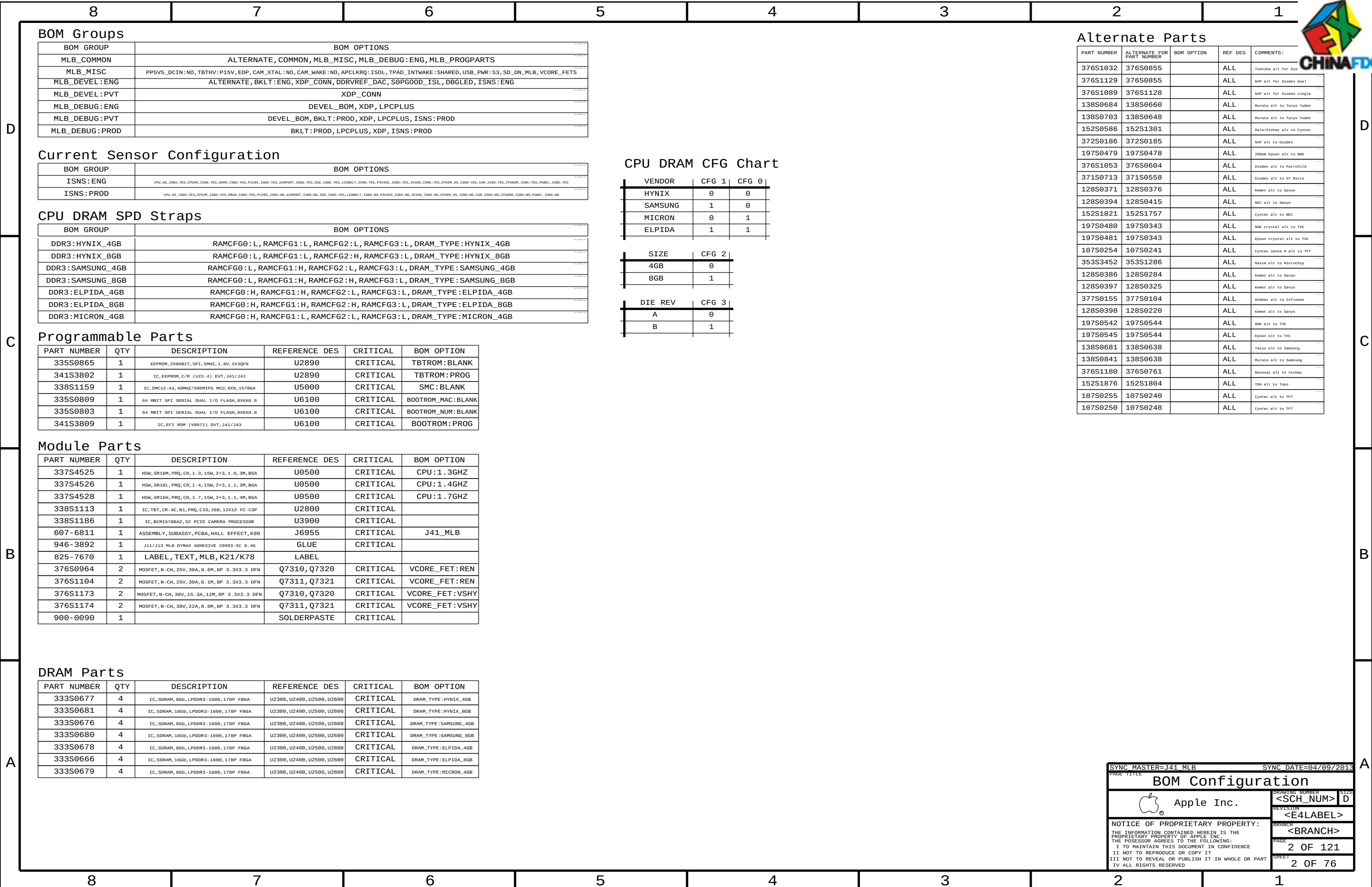
# ALIASES RESOLVED

Schematic / PCB #'s

| PART NUMBER | QTY | DESCRIPTION     | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-----------------|---------------|----------|------------|
| 051-9800    | 1   | SCHEM, MLB, J43 | SCH           | CRITICAL |            |
| 820-3437    | 1   | PCBF, MLB, J43  | PCB           | CRITICAL |            |

PRODUCT SAFETY REQUIREMENTS:  
PCB, UL RECOGNIZED, MIN. 130-C TEMP. RATING AND V-0 FLAME RATING PER UL 796 & UL 94.  
PCB TO BE SILK-SCREENED WITH UL/CUL RECOGNITION MARK, MANUFACTURER'S UL FILE  
NUMBER, UL PCB MATERIAL DESIGNATION, 130-C TEMP. RATING AND V-0 FLAME RATING.

|                                                                                                                                                                                                                                                                                                                                                                                                              |                                  |                |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------|----------|
| DRAWING TITLE                                                                                                                                                                                                                                                                                                                                                                                                |                                  |                |          |
| <p align="center">&lt;PART_DESCRIPTION&gt;</p>                                                                                                                                                                                                                                                                                                                                                               |                                  |                |          |
|                                                                                                                                                                                                                                                                                                                         | <p align="center">Apple Inc.</p> | DRAWING NUMBER | SIZE     |
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BOM Variants

NOTE: All the "GOOD" BOM Configs have been de-activated

| BOM NUMBER | BOM NAME                       | BOM OPTIONS                             |
|------------|--------------------------------|-----------------------------------------|
| 639-4146   | PCBA, MLB, GOOD, HY-4GB, J43   | MLB_CMNPTS, CPU:1.3GHZ, DDR3:HYNIX_4GB  |
| 639-4293   | PCBA, MLB, GOOD, HY-8GB, J43   | MLB_CMNPTS, CPU:1.3GHZ, DDR3:HYNIX_8GB  |
| 639-4294   | PCBA, MLB, GOOD, EL-4GB, J43   | MLB_CMNPTS, CPU:1.3GHZ, DDR3:ELPIDA_4GB |
| 639-4295   | PCBA, MLB, GOOD, EL-8GB, J43   | MLB_CMNPTS, CPU:1.3GHZ, DDR3:ELPIDA_8GB |
| 639-4745   | PCBA, MLB, GOOD, MI-4GB, J43   | MLB_CMNPTS, CPU:1.3GHZ, DDR3:MICRON_4GB |
| 639-4445   | PCBA, MLB, BETTER, HY-4GB, J43 | MLB_CMNPTS, CPU:1.3GHZ, DDR3:HYNIX_4GB  |
| 639-4446   | PCBA, MLB, BETTER, HY-8GB, J43 | MLB_CMNPTS, CPU:1.3GHZ, DDR3:HYNIX_8GB  |
| 639-4447   | PCBA, MLB, BETTER, EL-4GB, J43 | MLB_CMNPTS, CPU:1.3GHZ, DDR3:ELPIDA_4GB |
| 639-4448   | PCBA, MLB, BETTER, EL-8GB, J43 | MLB_CMNPTS, CPU:1.3GHZ, DDR3:ELPIDA_8GB |
| 639-4746   | PCBA, MLB, BETTER, MI-4GB, J43 | MLB_CMNPTS, CPU:1.3GHZ, DDR3:MICRON_4GB |
| 639-4755   | PCBA, MLB, BEST, HY-4GB, J43   | MLB_CMNPTS, CPU:1.7GHZ, DDR3:HYNIX_4GB  |
| 639-4756   | PCBA, MLB, BEST, HY-8GB, J43   | MLB_CMNPTS, CPU:1.7GHZ, DDR3:HYNIX_8GB  |
| 639-4757   | PCBA, MLB, BEST, EL-4GB, J43   | MLB_CMNPTS, CPU:1.7GHZ, DDR3:ELPIDA_4GB |
| 639-4758   | PCBA, MLB, BEST, EL-8GB, J43   | MLB_CMNPTS, CPU:1.7GHZ, DDR3:ELPIDA_8GB |
| 639-4759   | PCBA, MLB, BEST, MI-4GB, J43   | MLB_CMNPTS, CPU:1.7GHZ, DDR3:MICRON_4GB |
| 685-0025   | CMN PTS, PCBA, MLB, J43        | MLB_COMMON                              |
| 985-0018   | J43 MLB DEVELOPMENT BOM        | MLB_DEVEL:ENG                           |
| 685-0064   | VCORE FET, REN, J43            | VCORE_FET:REN                           |
| 685-0065   | VCORE FET, VSHY, J43           | VCORE_FET:VSHY                          |

Alternate Parts

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS:              |
|-------------|---------------------------|------------|---------|------------------------|
| 685-0064    | 685-0065                  |            | ALL     | Renesas alt for Vishay |

|          |          |  |     |                              |
|----------|----------|--|-----|------------------------------|
| 333S0704 | 333S0700 |  | ALL | Elpida CAM DRAW alt to Hynix |
|----------|----------|--|-----|------------------------------|

Programmable Parts

| PART NUMBER | QTY | DESCRIPTION                                   | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-----------------------------------------------|---------------|----------|------------|
| 341S3758    | 1   | IC, SMC-A3 SCPL, EXT, V22.12a19, PROTO 1, J43 | U5000         | CRITICAL | SMC:PROG   |

BOM Groups

| BOM GROUP     | BOM OPTIONS                         |
|---------------|-------------------------------------|
| MLB_PROGPARTS | BOOTROM:PROG, SMC:PROG, TBTRON:PROG |

Module Parts

| PART NUMBER | QTY | DESCRIPTION                                | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|--------------------------------------------|---------------|----------|------------|
| 338S1215    | 1   | IC, GL3219, USB3 SD CARD READER, 46P, LQFN | U4500         | CRITICAL |            |

Sub-BOMs

| PART NUMBER | QTY | DESCRIPTION             | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------|---------------|----------|------------|
| 985-0018    | 1   | J43 MLB DEVELOPMENT BOM | DEVEL         | CRITICAL | DEVEL_BOM  |
| 685-0025    | 1   | CMN PTS, PCBA, MLB, J43 | CMNPTS        | CRITICAL | MLB_CMNPTS |
| 685-0065    | 1   | VCORE FET, VSHY, J43    | VCOREFETS     | CRITICAL | VCORE_FETS |

SYNC MASTER=K21 MLB

SYNC DATE=11/16/2016

BOM Variants

Apple Inc.

DRAWING NUMBER

<SCH\_NUM>

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CHINA

FDX

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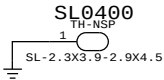
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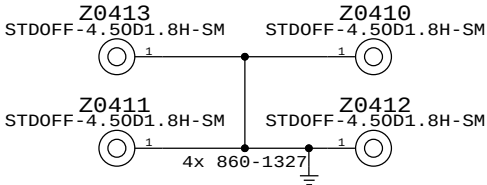
PD Module Parts

| PART NUMBER | QTY | DESCRIPTION                       | REFERENCE DES       | CRITICAL | BOM OPTION |
|-------------|-----|-----------------------------------|---------------------|----------|------------|
| 806-5107    | 1   | CAN, TOPSIDE, ALT, J41/J43        | TBTTOPSIDE_2P_FENCE | CRITICAL |            |
| 806-5108    | 1   | CAN, TOPSIDE, COVER, ALT, J41/J43 | TBTTOPSIDE_2P_COVER | CRITICAL |            |
| 806-3142    | 1   | CAN, TBT, J11/J13                 | TBTFENCE            | CRITICAL |            |
| 806-3215    | 1   | CAN, COVER, TBT, J11/J13          | TBTCOVER            | CRITICAL |            |
| 806-3216    | 1   | CAN, MDP, J11/J13                 | MDPCAN              | CRITICAL |            |
| 806-3083    | 1   | SHLD, USB, MLB, J11/J13           | USBCAN              | CRITICAL |            |
| 725-1792    | 1   | INSULATOR, CPU, J41/J43           | CPU_INSULATOR       | CRITICAL |            |

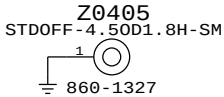
Plated Board Slot



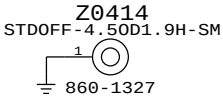
CPU Heat Sink Mounting Bosses



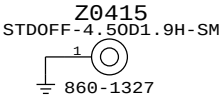
Fan Boss



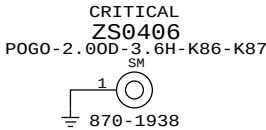
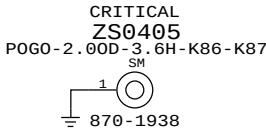
X21 Boss



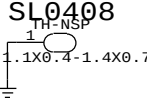
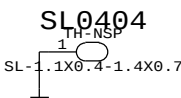
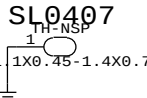
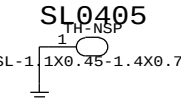
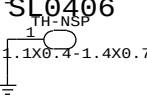
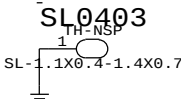
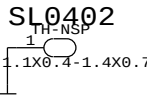
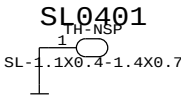
SSD Boss



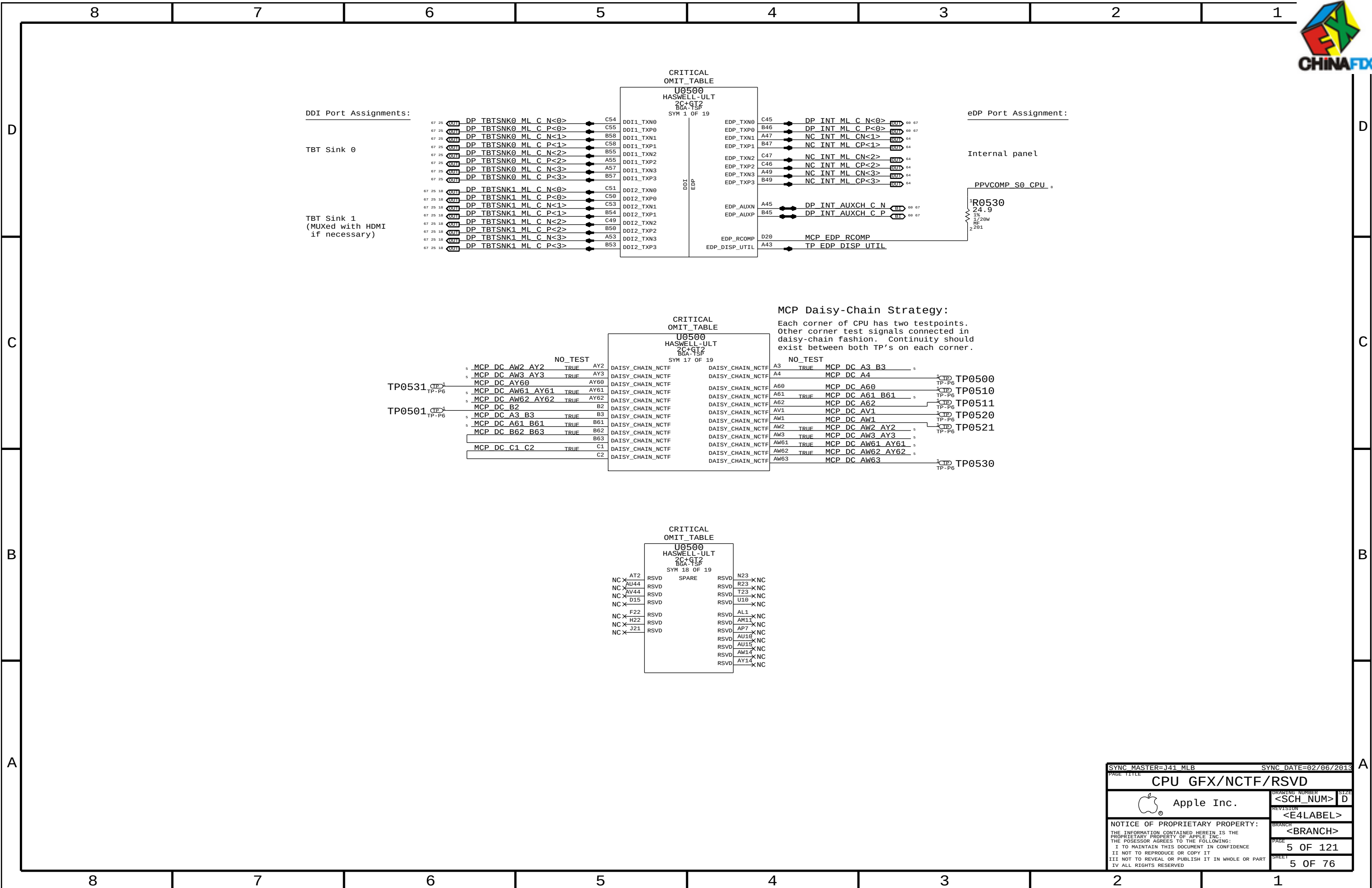
EMI I/O Pogo Pins  
DisplayPort PogoUSB/SD Card Pogo



Can Slots



|                                                                                                                                                                                                                                                                                                                         |  |                  |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------|----------|
| SYNC MASTER=MASTER                                                                                                                                                                                                                                                                                                      |  | SYNC DATE=MASTER |          |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  | PD PARTS         |          |
|  Apple Inc.                                                                                                                                                                                                                        |  | DRAWING NUMBER   | SIZE     |
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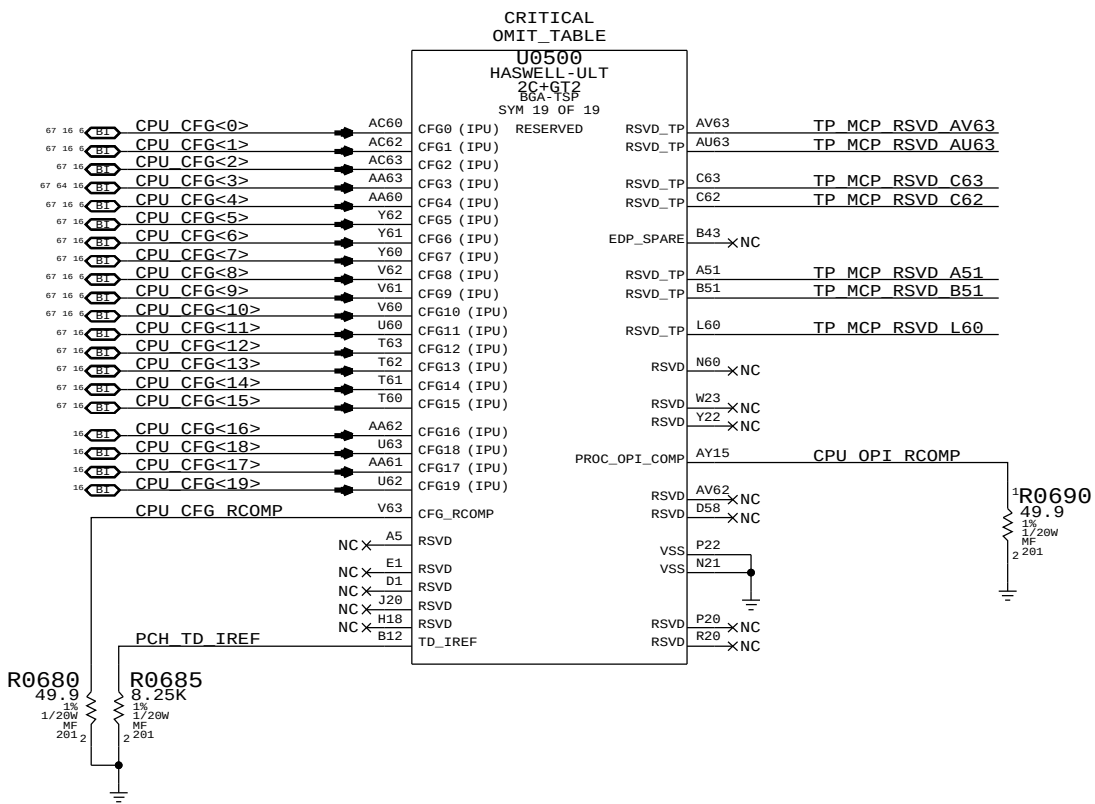
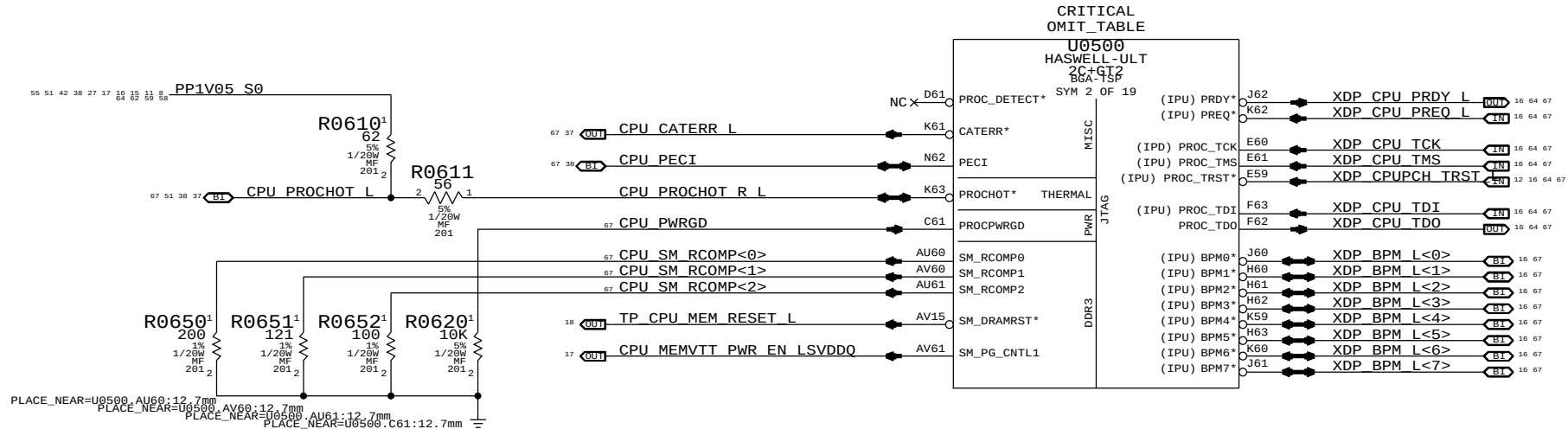
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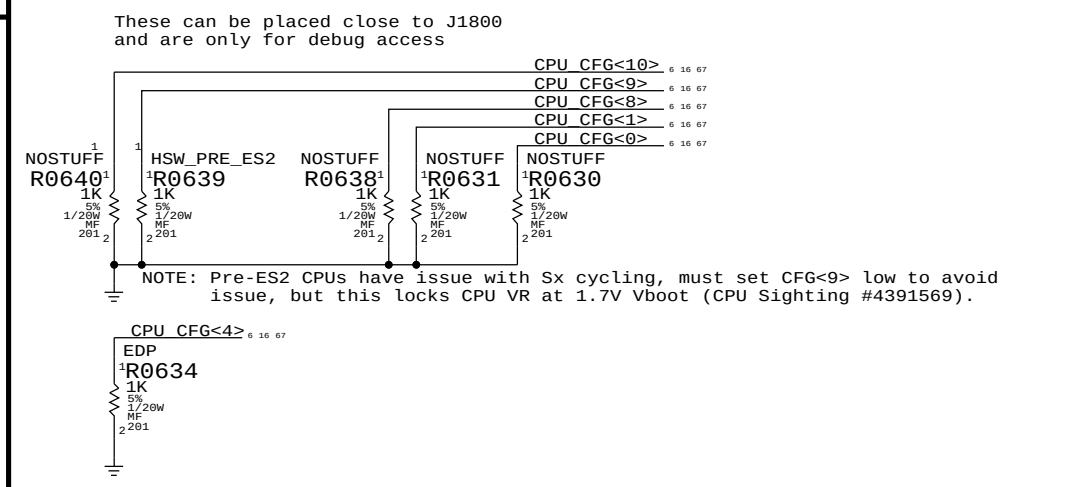
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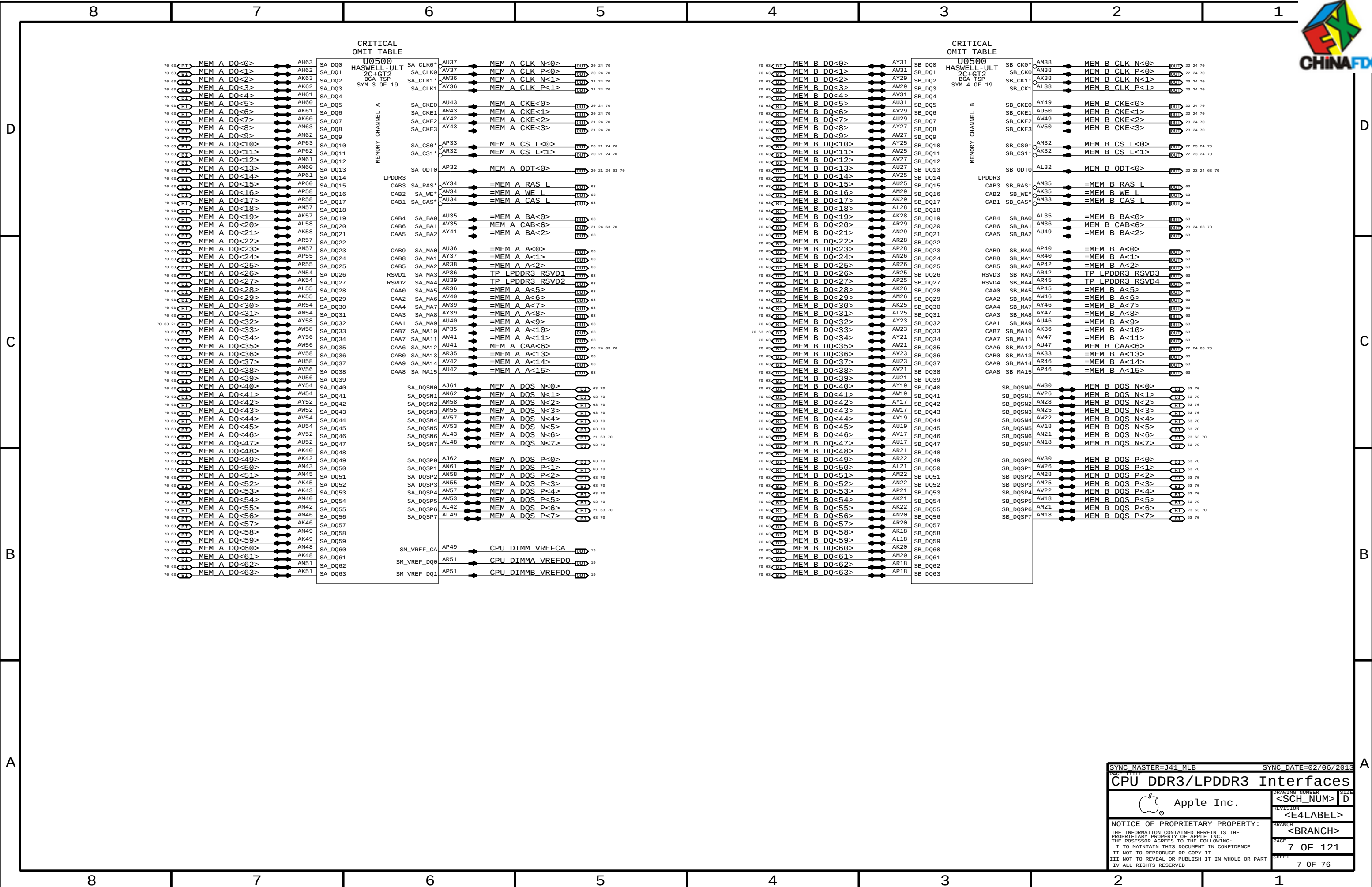
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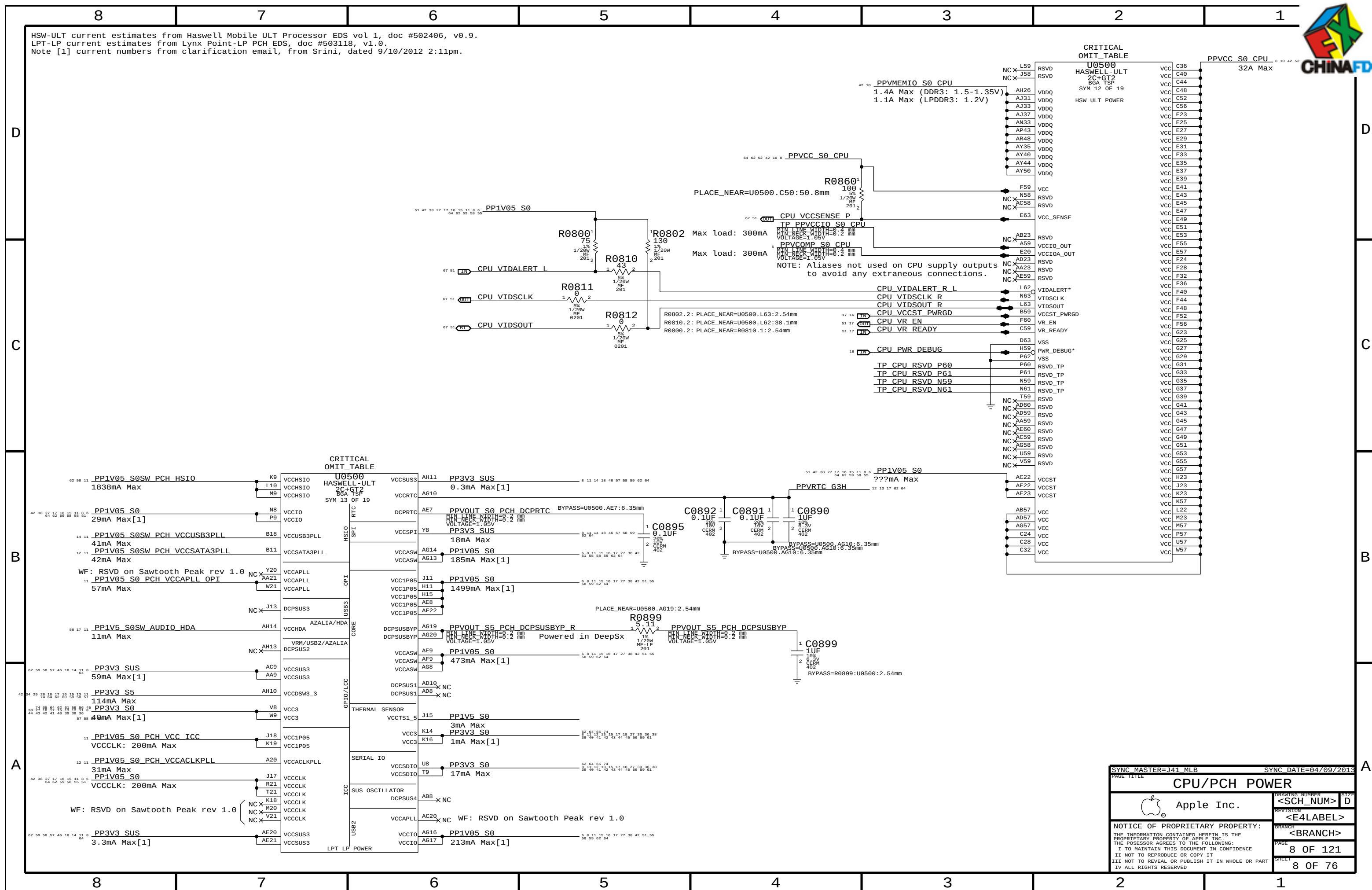


CFG<10>:SAFE MODE BOOT 1 = NORMAL OPERATION 0 = POWER FEATURES NOT ACTIVE  
CFG<9> :NO SVID-CAPABLE VR 1 = VR SUPPORTS SVID 0 = VR DOES NOT SUPPORT SVID  
CFG<8> :ALLOW NOA ON LOCKED UNITS 1 = NORMAL OPERATION 0 = NOA ALWAYS UNLOCKED  
CFG<4> :eDP ENABLE/DISABLE 1 = DISABLED 0 = ENABLED  
CFG<1> :PCH-LESS MODE 1 = NORMAL OPERATION 0 = PCH-LESS MODE  
CFG<0> :RESET SEQUENCE STALL 1 = NORMAL OPERATION 0 = STALL AFTER PCU PLL LOCK

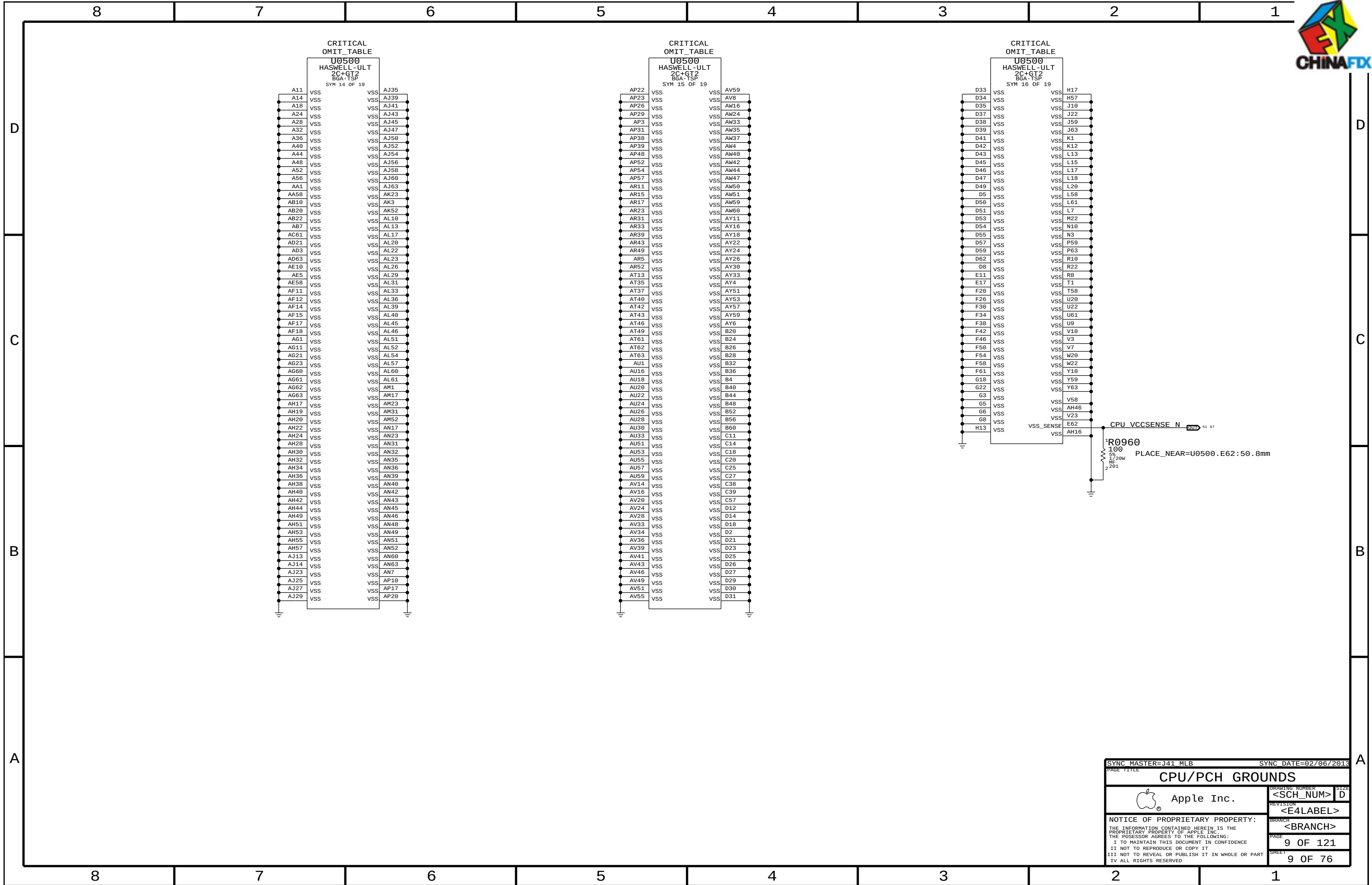














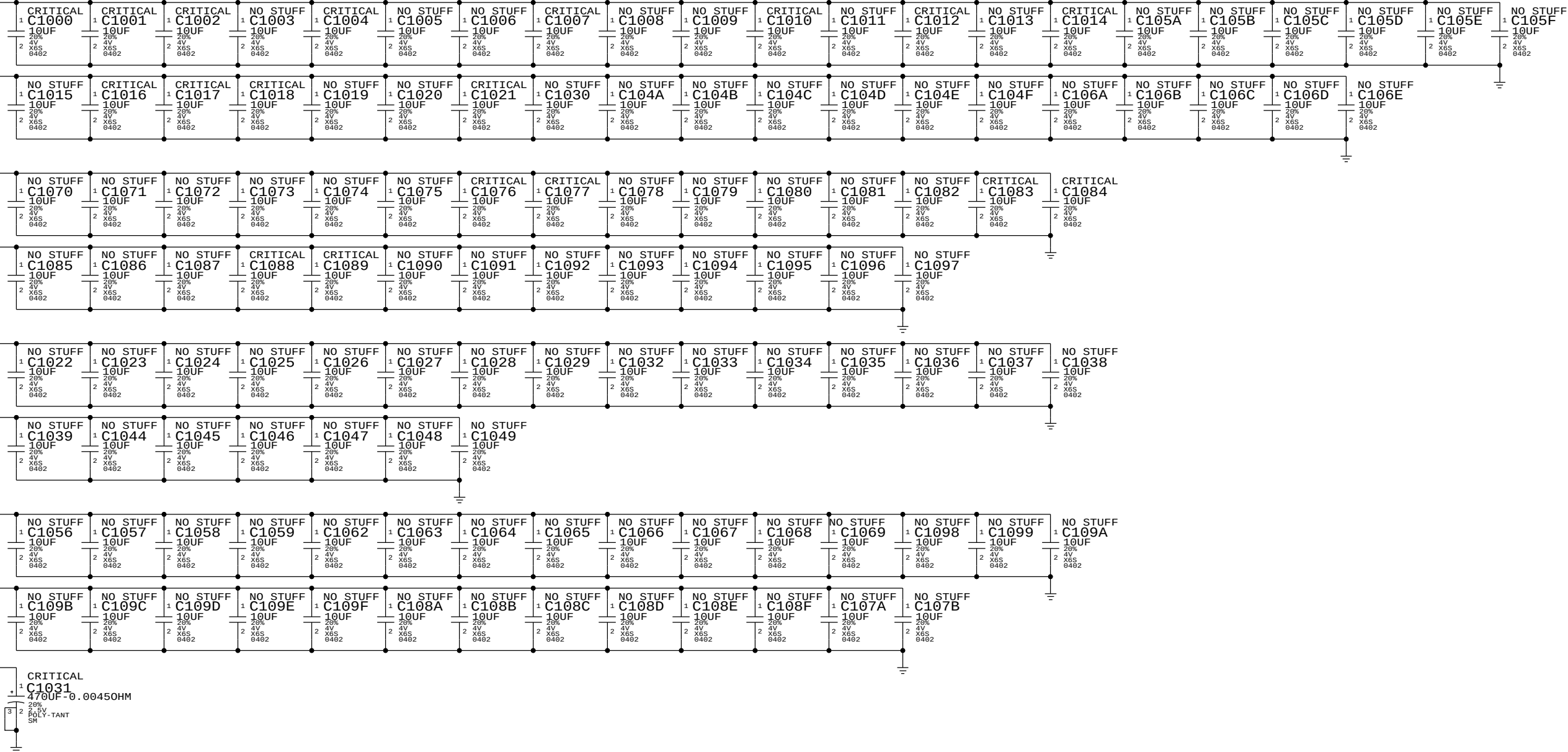
All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 1.0 unless stated otherwise

## CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff

Apple implementation : 18x 10uF 0402 mirrored stuff, 1x 470uF stuff, 50x 10uF mirrored no stuff, 50x 10uF single sided no stuff

64 62 52 42 8 PPVCC\_S0\_CPU

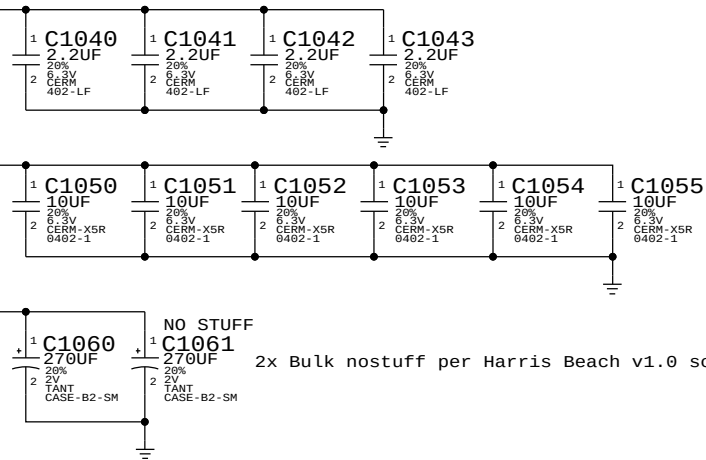


## CPU VDDQ DECOUPLING

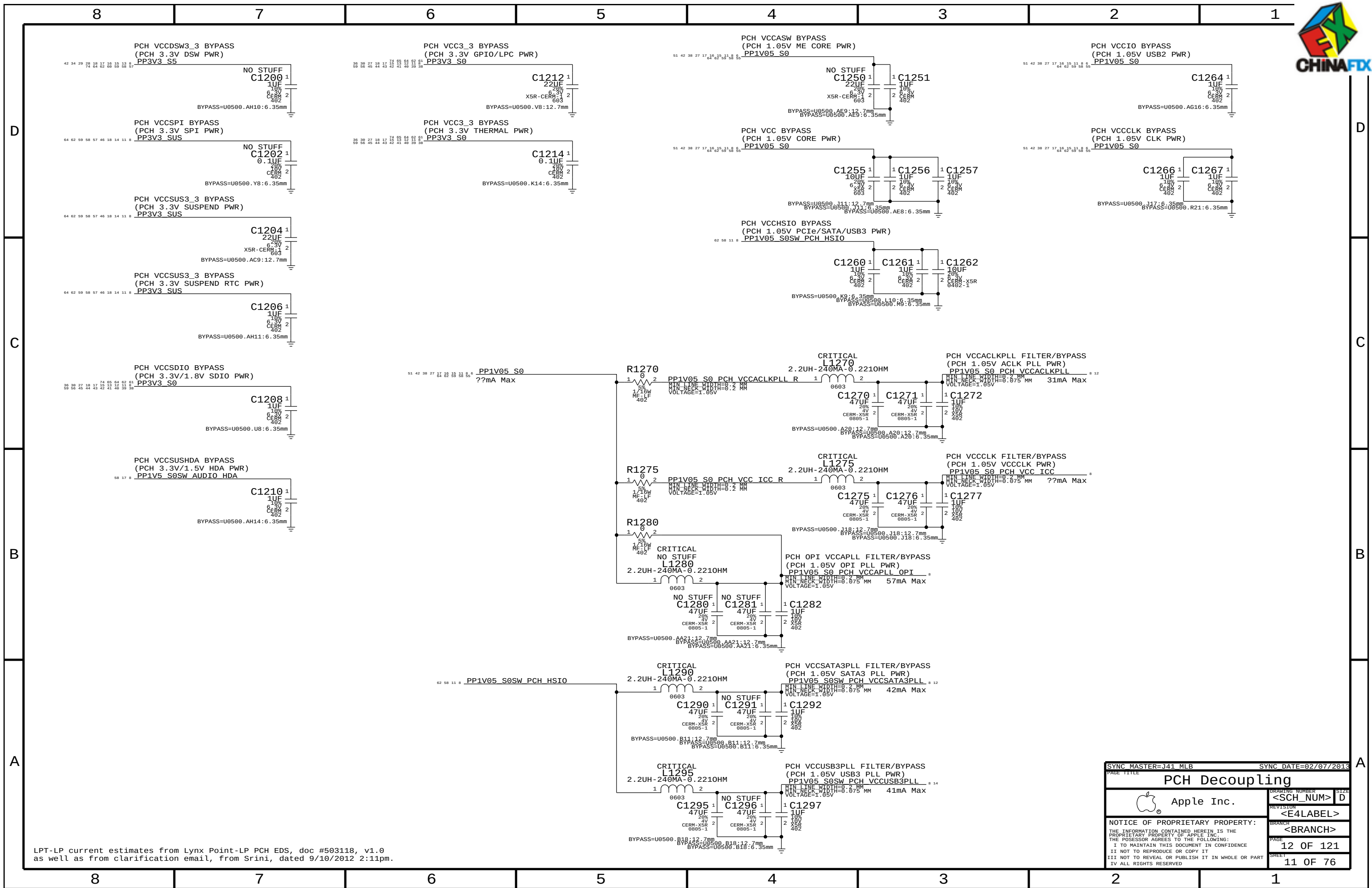
Intel recommendation (Table 5-4): 4x 2.2uF 0402, 6x 10uF 0603

Apple implementation : 4x 2.2uF 0402, 6x 10uF 0402, 2x 270uF B2 no stuff

42 8 PPVMEMIO\_S0\_CPU

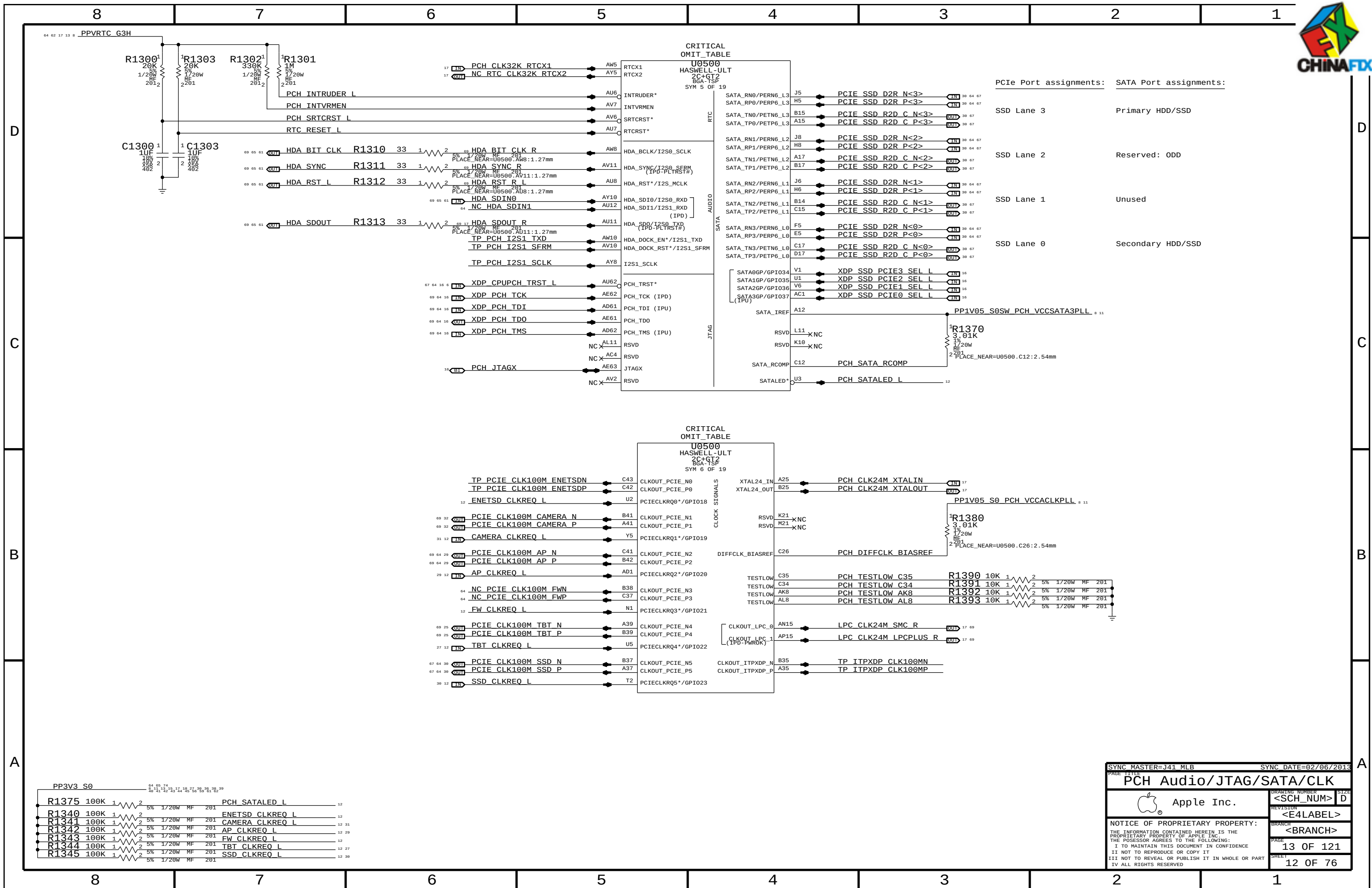


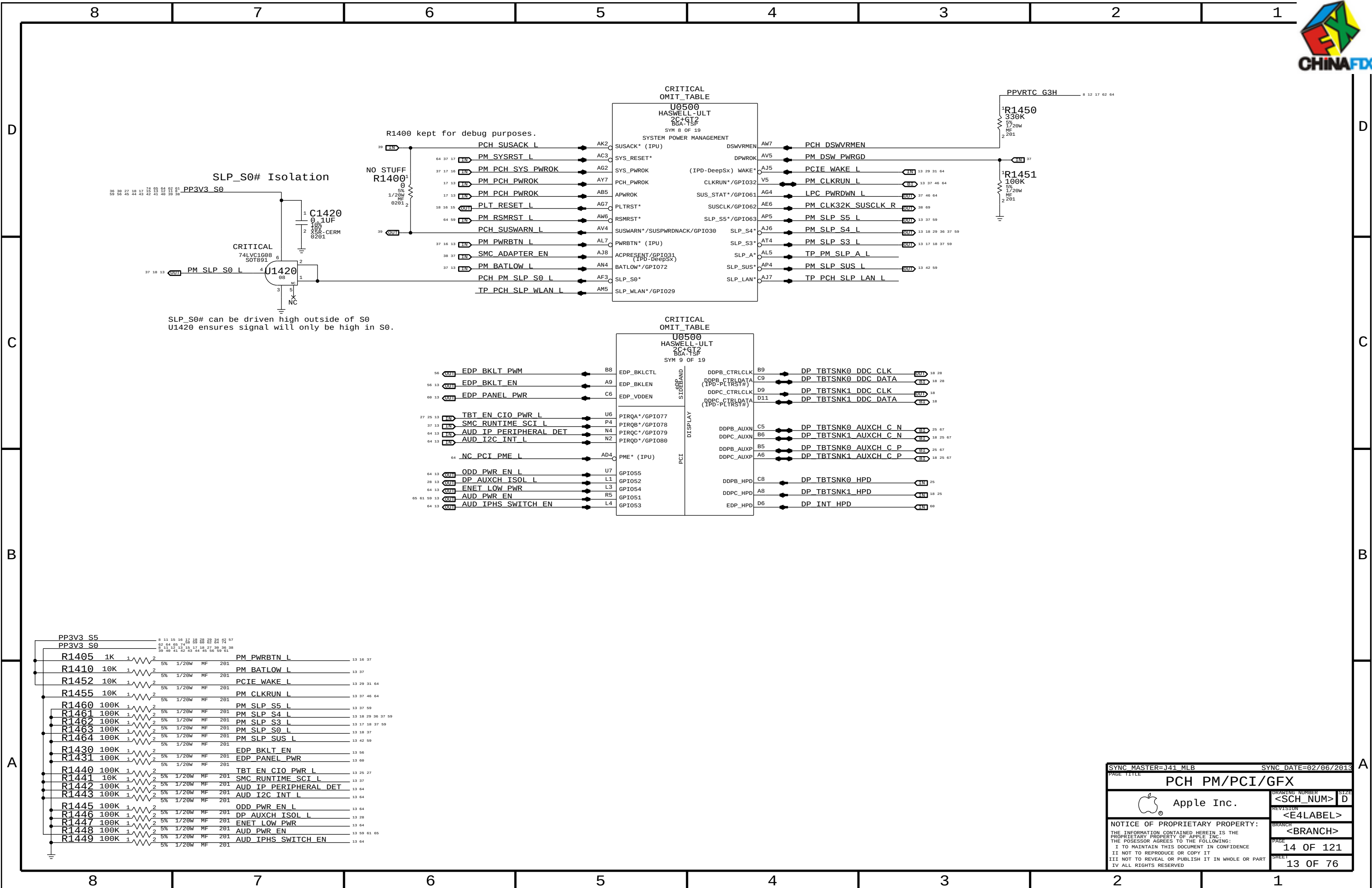
|                                                                                                                                                                                                                                                                                                                   |  |                      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=WILL J43                                                                                                                                                                                                                                                                                              |  | SYNC DATE=01/08/2013 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |  | CPU Decoupling       |           |
| Apple Inc.                                                                                                                                                                                                                                                                                                        |  | DRAWING NUMBER       | SIZE      |
|                                                                                                                                                                                                                                                                                                                   |  | <SCH_NUM>            | D         |
|                                                                                                                                                                                                                                                                                                                   |  | REVISION             |           |
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|                                                                                                                                                                                                                                                                                                                   |  | PAGE                 | 10 OF 121 |
|                                                                                                                                                                                                                                                                                                                   |  | SHEET                | 10 OF 76  |
|                                                                                                                                                                                                                                                                                                                   |  |                      |           |



LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0 as well as from clarification email, from Srin, dated 9/10/2012 2:11pm.

|                                                                                                                   |  |                      |           |
|-------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=J41 MLB                                                                                               |  | SYNC DATE=02/07/2013 |           |
| PAGE TITLE                                                                                                        |  | PCH Decoupling       |           |
| Apple Inc.                                                                                                        |  | <SCH_NUM>            | SIZE      |
|                                                                                                                   |  | <E4LABEL>            | D         |
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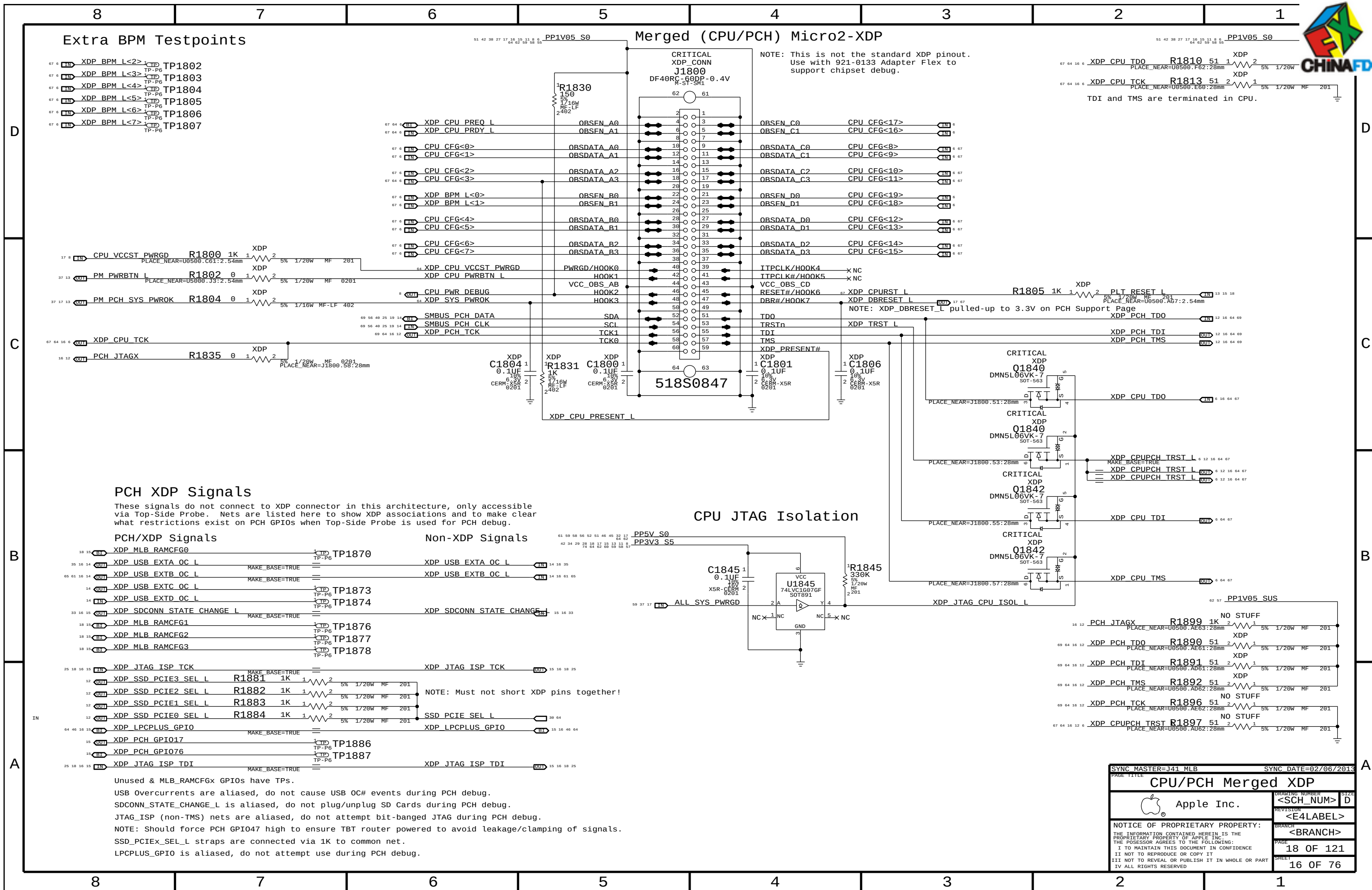












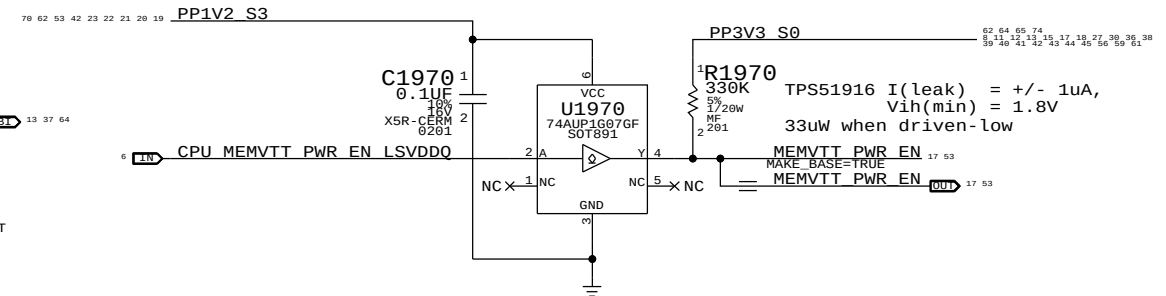
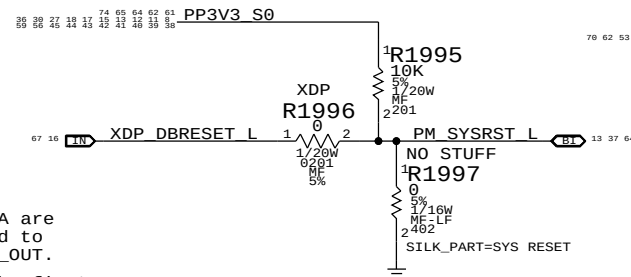
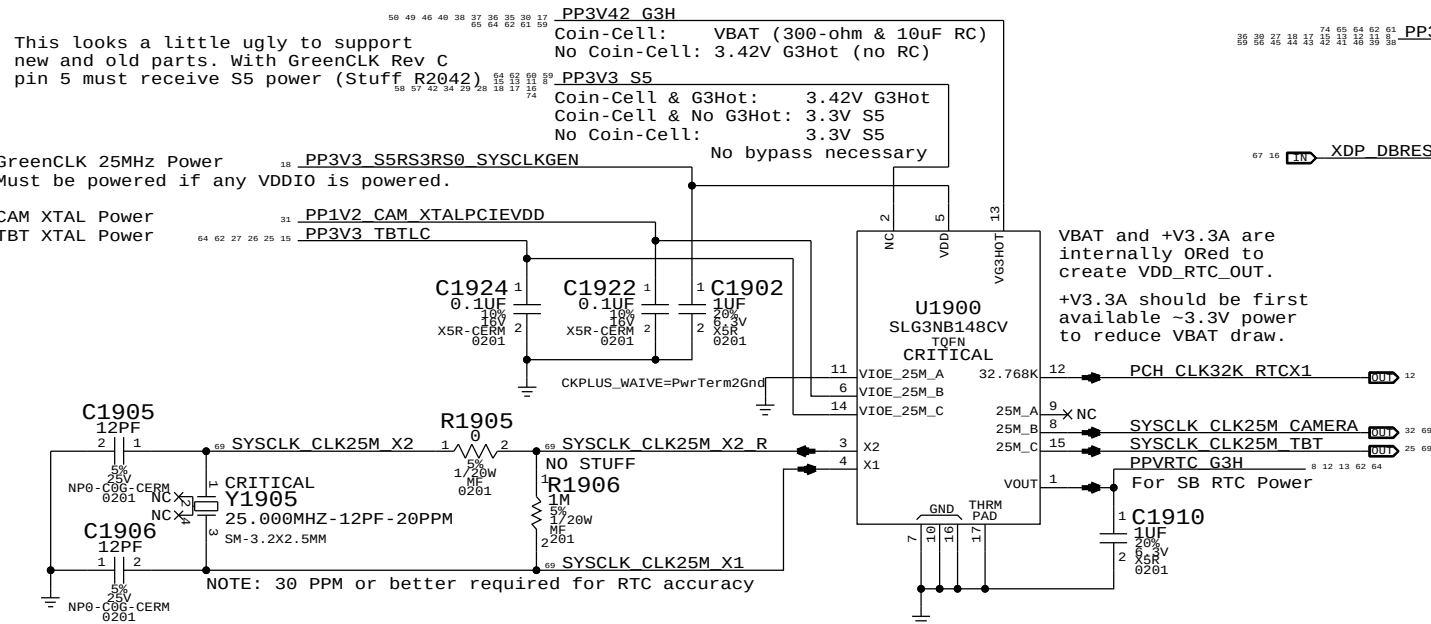




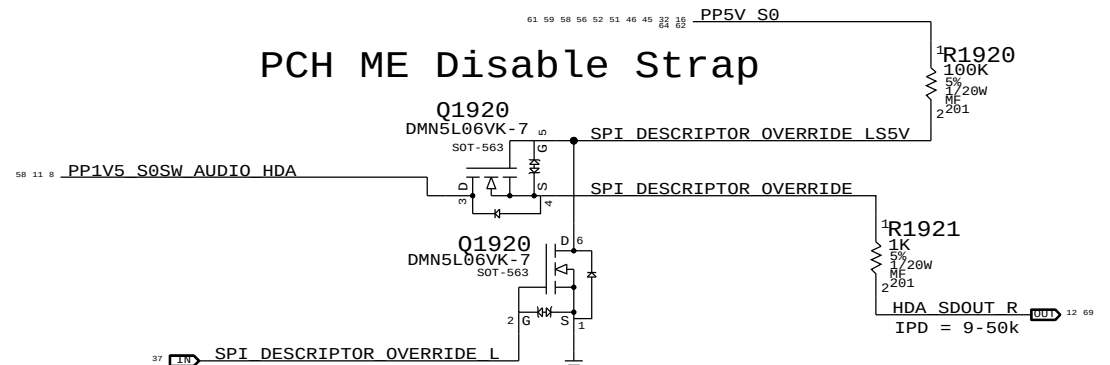
## Memory VTT Enable Level-Shifter

Chipset uses 24MHz crystal, GreenCLK kept to save 1x 25MHz crystal & 1x 32kHz crystal

CPU output is on VDDQ rail (1.2V), TPS51916 has 1.8V Vih(min).

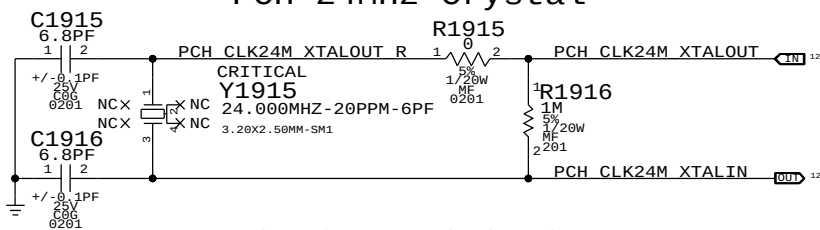


## PCH ME Disable Strap

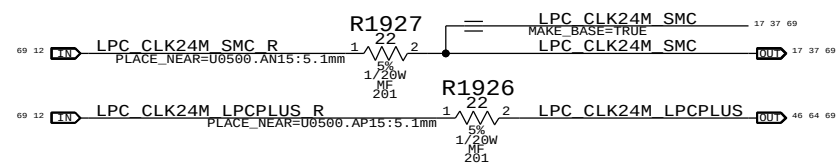


PCH uses HDA\_SDO as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. P1920 & 5V pull-up allows circuit to work regardless of HDA voltage.

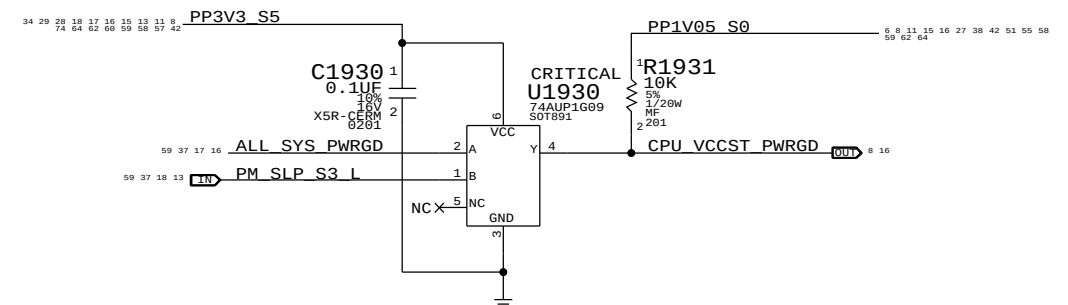
## PCH 24MHz Crystal



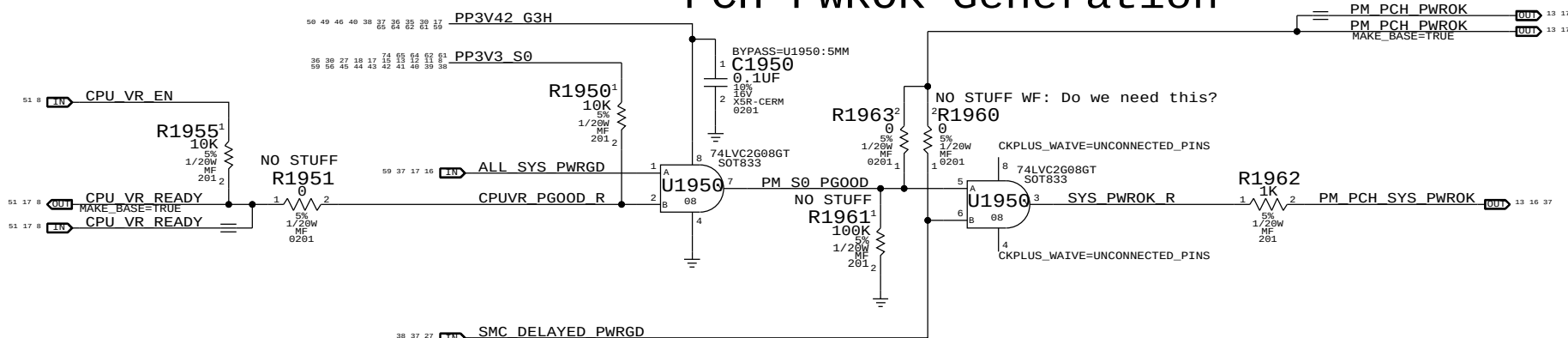
## PCH 24MHz Outputs



VCCST (1.05V S0) PWRGD



## PCH PWR0K Generation



|                                                                                                                                                                                                                                                                                                                     |            |                                    |                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------|------------------------------------------|
| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                                                 |            | SYNC DATE=02/06/2013               |                                          |
| PAGE TITLE                                                                                                                                                                                                                                                                                                          |            |                                    |                                          |
| Chipset Support                                                                                                                                                                                                                                                                                                     |            |                                    |                                          |
|                                                                                                                                                                                                                                | Apple Inc. |                                    | DRAWING NUMBER<br><b>&lt;SCH_NUM&gt;</b> |
|                                                                                                                                                                                                                                                                                                                     |            |                                    | SIZE<br><b>D</b>                         |
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|                                                                                                                                                                                                                                                                                                                     |            | PAGE<br><b>19 OF 121</b>           |                                          |
|                                                                                                                                                                                                                                                                                                                     |            | SHEET<br><b>17 OF 76</b>           |                                          |



## Platform Reset Connections

## GreenCLK 25MHz Power

## DDC Pull-Ups

Unbuffered

Scrub for Layout Optimization

Buffered

NO STUFF

NO STUFF

NO STUFF

2.2k pull-ups are required by PCH to indicate active display interface.  
DP++ spec violation, should remove!

## Thunderbolt Pull-up/downs

Cactus Ridge G02SX signal pulled-up to SUS rail

## Power State Debug LEDs

(For development only)

## RAM Configuration Straps

Pull-downs for chip-down RAM systems

## LPDDR3 Alias Support

## TBT Aliases

|                                                                                                                   |  |                      |  |
|-------------------------------------------------------------------------------------------------------------------|--|----------------------|--|
| SYNC MASTER=J41 MLB                                                                                               |  | SYNC DATE=02/15/2013 |  |
| PAGE TITLE                                                                                                        |  |                      |  |
| Project Chipset Support                                                                                           |  |                      |  |
| Apple Inc.                                                                                                        |  | DRAWING NUMBER       |  |
| Apple Inc.                                                                                                        |  | <SCH_NUM>            |  |
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|                                                                                                                   |  | SHEET                |  |
|                                                                                                                   |  | 18 OF 76             |  |





## CPU-Based Margining

- =PP3V3\_S3\_VREFMRGN
- =PPDDR\_S3\_MEMVREF

- =I2C\_VREFDACS\_SCL
- =I2C\_VREFDACS\_SDA
- =I2C\_PCA9557D\_SCL
- =I2C\_PCA9557D\_SDA

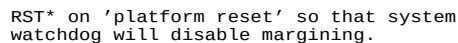
- DDRVREF\_DAC - Stuffs DAC margining circuit.

NOTE: CPU DAC output step sizes:

|        |         |        |          |
|--------|---------|--------|----------|
| DDR3   | (1.5V)  | 7.70mV | per step |
| DDR3L  | (1.35V) | 6.99mV | per step |
| LPDDR3 | (1.2V)  | ?.??mV | per step |

NOTE: CPU has single output for VREFCA. Split into two signals for independent DAC margining support. When DAC margining VREFCA ensure VREFMRGN\_CPU\_EN is low to remove short due to CPU.

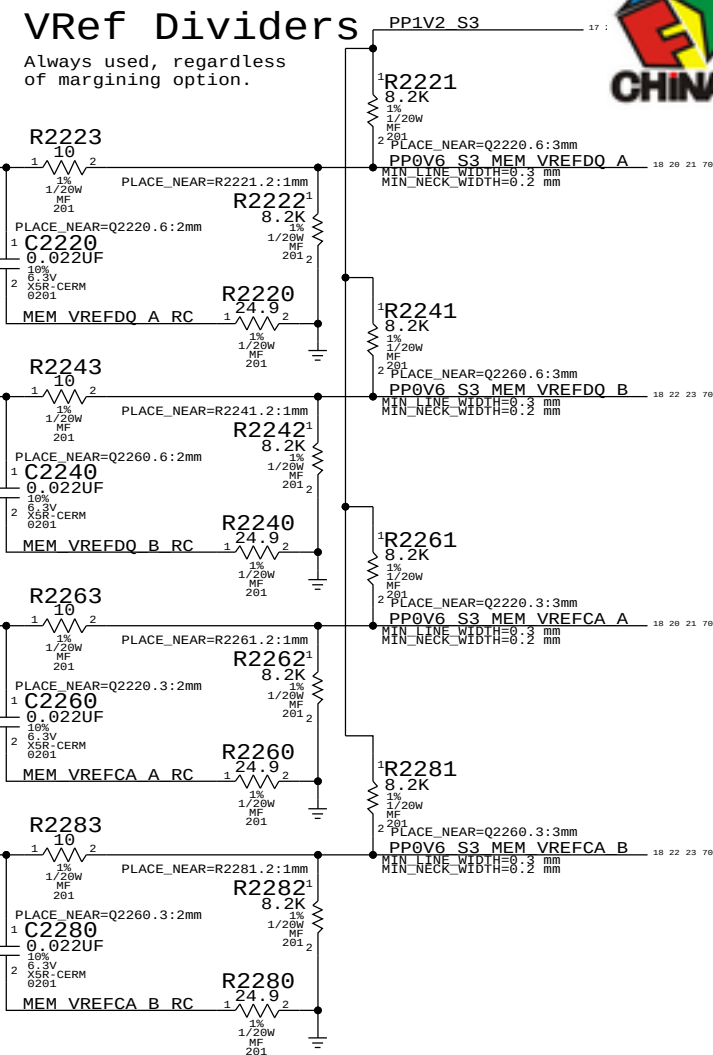
DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.



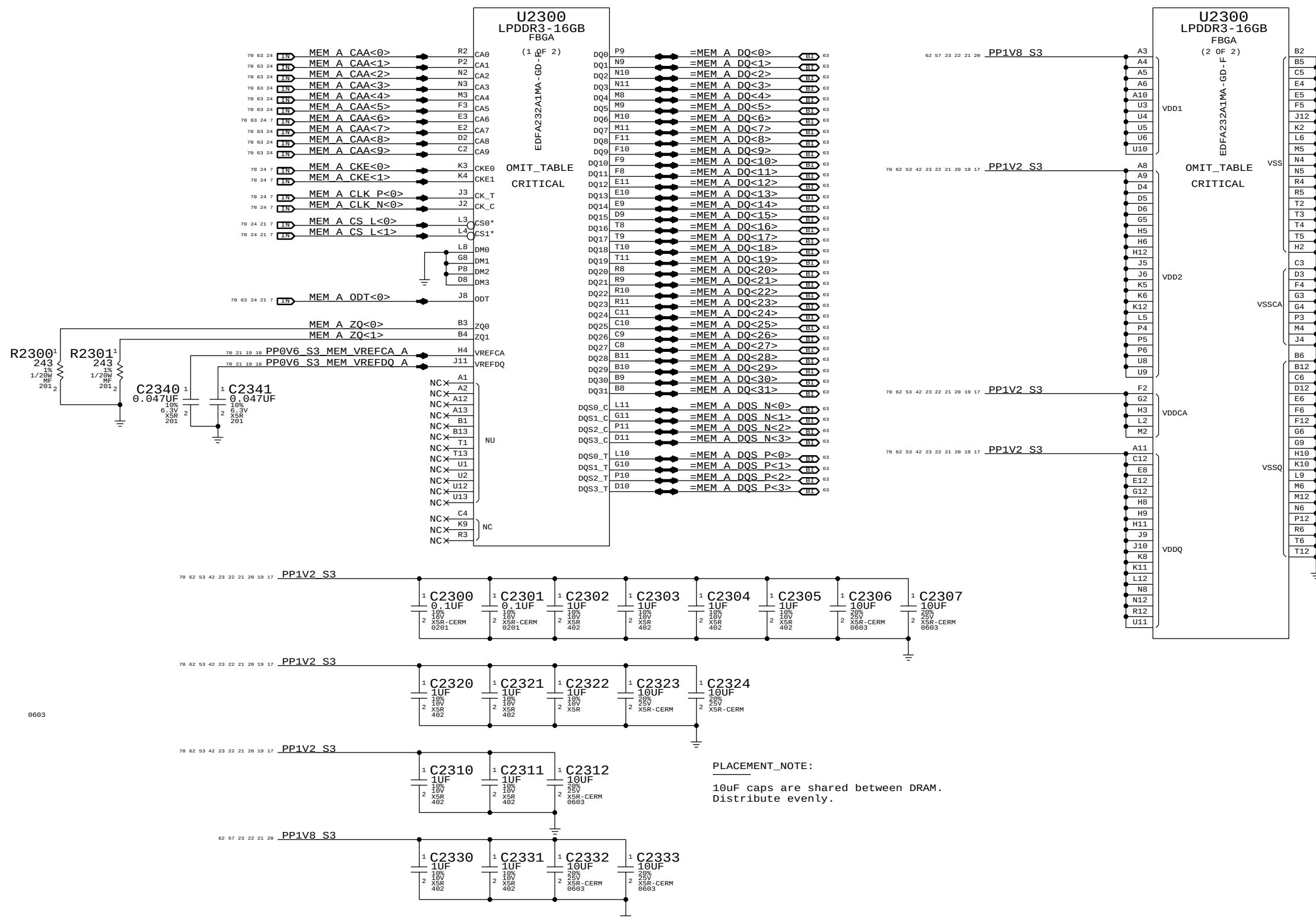
NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider  
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

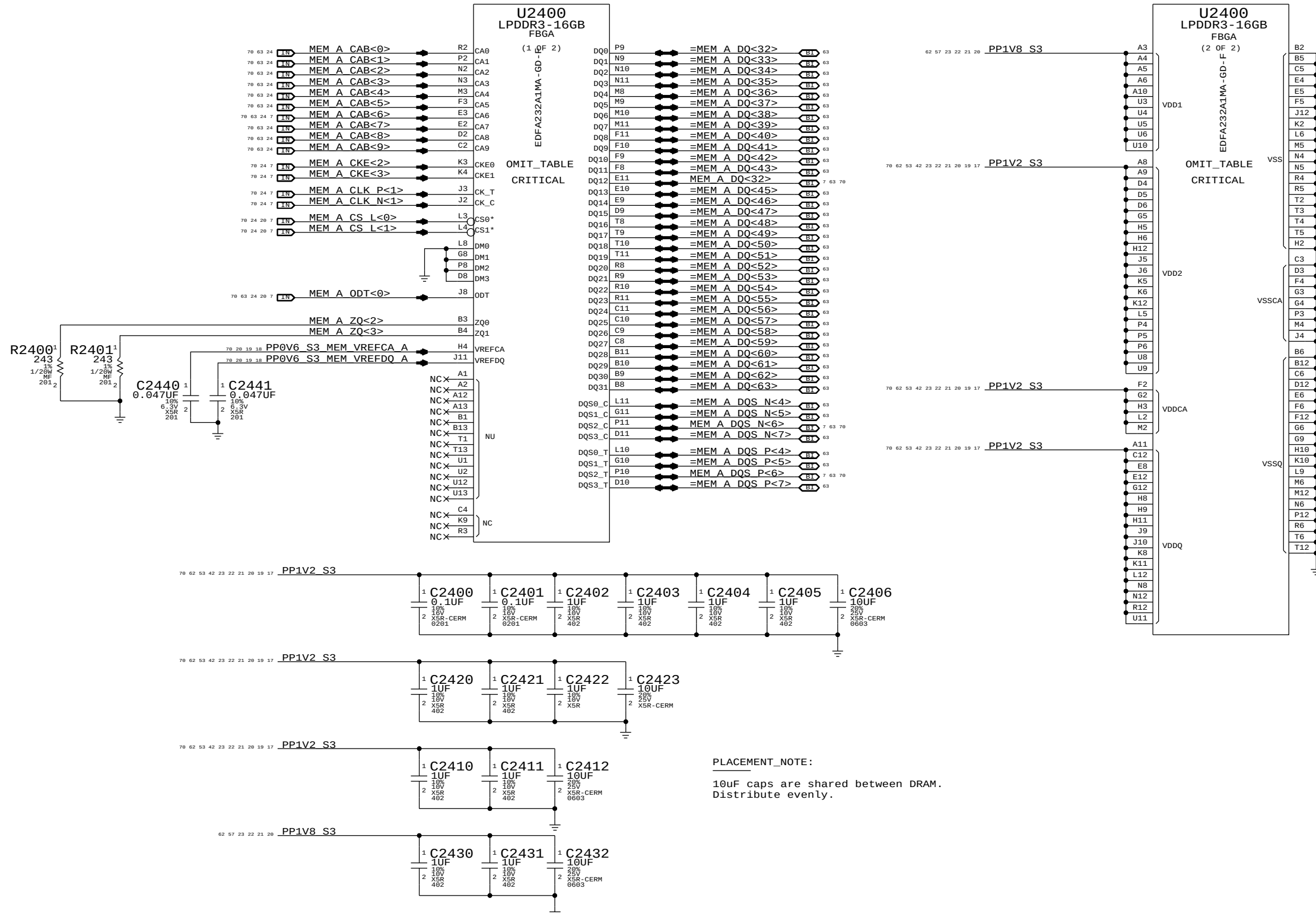
Always used, regardless of margining option.

A

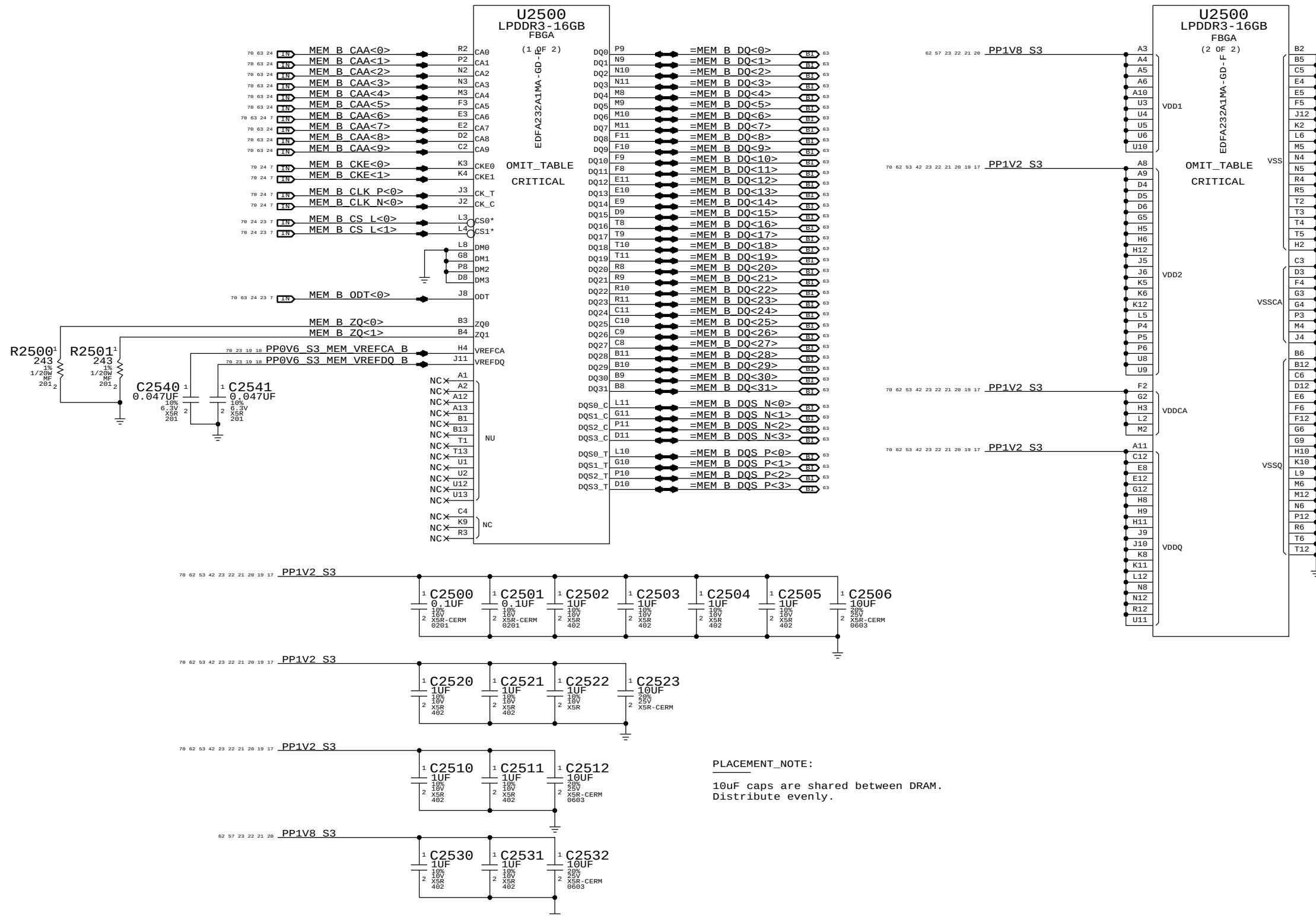
# LPDDR3 CHANNEL A (0-31)



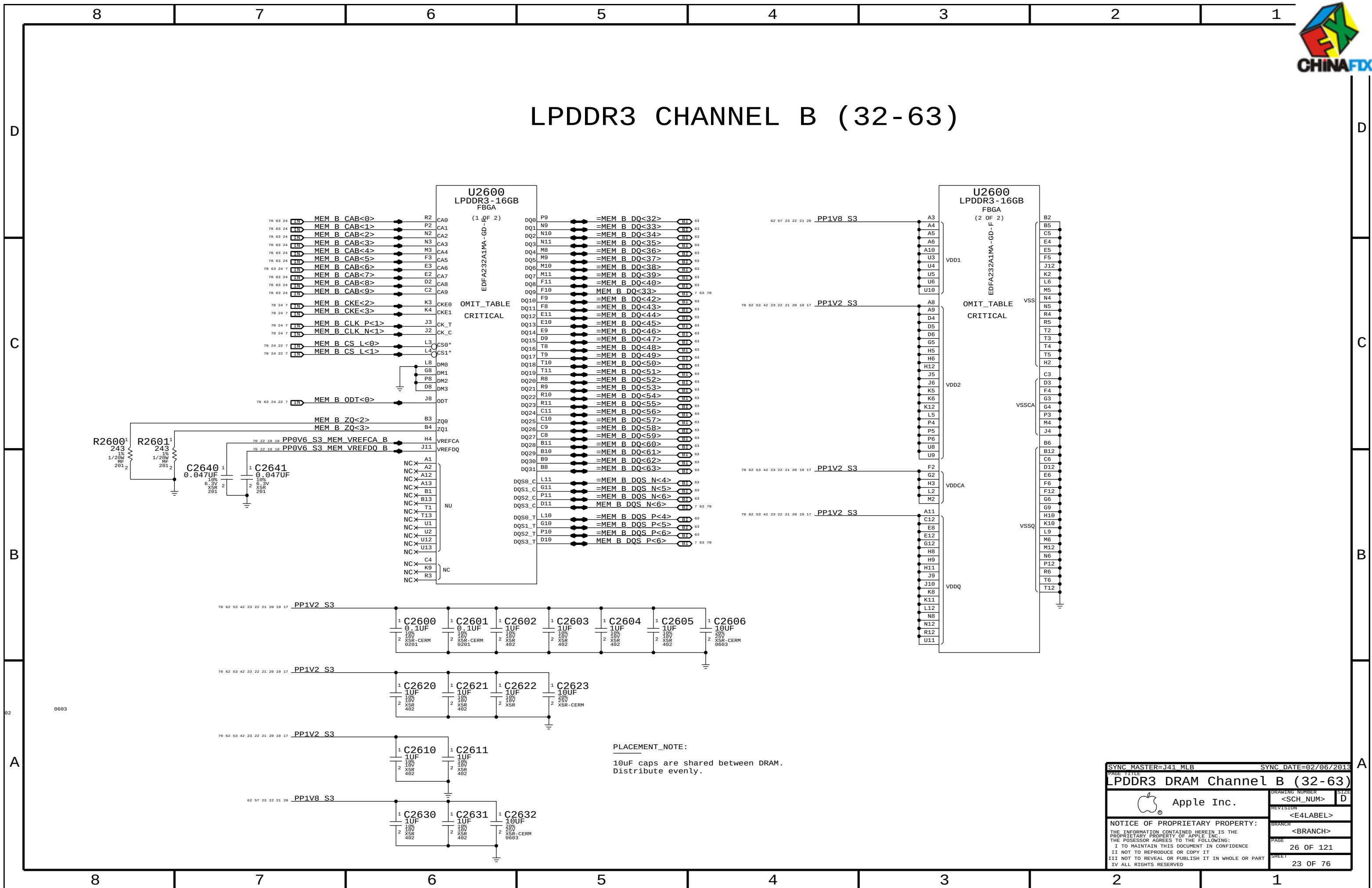
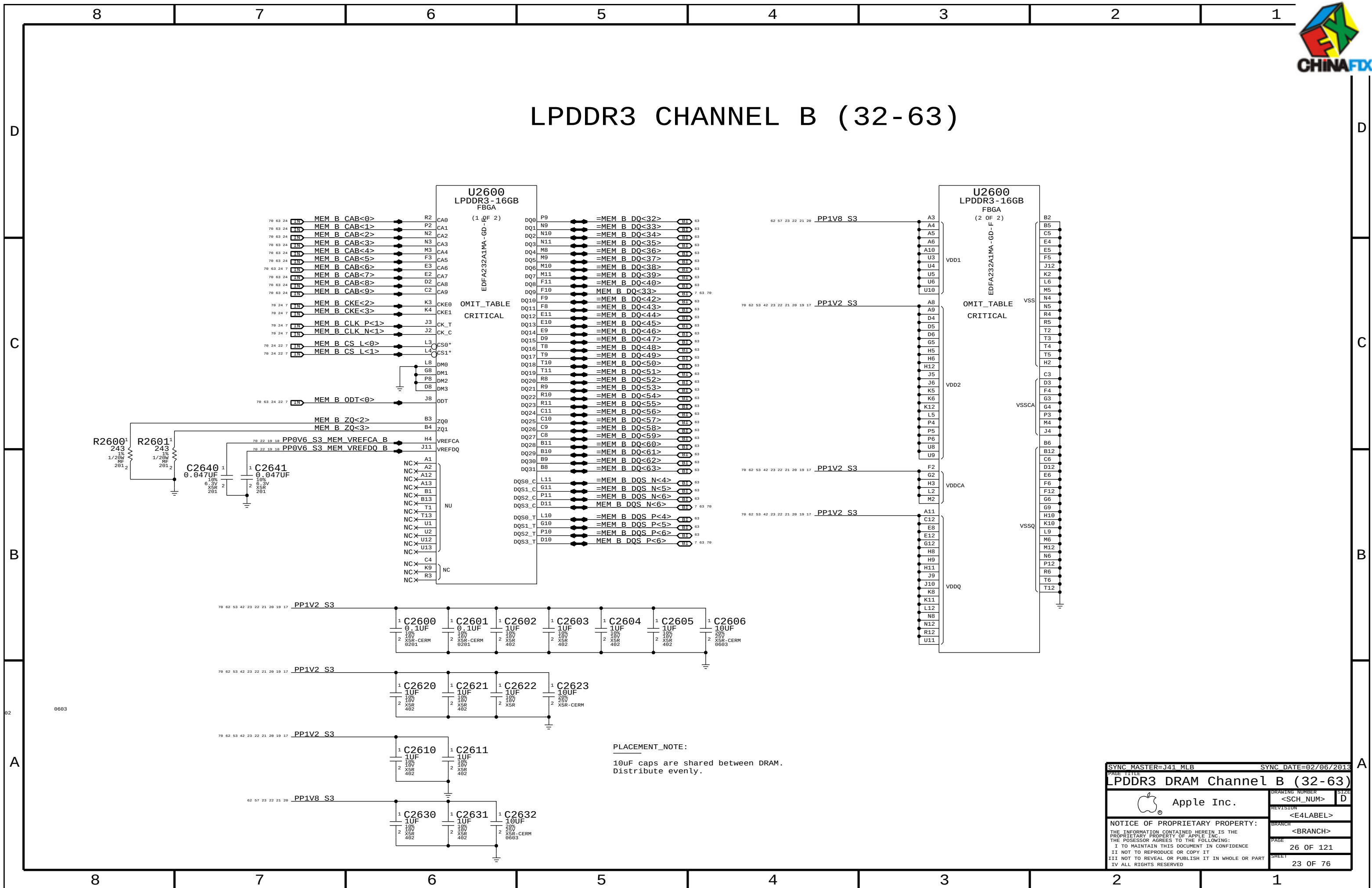
# LPDDR3 CHANNEL A (32-63)



# LPDDR3 CHANNEL B (0-31)

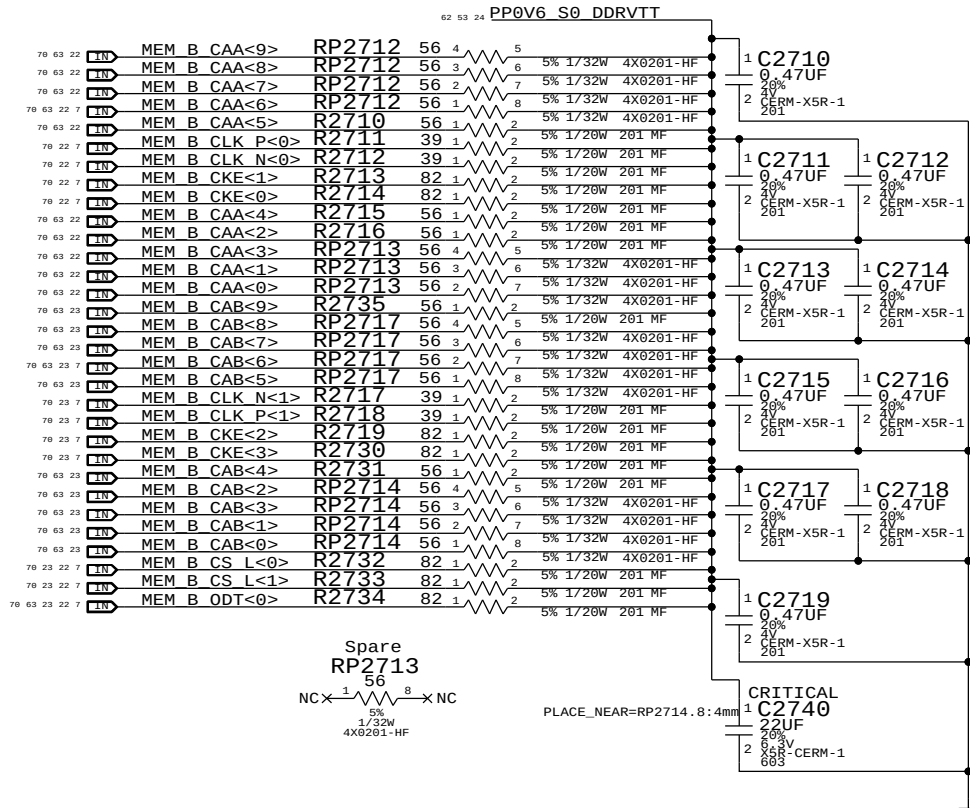
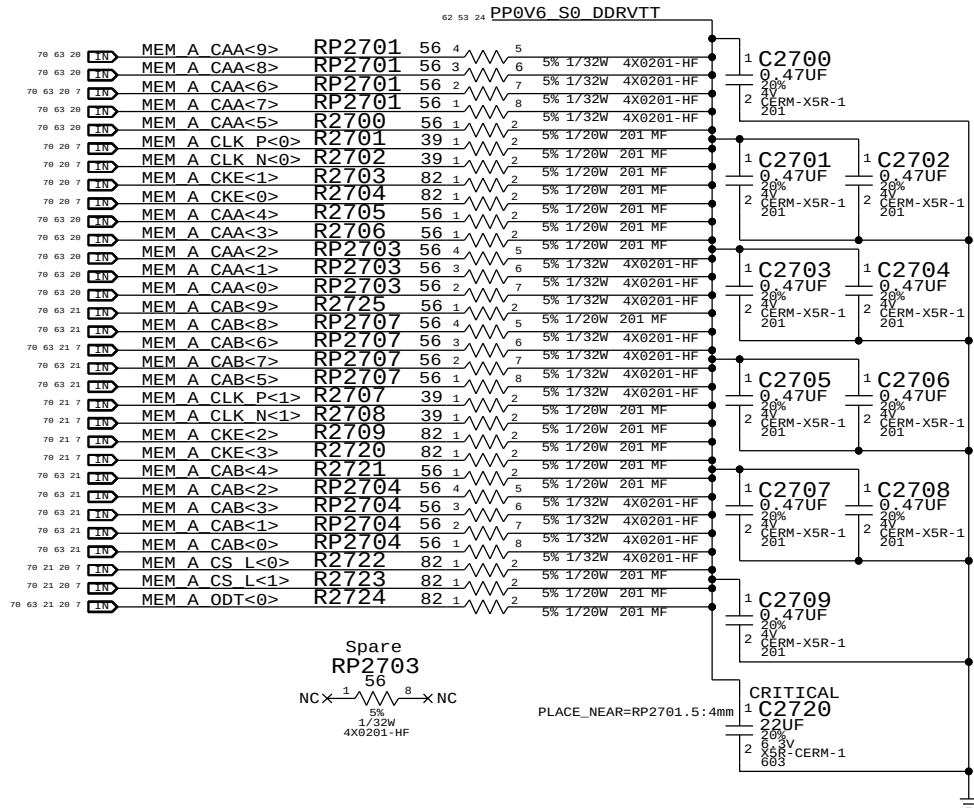




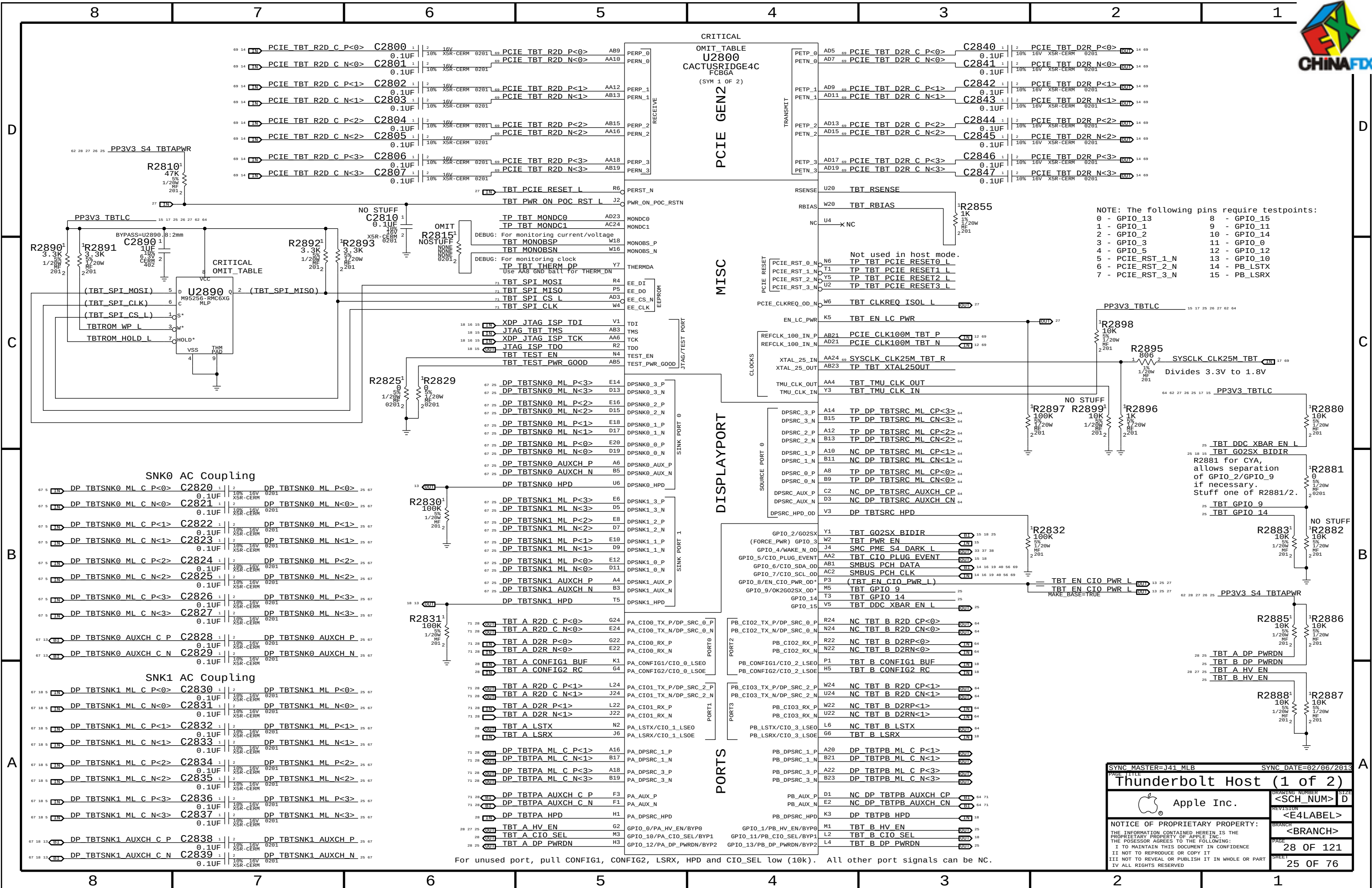
[illegible][illegible][illegible]

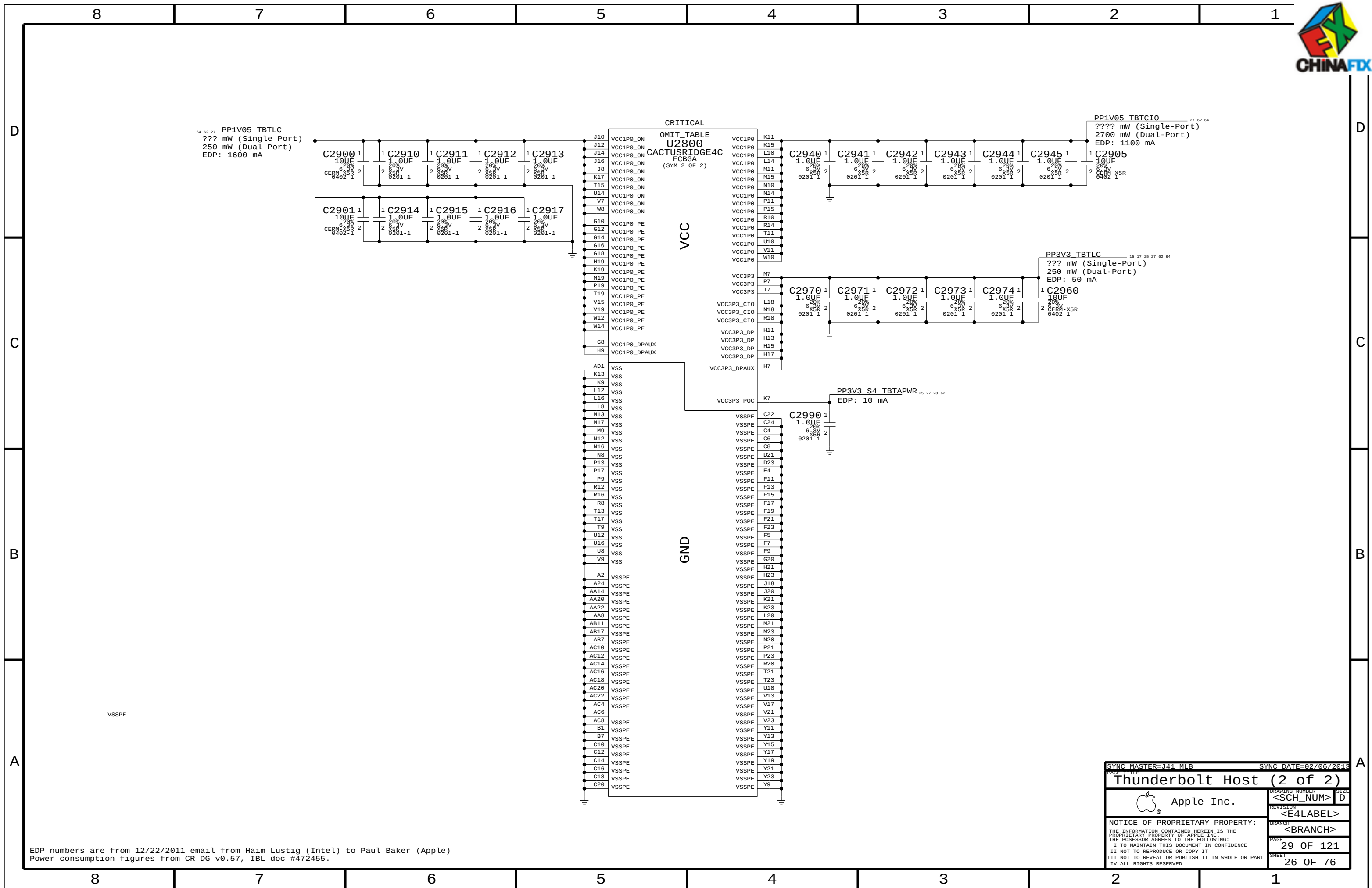


Intel recommends 55 Ohm for CMD/ADDR, 80 Ohm for CTRL/CKE, 38 Ohm for CLK



|                                                                                                                                                                                                                                                                                      |  |                      |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                  |  | SYNC DATE=02/06/2013 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                           |  |                      |           |
| LPDDR3 DRAM Termination                                                                                                                                                                                                                                                              |  |                      |           |
|  Apple Inc.                                                                                                                                                                                     |  | DRAWING NUMBER       | SIZE      |
|                                                                                                                                                                                                                                                                                      |  | <SCH_NUM>            | D         |
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EDP numbers are from 12/22/2011 email from Haim Lustig (Intel) to Paul Baker (Apple)  
Power consumption figures from CR DG v0.57, IBL doc #472455.

|                                                                                                                                                                                                                                                                                                                         |  |                           |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------|-----------|
| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                                                     |  | SYNC DATE=02/06/2013      |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  | Thunderbolt Host (2 of 2) |           |
| Apple Inc.                                                                                                                                                                                                                                                                                                              |  | DRAWING NUMBER            | SIZE      |
|                                                                                                                                                                                                                                                                                                                         |  | <SCH_NUM>                 | D         |
|                                                                                                                                                                                                                                                                                                                         |  | REVISION                  |           |
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|                                                                                                                                                                                                                                                                                                                         |  |                           |           |





## Page Notes

Power aliases required by this page:

- PPVIN\_SW\_TBTBST (8-13V Boost Input)
- PP15V\_TBT\_REG (15V Boost Output)
- PP3V3\_TBT\_P3V3TBTFTET (3.3V FET Input)
- PP3V3\_TBT\_FET (3.3V FET Output)
- PP3V3\_S0\_TBT\_PWRCTL (1.05V FET Input)
- PP1V05\_TBT\_P1V05TBTFTET (1.05V FET Input)
- PP1V05\_TBT\_FET (1.05V FET Output)

Signal aliases required by this page:

- TBT\_CLKREQ\_L
- TBT\_RESET\_L

BOM options provided by this page:

(NONE)

## TBT 15V Boost Regulator

SI8409DB:  
Vds(max): -30V  
Vgs(max): +/-12V  
Vgs(th): -1.4V  
Rds(on): 46mOhm @ 4.5V Vgs  
Id(max): 3.7A @ 70C

CRITICAL  
Q3080  
SI8409DB

PPVIN\_S4SW\_TBTBST FET  
Voltage not specified here,  
add property on another page.

CRITICAL  
L3095  
6.8UH-4.0A

CRITICAL  
D3095  
POWERDI-123

PP15V TBT  
Vout = 15.1V  
Max Current = 1.0A  
Freq = 300KHz

## Supervisor & CLKREQ# Isolation

## TBT "POC" Power-up Reset

## 3.3V TBT "LC" Switch

U3010  
TPS22924  
CSP  
Max Current = 2A (85C)

| Part  | TPS22924C                      |
|-------|--------------------------------|
| Type  | Load Switch                    |
| R(on) | 18.5 mOhm Typ<br>25.8 mOhm Max |

## 1.05V TBT "LC" Switch

U3015  
TPS22920  
CSP  
Max Current = 4A (85C)

| Part  | TPS22920                      |
|-------|-------------------------------|
| Type  | Load Switch                   |
| R(on) | 6.1 mOhm Typ<br>10.4 mOhm Max |

## 1.05V TBT "CIO" Switch

U3020  
TPS22920  
CSP  
Max Current = 4A (85C)

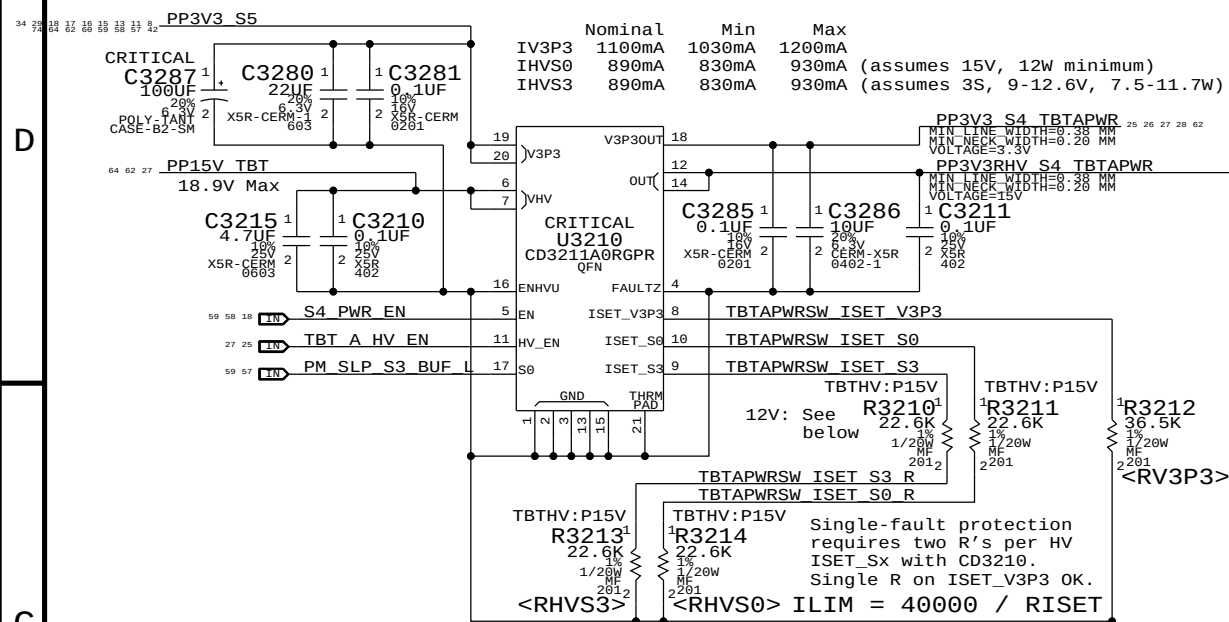
| Part  | TPS22920                      |
|-------|-------------------------------|
| Type  | Load Switch                   |
| R(on) | 6.1 mOhm Typ<br>10.4 mOhm Max |

|                                                                                                                                                                                                                                                                                      |  |                      |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                  |  | SYNC DATE=02/06/2013 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                           |  |                      |           |
| TBT Power Support                                                                                                                                                                                                                                                                    |  |                      |           |
| Apple Inc.                                                                                                                                                                                                                                                                           |  | DRAWING NUMBER       | SIZE      |
|                                                                                                                                                                                                                                                                                      |  | <SCH_NUM>            | D         |
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|                                                                                                                                                                                                                                                                                      |  | SHEET                | 27 OF 76  |



### 3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt device attach.

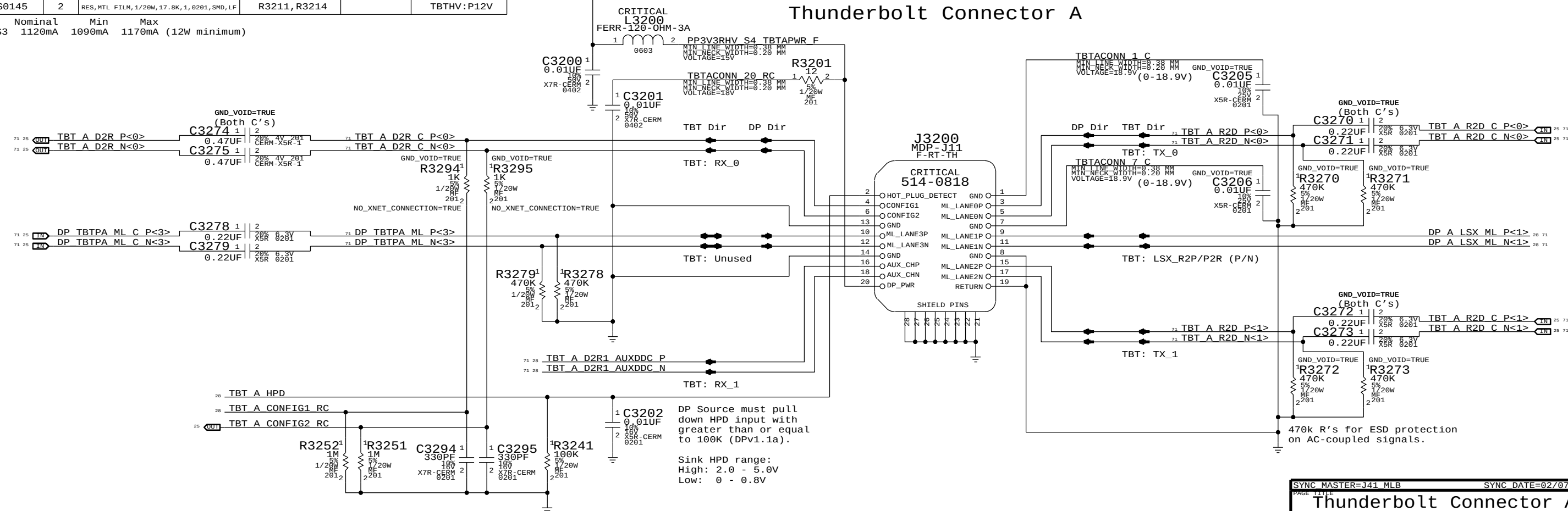


For 12V systems:

| PART NUMBER | QTY | DESCRIPTION                                   | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-----------------------------------------------|---------------|----------|------------|
| 118S0145    | 2   | RES, MTL FILM, 1/20W, 17.8K, 1, 0201, SMD, LF | R3210, R3213  |          | TBTHV:P12V |
| 118S0145    | 2   | RES, MTL FILM, 1/20W, 17.8K, 1, 0201, SMD, LF | R3211, R3214  |          | TBTHV:P12V |

|          |         |        |                      |
|----------|---------|--------|----------------------|
|          | Nominal | Min    | Max                  |
| IHVS0/S3 | 1120mA  | 1090mA | 1170mA (12W minimum) |

### Thunderbolt Connector A

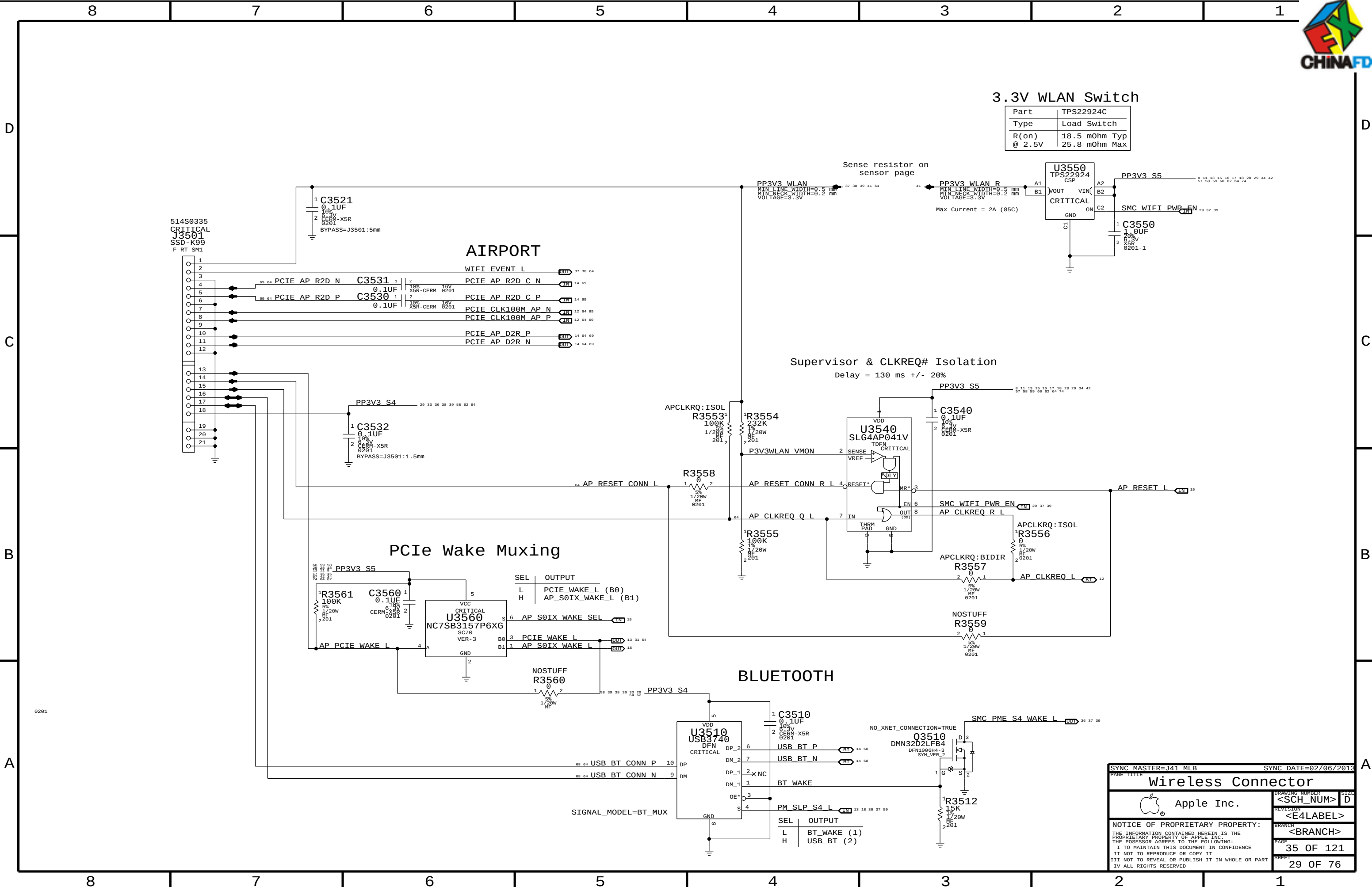


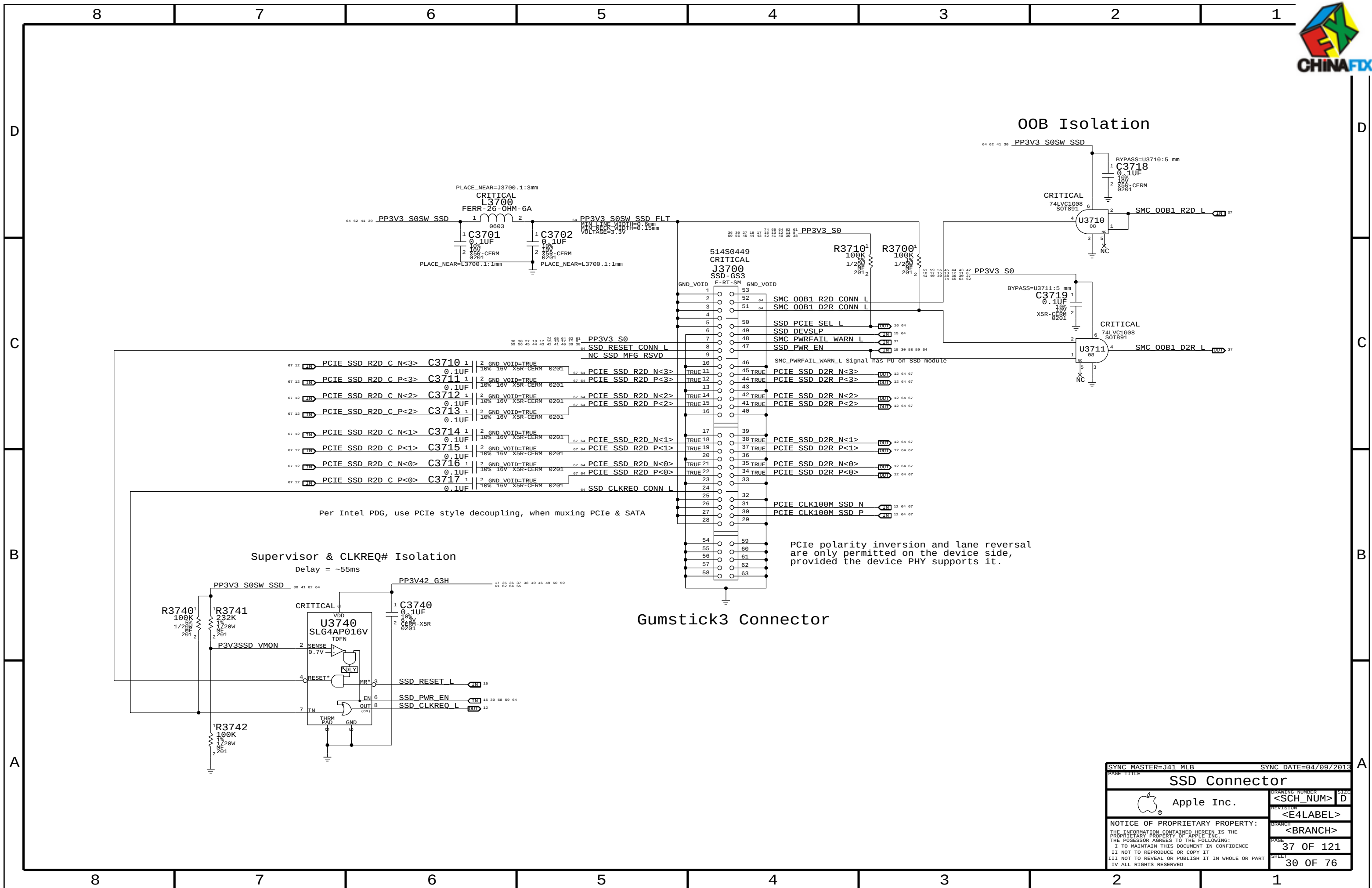
DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:  
High: 2.0 - 5.0V  
Low: 0 - 0.8V

|                                                                                                  |  |                      |      |
|--------------------------------------------------------------------------------------------------|--|----------------------|------|
| SYNC MASTER=J41 MLB                                                                              |  | SYNC DATE=02/07/2013 |      |
| PAGE 111                                                                                         |  |                      |      |
| Thunderbolt Connector A                                                                          |  |                      |      |
|  Apple Inc. |  | DRAWING NUMBER       | SIZE |
|                                                                                                  |  | <SCH_NUM>            | D    |
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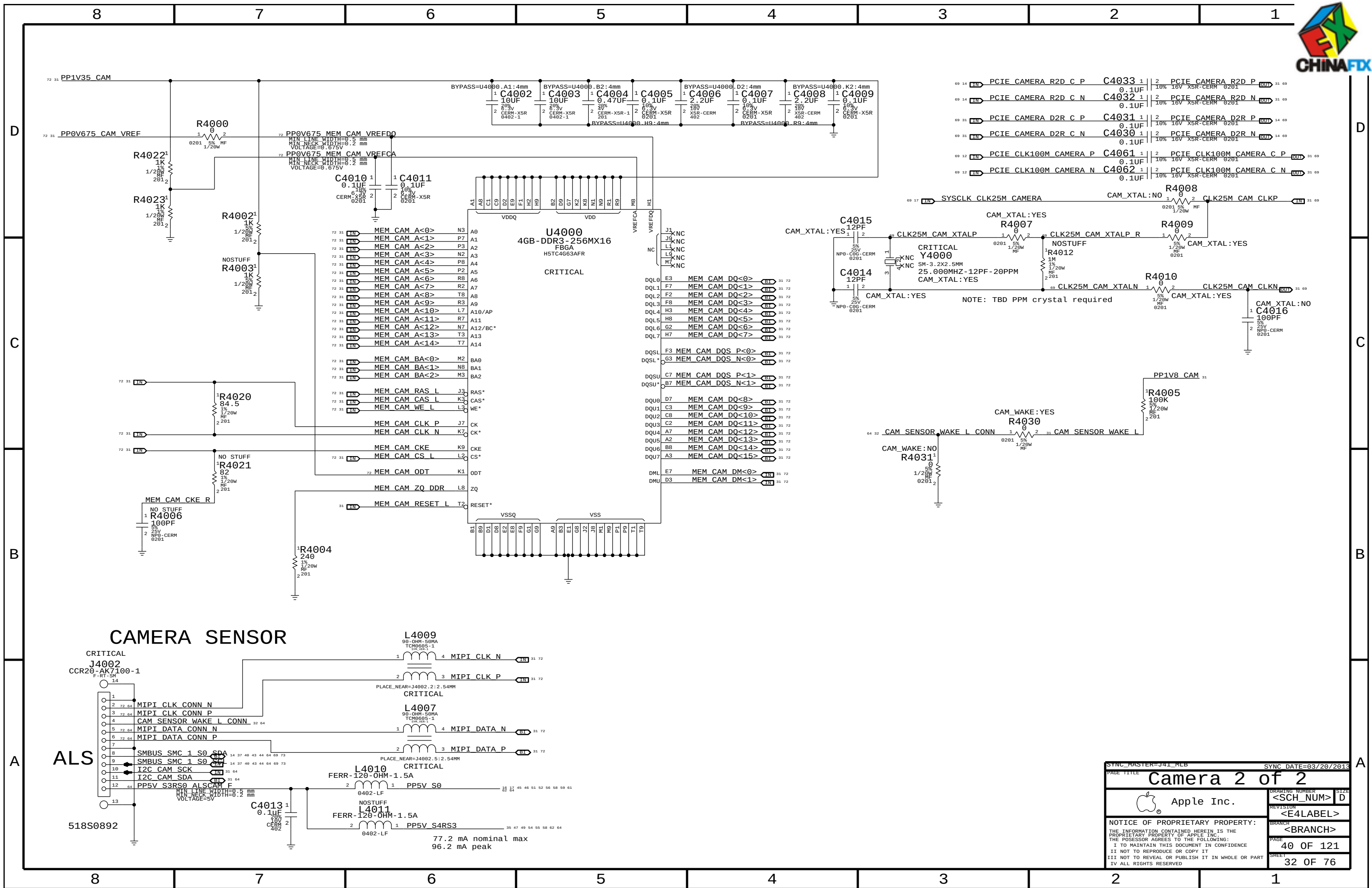
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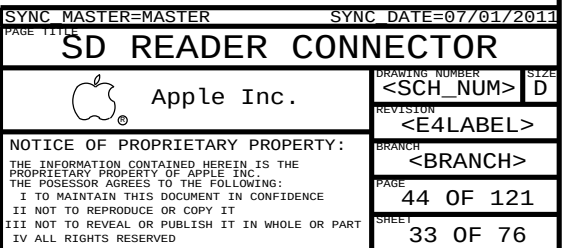












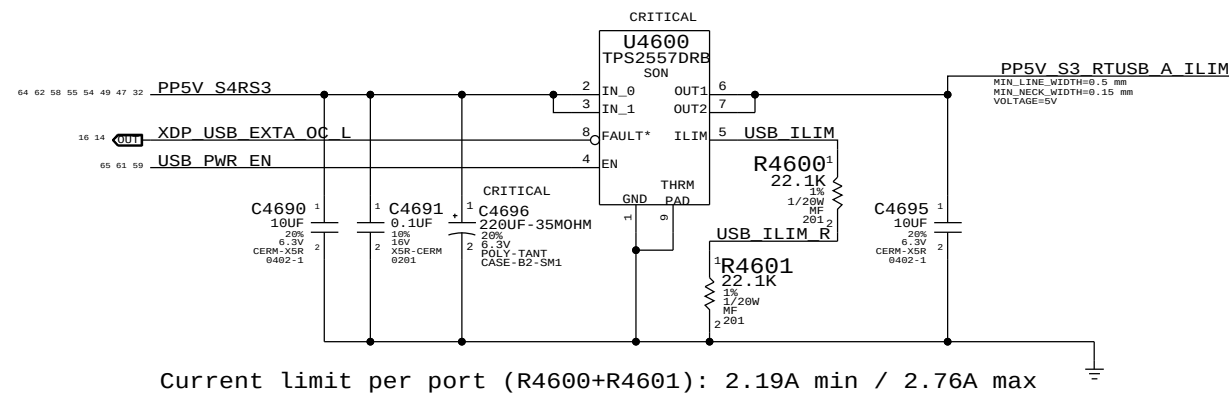




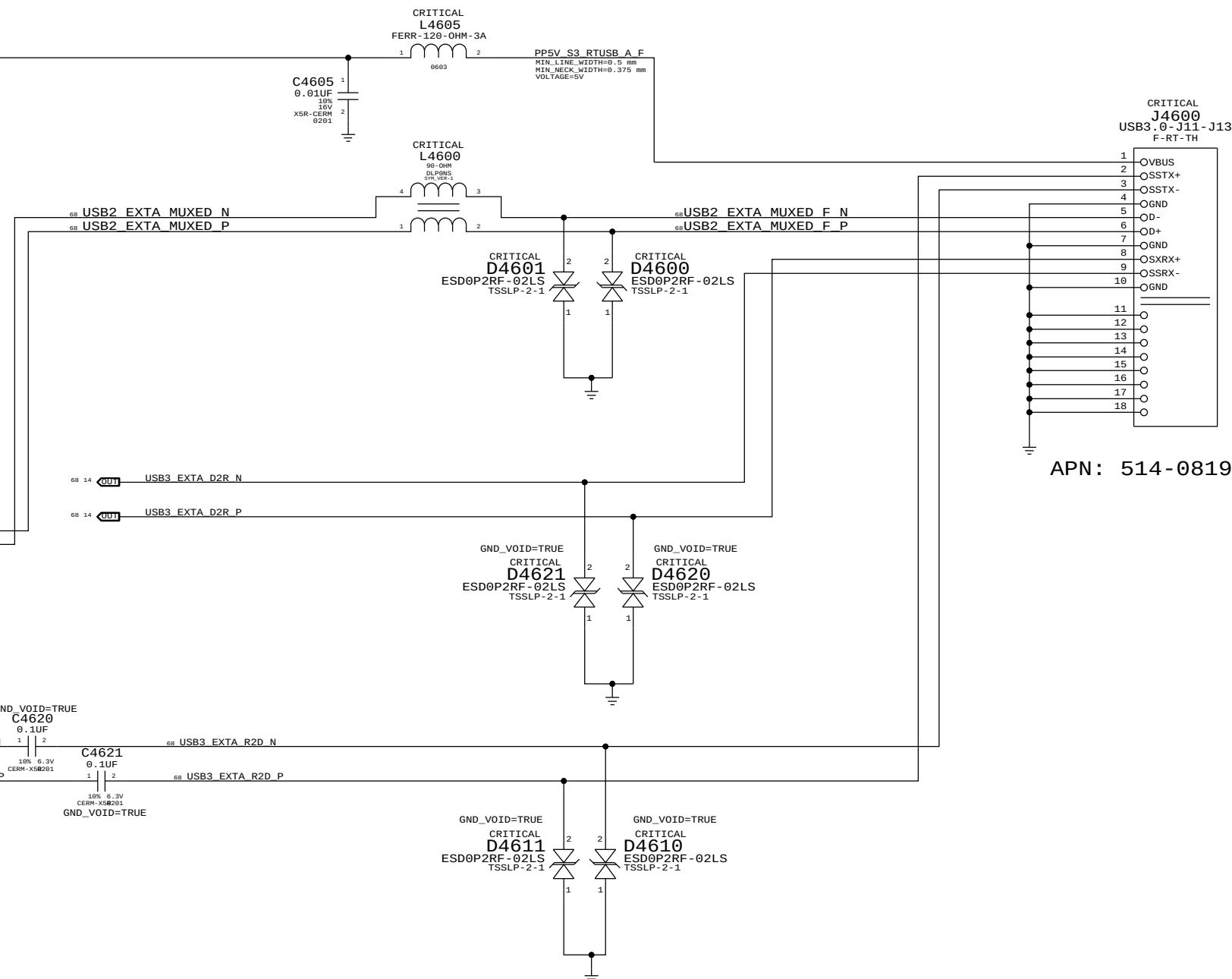
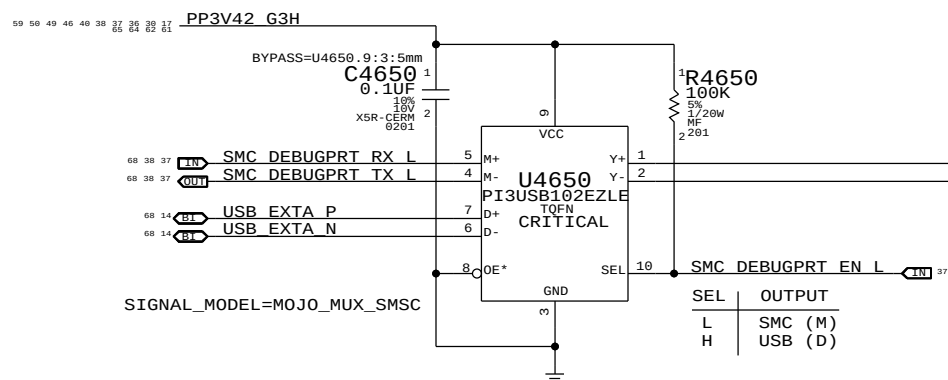


# Right USB Port A

## USB Port Power Switch



## Mojo SMC Debug Mux



APN: 514-0819

|                                                  |  |                      |  |
|--------------------------------------------------|--|----------------------|--|
| SYNC_MASTER=J41_MLB                              |  | SYNC_DATE=02/07/2013 |  |
| External A USB3 Connector                        |  | DRAWING NUMBER       |  |
| Apple Inc.                                       |  | <SCH_NUM> D          |  |
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# IPD Flex Connector

D

C

B

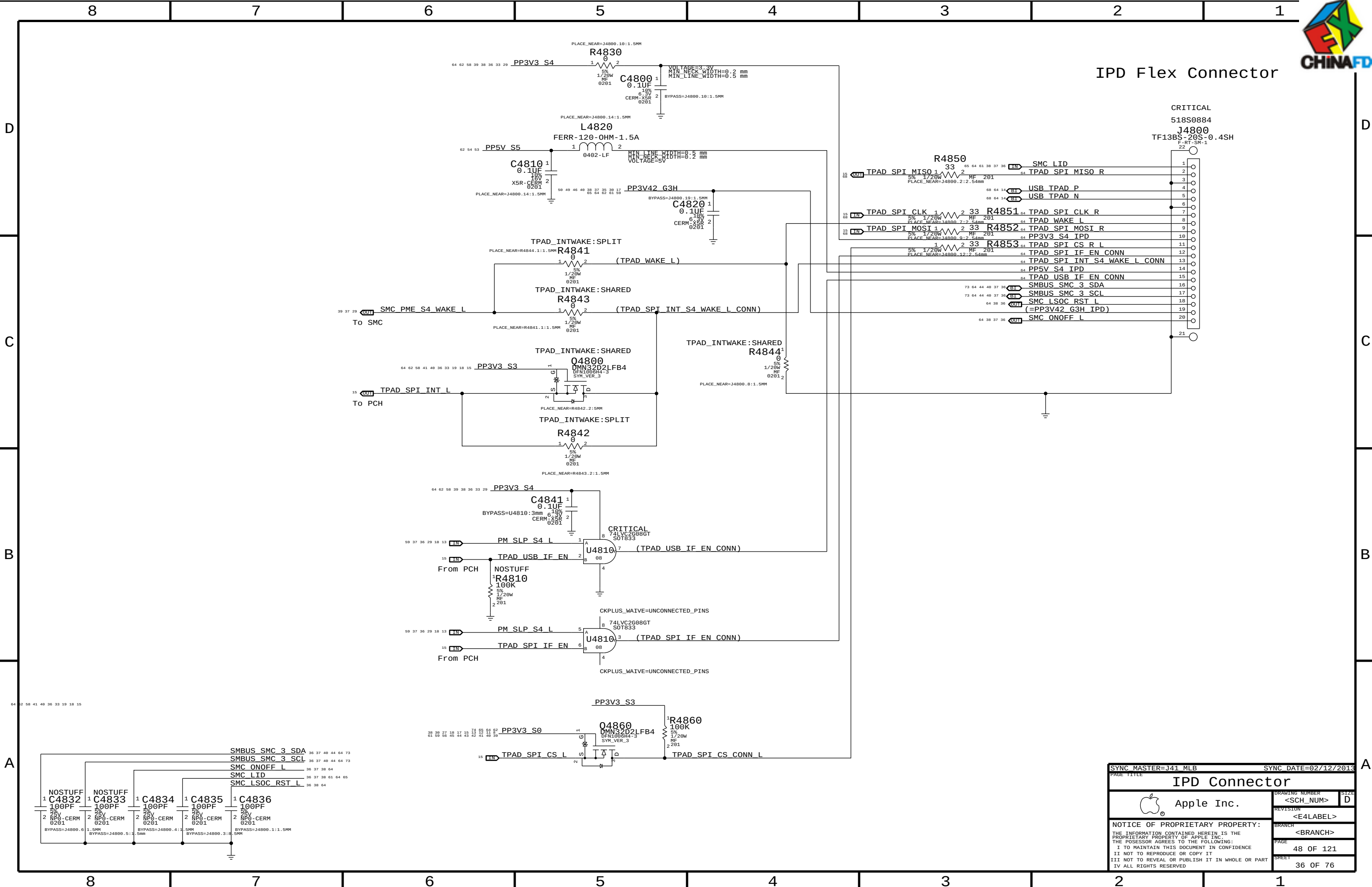
A

D

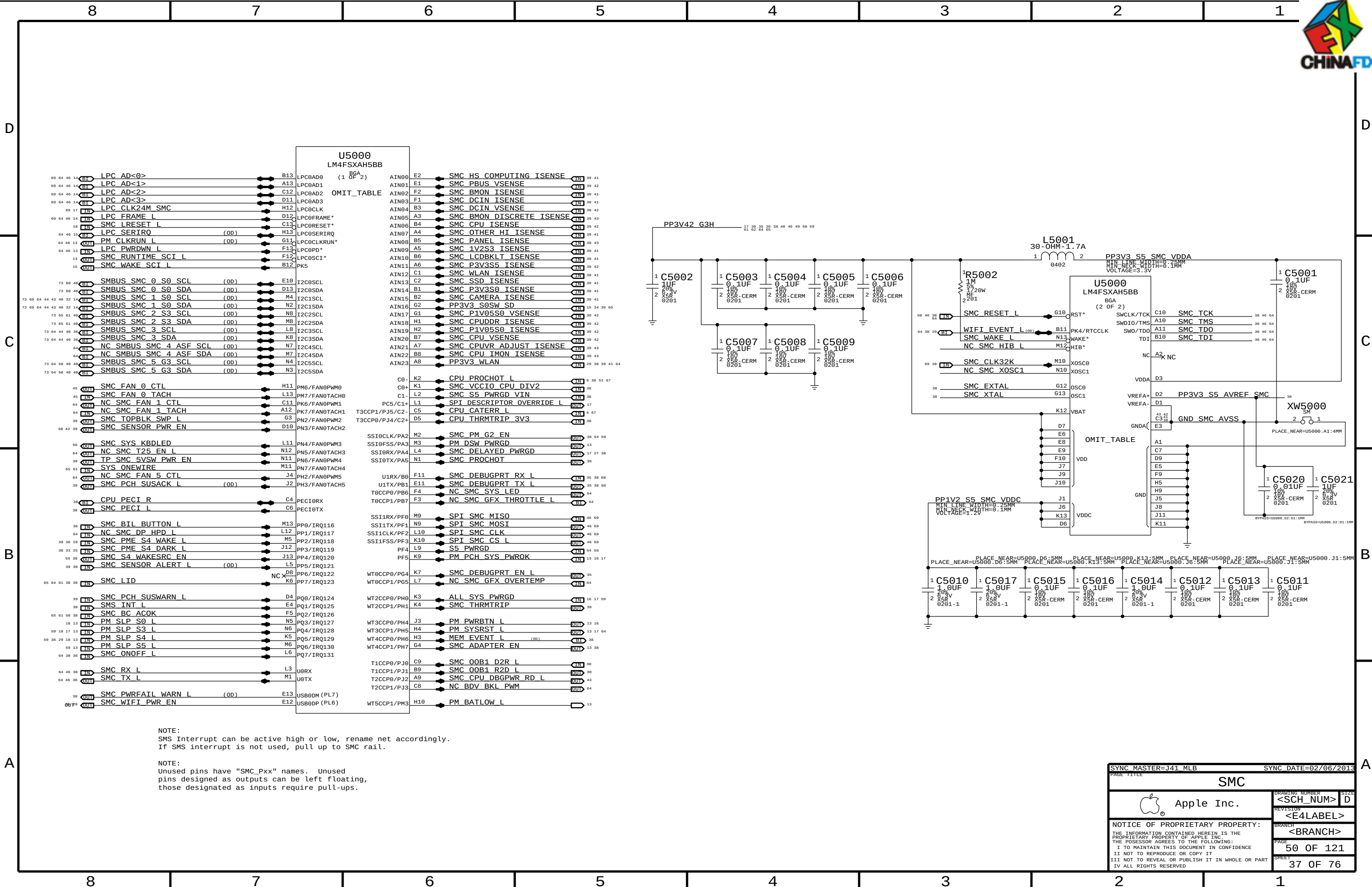
C

B

A



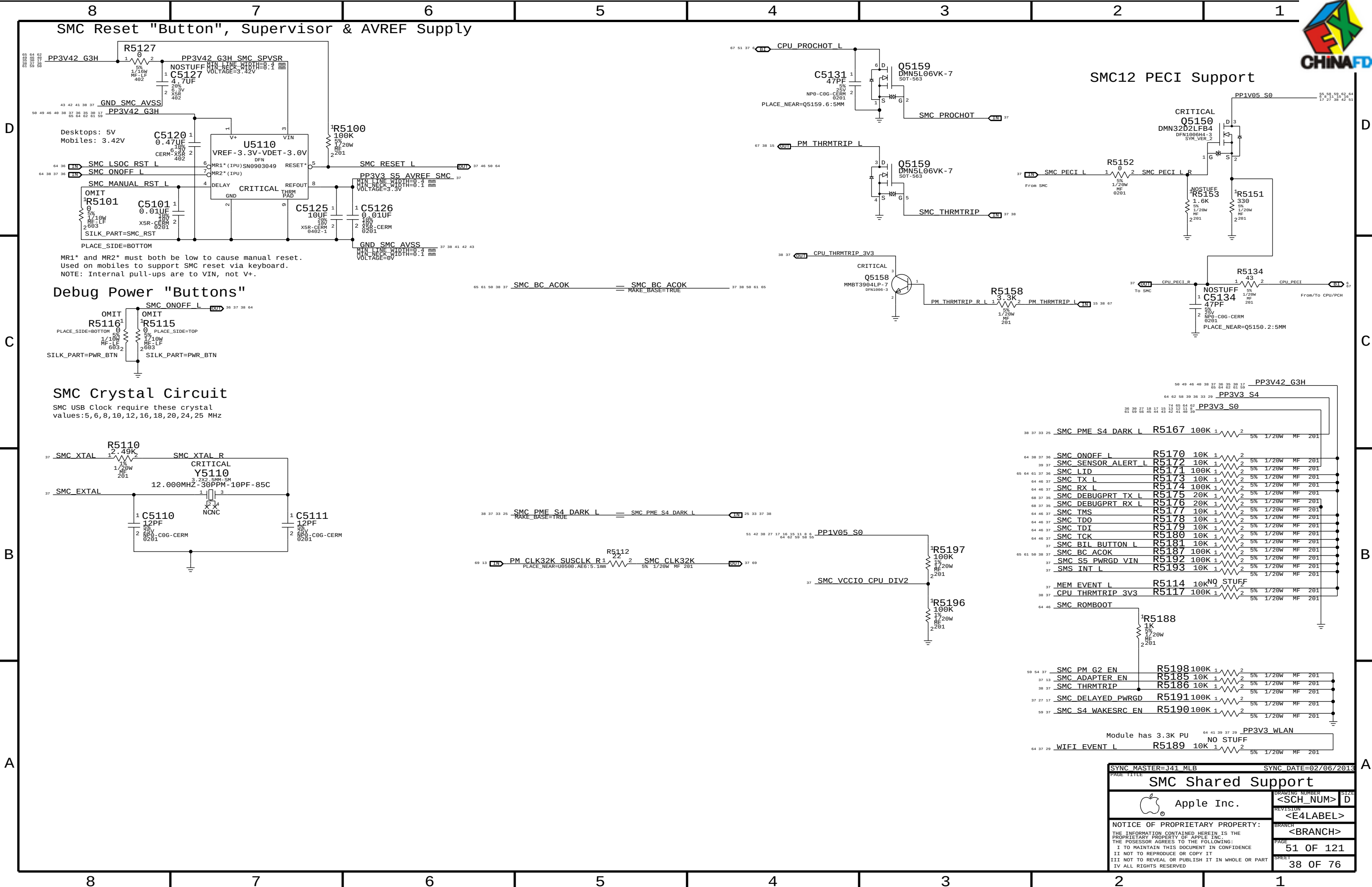
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| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                                                     |  | SYNC DATE=02/12/2013 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  |                      |           |
| IPD Connector                                                                                                                                                                                                                                                                                                           |  |                      |           |
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NOTE:  
SMS Interrupt can be active high or low, rename net accordingly.  
If SMS interrupt is not used, pull up to SMC rail.

NOTE:  
Unused pins have "SMC\_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

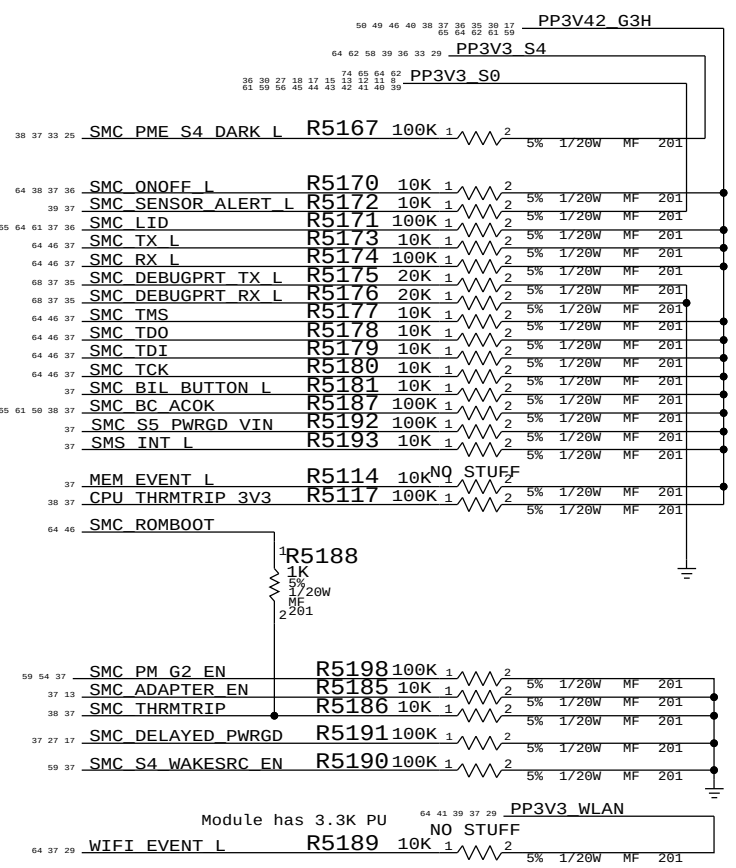
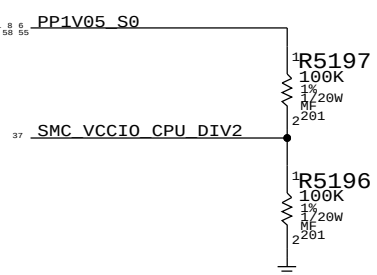
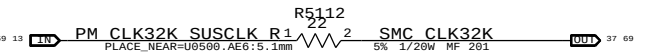
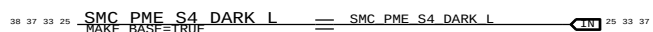
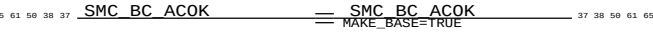
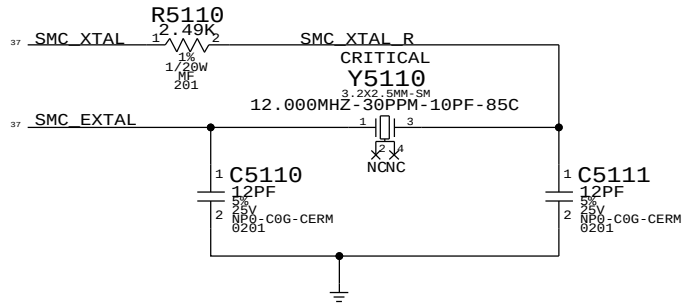
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MR1\* and MR2\* must both be low to cause manual reset.  
Used on internals to support SMC reset via keyboard.  
NOTE: Internal pull-ups are to VIN, not V+.

[illegible]

SMC USB Clock require these crystal values:5,6,8,10,12,16,18,20,24,25 MHz

[illegible]

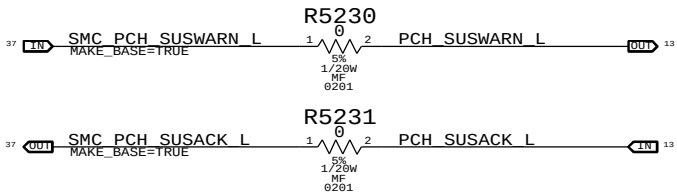
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|                                                                                                                                                                                                                                                                                       |  | <div> <div>SHEET</div> <div>38 OF 76</div> </div>            |  |



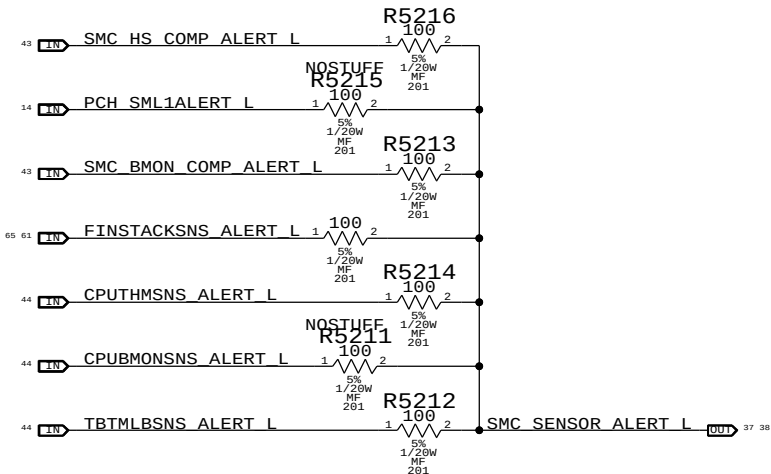
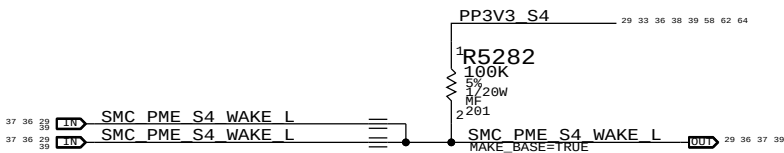
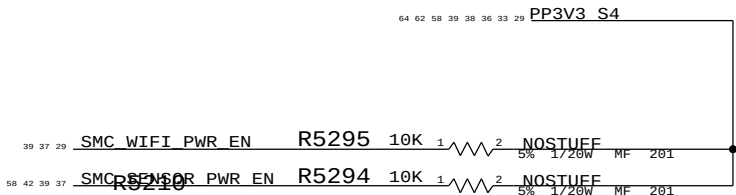
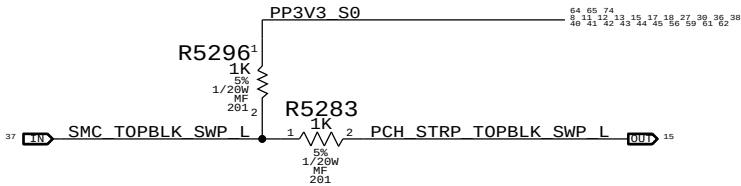


41 39 37 SMC HS COMPUTING ISENSE SMC HS COMPUTING ISENSE 37 39 41  
42 39 37 SMC PBUS VSENSE SMC PBUS VSENSE 37 39 42  
41 39 37 SMC BMON ISENSE SMC BMON ISENSE 37 39 41  
41 39 37 SMC DCIN ISENSE SMC DCIN ISENSE 37 39 41  
42 39 37 SMC DCIN VSENSE SMC DCIN VSENSE 37 39 42  
43 39 37 SMC BMON DISCRETE ISENSE SMC BMON DISCRETE ISENSE 37 39 43  
42 39 37 SMC CPU ISENSE SMC CPU ISENSE 37 39 42  
41 39 37 SMC OTHER HI ISENSE SMC OTHER HI ISENSE 37 39 41  
43 39 37 SMC PANEL ISENSE SMC PANEL ISENSE 37 39 43  
41 39 37 SMC 1V2S3 ISENSE SMC 1V2S3 ISENSE 37 39 41  
41 39 37 SMC LCDBKLT ISENSE SMC LCDBKLT ISENSE 37 39 41  
42 39 37 SMC P3V3S5 ISENSE SMC P3V3S5 ISENSE 37 39 42  
41 39 37 SMC WLAN ISENSE SMC WLAN ISENSE 37 39 41  
41 39 37 SMC SSD ISENSE SMC SSD ISENSE 37 39 41  
41 39 37 SMC P3V3S0 ISENSE SMC P3V3S0 ISENSE 37 39 41  
41 39 37 SMC CAMERA ISENSE SMC CAMERA ISENSE 37 39 41  
PP3V3 S0SW SD SD alias on page 103  
42 39 37 SMC P1V05S0 VSENSE SMC P1V05S0 VSENSE 37 39 42  
42 39 37 SMC CPUDDR ISENSE SMC CPUDDR ISENSE 37 39 42  
42 39 37 SMC P1V05S0 ISENSE SMC P1V05S0 ISENSE 37 39 42  
42 39 37 SMC CPU VSENSE SMC CPU VSENSE 37 39 42  
43 39 37 SMC CPUVR ADJUST ISENSE SMC CPUVR ADJUST ISENSE 37 39 43  
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64 41 39 38 37 29 PP3V3 WLAN 29 37 38 39 41 64

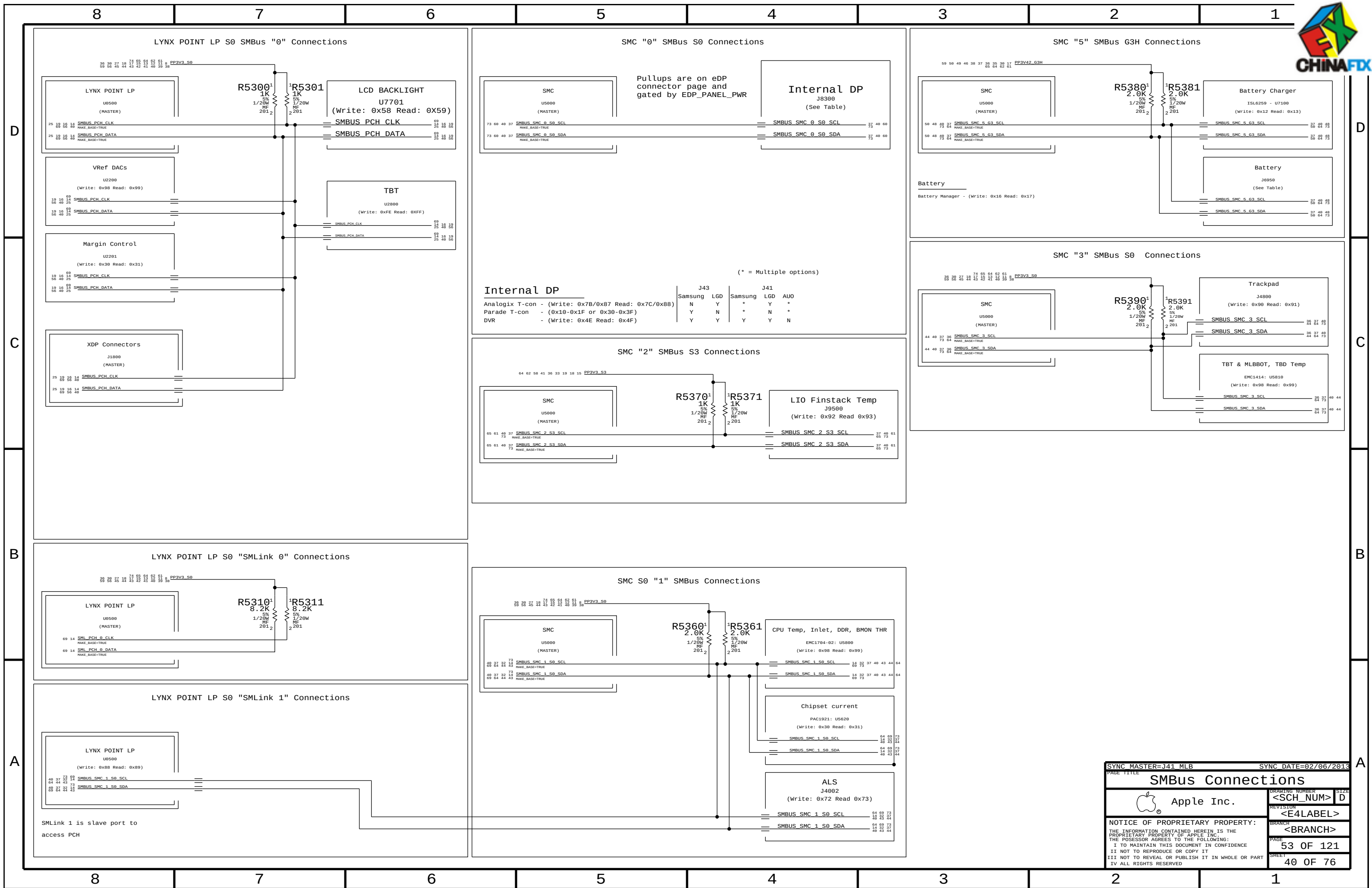
58 42 39 37 SMC SENSOR PWR EN SMC SENSOR PWR EN 37 39 42 58  
39 37 29 SMC WIFI PWR EN SMC WIFI PWR EN 29 37 39  
39 37 TP SMC 5VSW PWR EN TP SMC 5VSW PWR EN 37 39



### Top-Block Swap



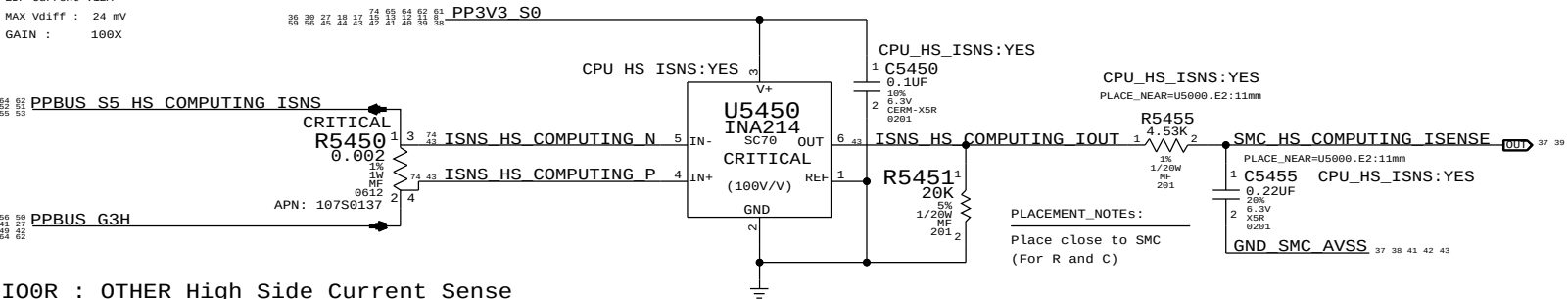
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| SMC Project Support                                                                                                                                                                                                                                                                                               |  |                      |           |
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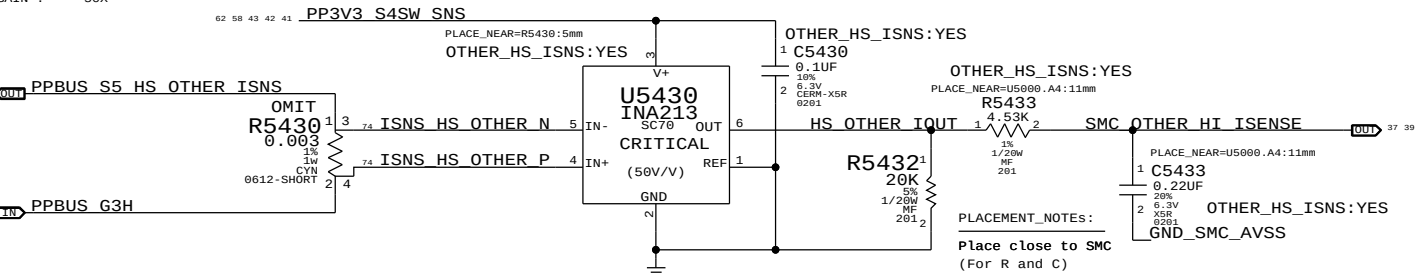
### IC0R : COMPUTING High Side Current Sense

EDP Current :12A  
MAX Vdiff : 24 mV  
GAIN : 100X



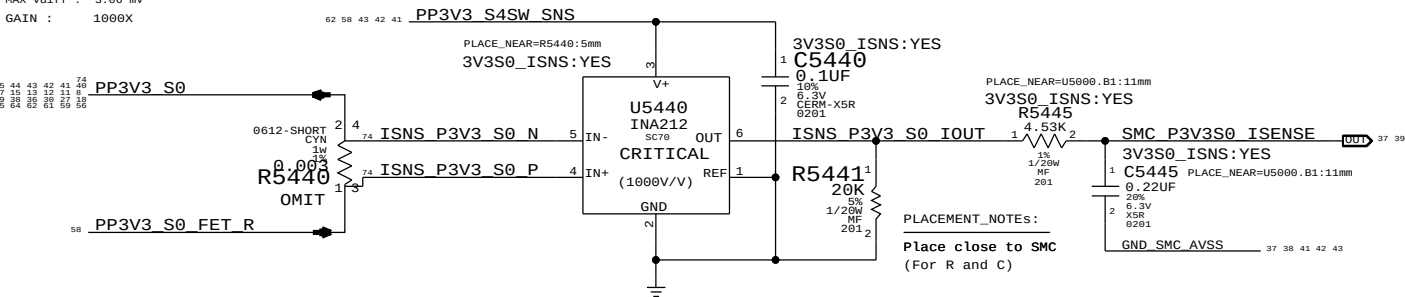
### I00R : OTHER High Side Current Sense

EDP Current :10.75A  
MAX Vdiff : 53.75 mV  
GAIN : 50X



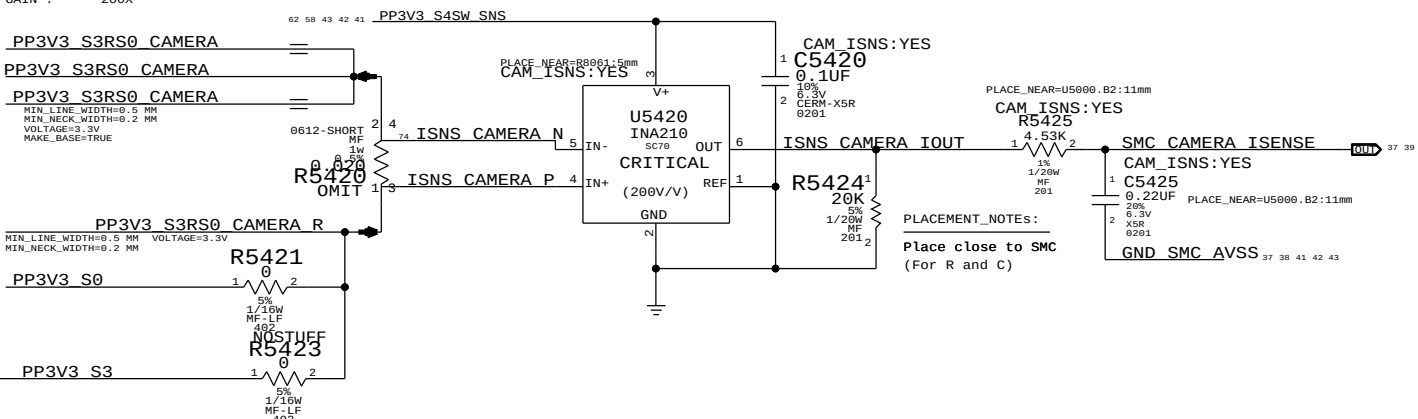
### IR0C : 3.3V S0 FET Current Sense

EDP Current :1.02A  
MAX Vdiff : 3.06 mV  
GAIN : 1000X

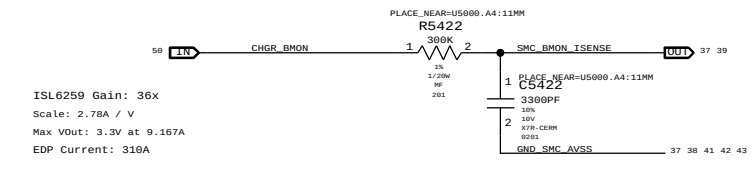


### IS2C : 3.3V Camera Current Sense

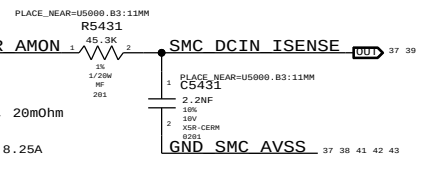
EDP Current : 0.82A  
MAX Vdiff : 16.36 mV  
GAIN : 200X



### CHARGER BMON High Side Current Sense



### DC-IN (AMON) Current Sense

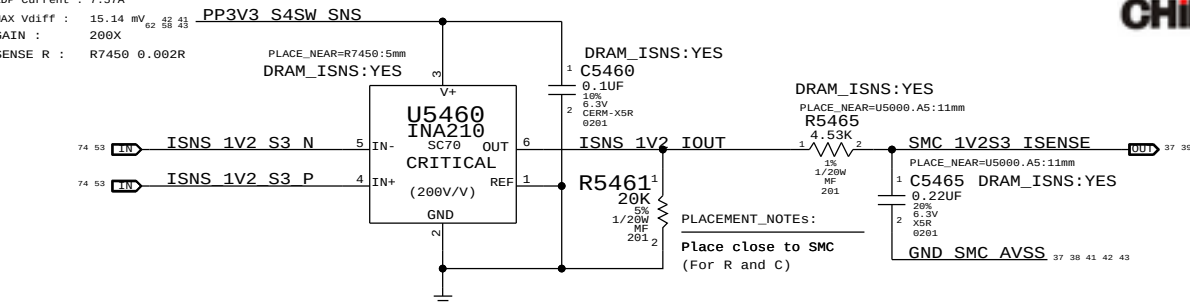


### Replacing caps with 100K PD on ISENSE SMC inputs

| PART NUMBER | QTY | DESCRIPTION                      | REFERENCE DES | CRITICAL | BOM OPTION       |
|-------------|-----|----------------------------------|---------------|----------|------------------|
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5455         |          | CPU_HS_ISNS:NO   |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5465         |          | DRAM_ISNS:NO     |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5475         |          | AIRPORT_ISNS:NO  |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5485         |          | SSD_ISNS:NO      |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5495         |          | LCDBKLT_ISNS:NO  |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5433         |          | OTHER_HS_ISNS:NO |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5425         |          | CAM_ISNS:NO      |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5445         |          | 3V3S0_ISNS:NO    |

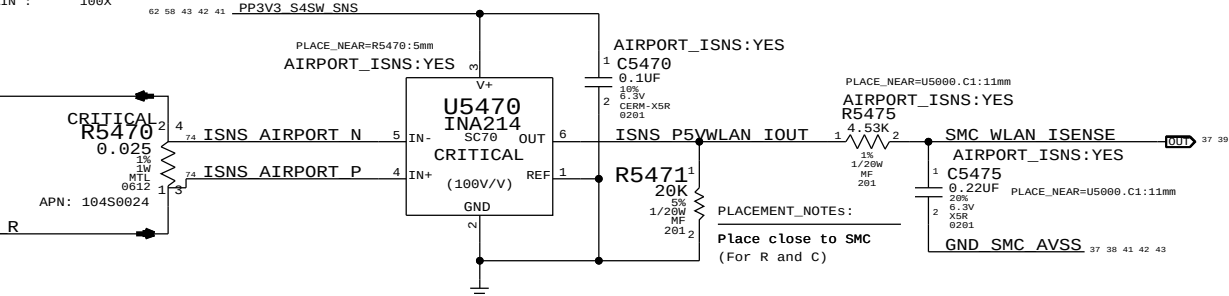
### IM3C :DDR 1V2 Current Sense (LPDDR + CPUDDR)

EDP Current : 7.57A  
MAX Vdiff : 15.14 mV  
GAIN : 200X  
SENSE R : R7450 0.002R



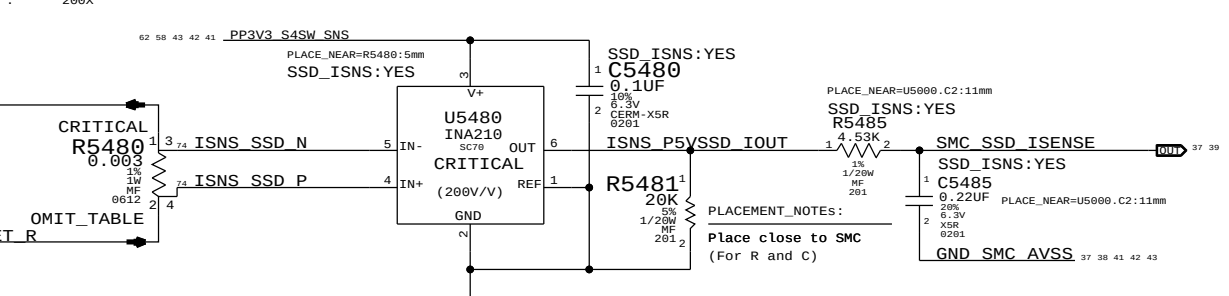
### IAPC :AirPort Current Sense

EDP Current : 1.00A  
MAX Vdiff : 25 mV  
GAIN : 100X



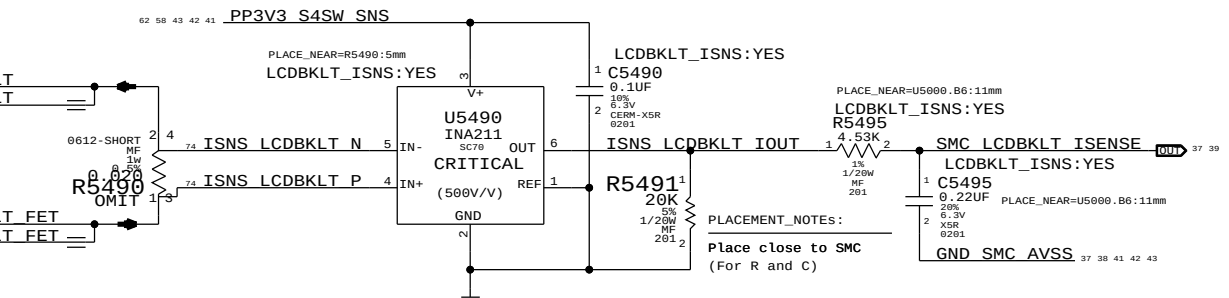
### ISDC : SSD Current Sense

EDP Current : 3.00A  
MAX Vdiff : 15 mV  
GAIN : 200X



### IBLC : LCD Backlight Driver Input Current Sense

EDP Current : 0.67A  
MAX Vdiff : 0.06 mV  
GAIN : 500X

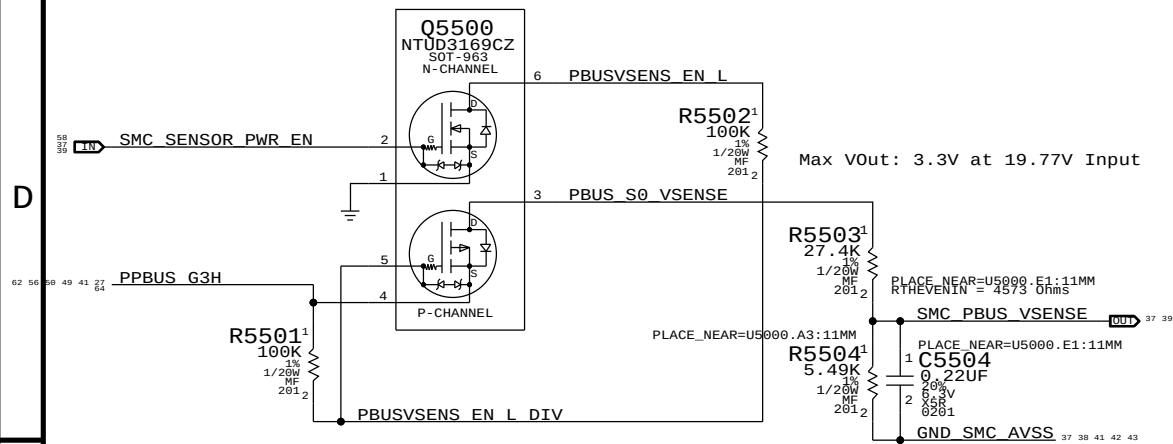


| PART NUMBER | QTY | DESCRIPTION                              | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|------------------------------------------|---------------|----------|------------|
| 107S0248    | 1   | RES,SENSE,0.0030HM,1W,4-TERM,1%,0612,TFT | R5480         | CRITICAL |            |

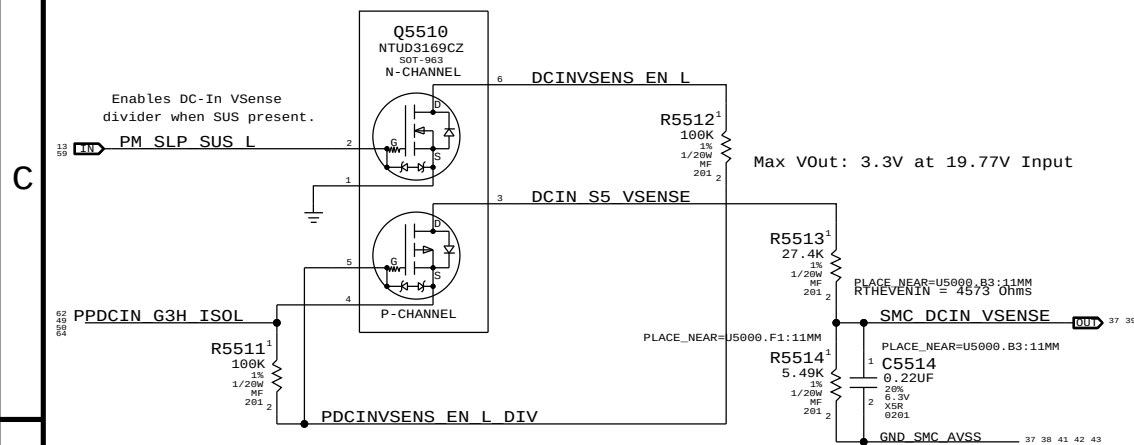
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| SYNC MASTER=J41 MLB                                                                                               |  | SYNC DATE=03/28/2013      |           |
| PAGE 1/11                                                                                                         |  | High Side Current Sensing |           |
| Apple Inc.                                                                                                        |  | DRAWING NUMBER            | SIZE      |
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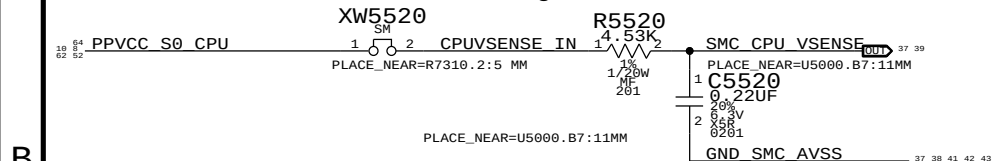
## VP0R: PBUS Voltage Sense Enable & Filter



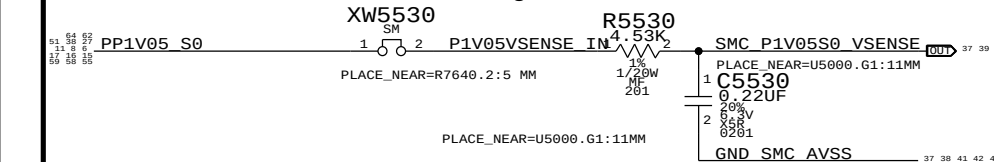
## VD0R: DC-In Voltage Sense Enable & Filter



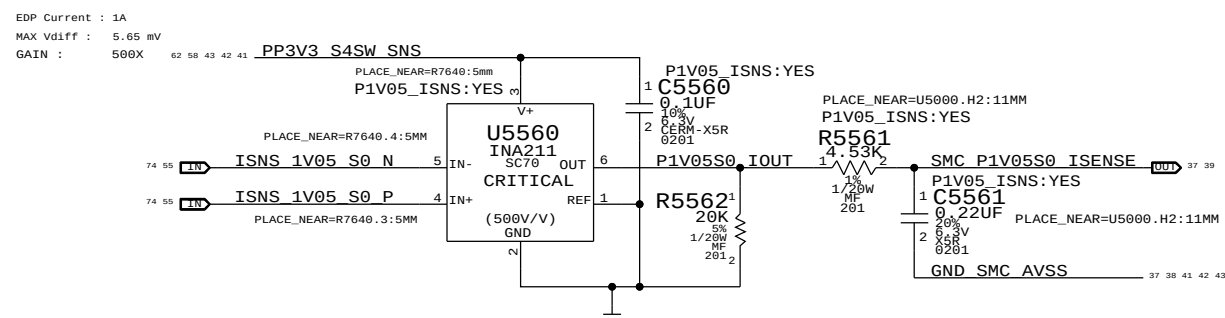
## CPU Vcore Voltage Sense / Filter



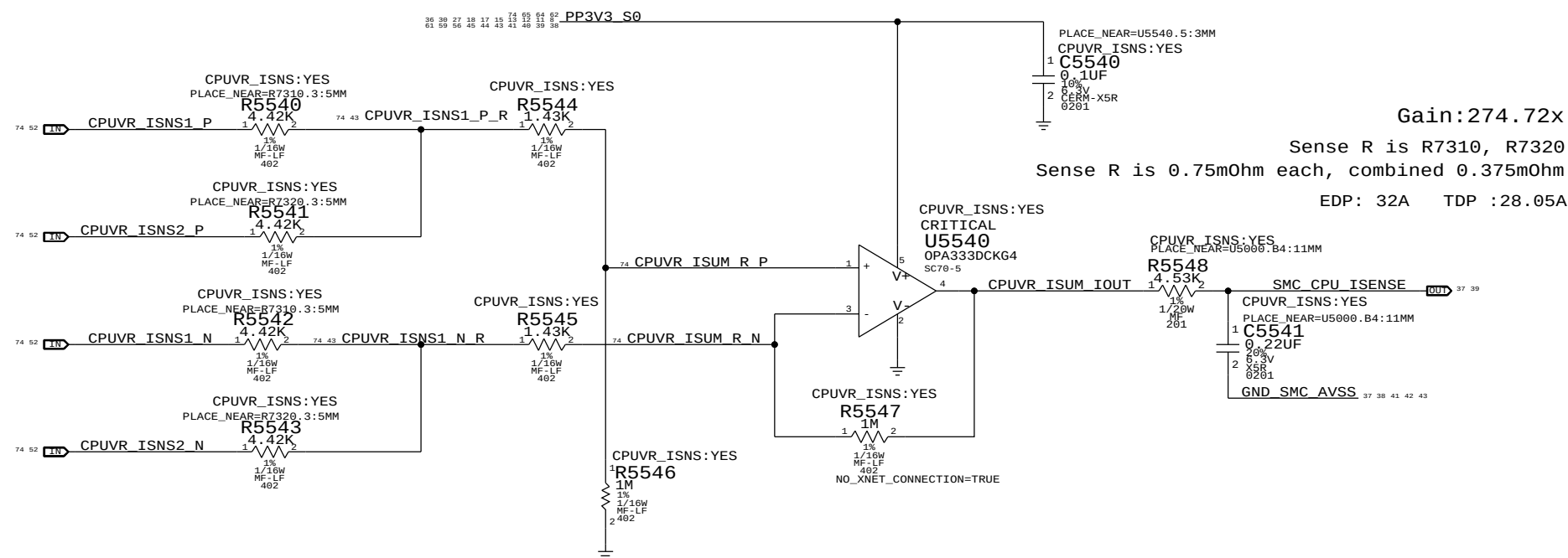
## 1.05V Voltage Sense / Filter



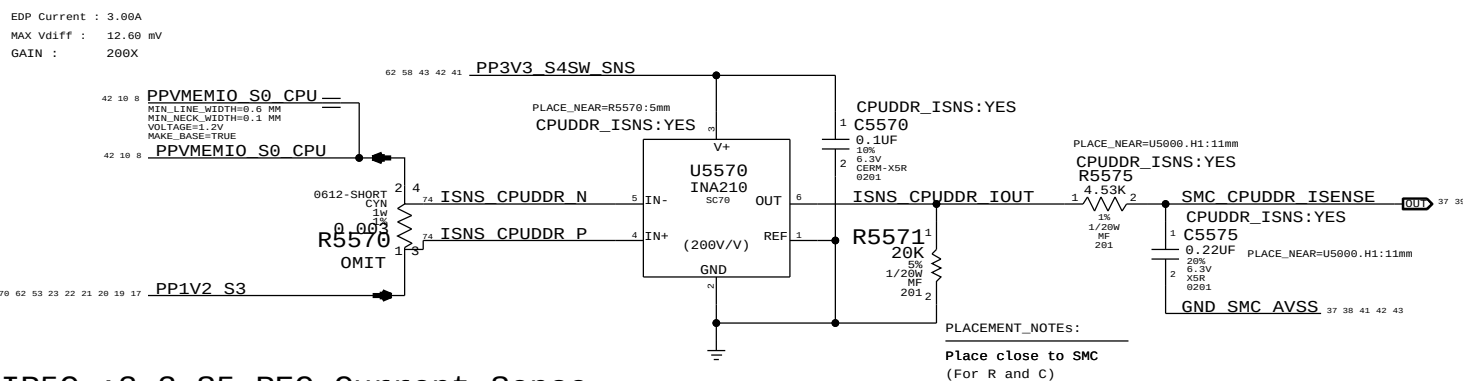
## IC1C: 1.05V S0 CURRENT SENSE / FILTER



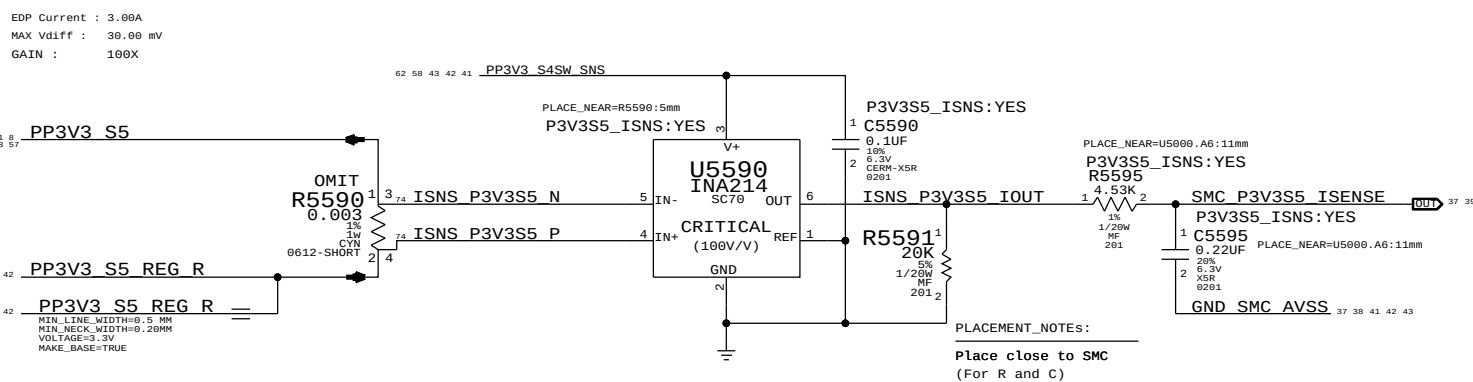
## ICS0 : CPU VCore Load Side Current Sense



## IM0C : CPU DDR Current Sense



## IR5C :3.3 S5 REG Current Sense



## Replacing caps with 100K PD on ISENSE SMC inputs

| PART NUMBER | QTY | DESCRIPTION                      | REFERENCE DES | CRITICAL | BOM OPTION     |
|-------------|-----|----------------------------------|---------------|----------|----------------|
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5541         |          | CPUVR_ISNS:NO  |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5561         |          | P1V05_ISNS:NO  |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5595         |          | P3V3S5_ISNS:NO |
| 117S0008    | 1   | RES,MF,1/20W,100K OHM,5,0201,SMD | C5575         |          | CPUDDR_ISNS:NO |

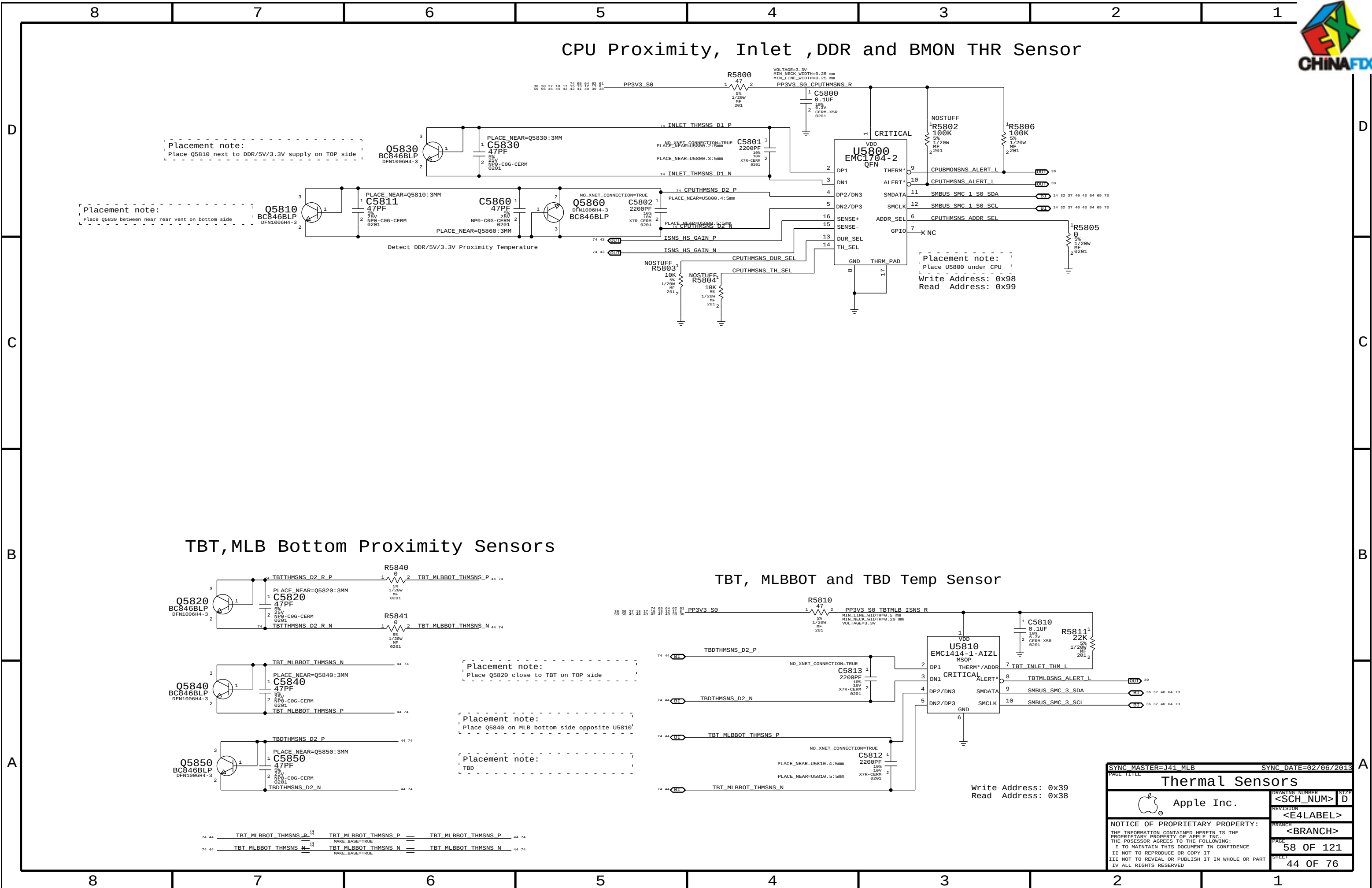
## Voltage & Load Side Current Sensing

|                     |  |                      |           |
|---------------------|--|----------------------|-----------|
| SYNC MASTER=J41 MLB |  | SYNC DATE=03/28/2013 |           |
| Apple Inc.          |  | DRAWING NUMBER       | SIZE      |
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|                     |  | REVISION             | <E4LABEL> |
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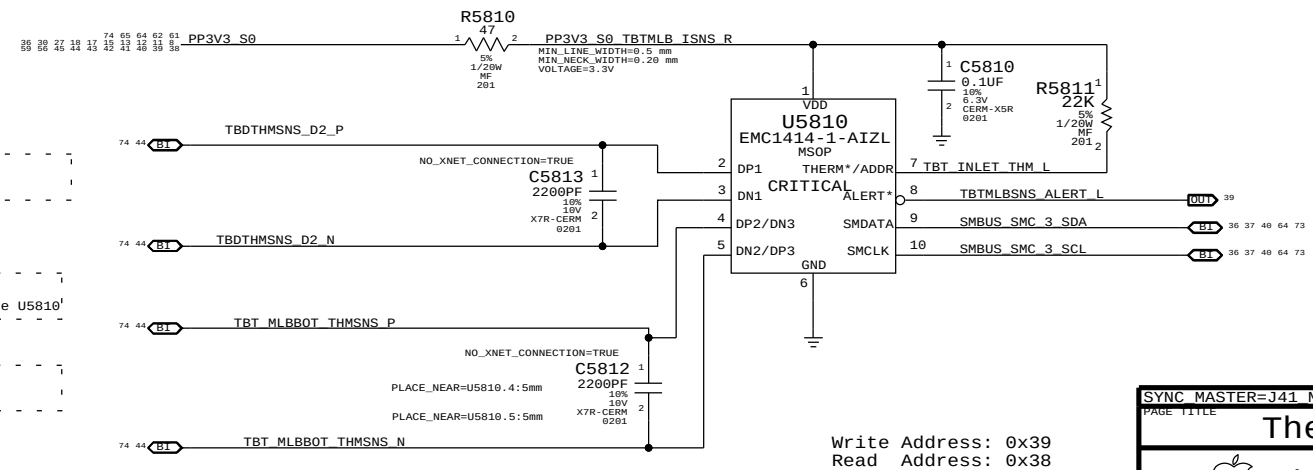
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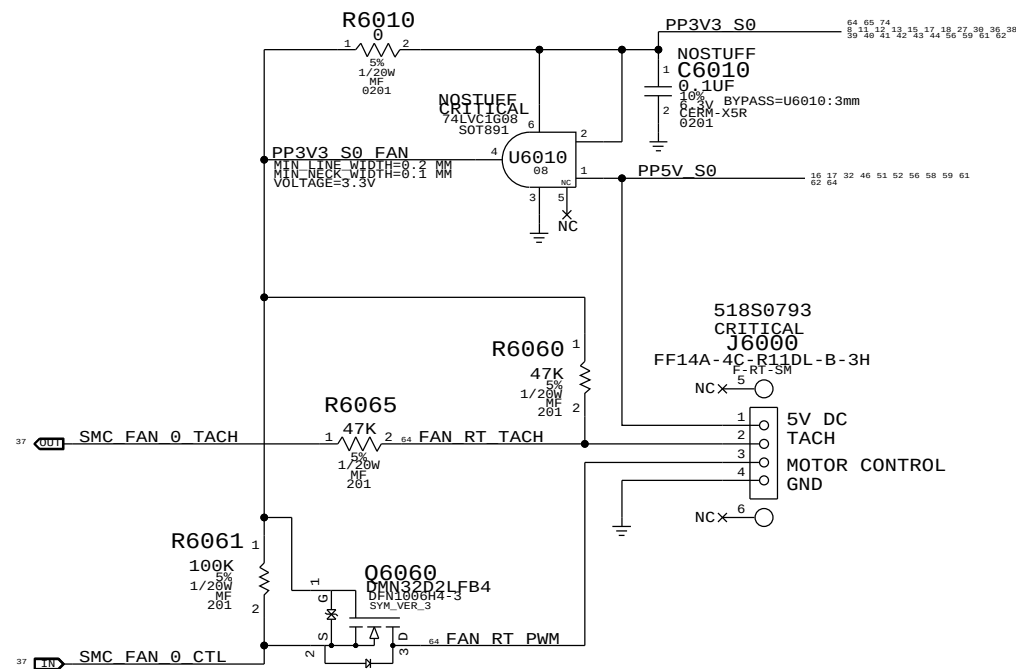


## TBT, MLBBOT and TBD Temp Sensor

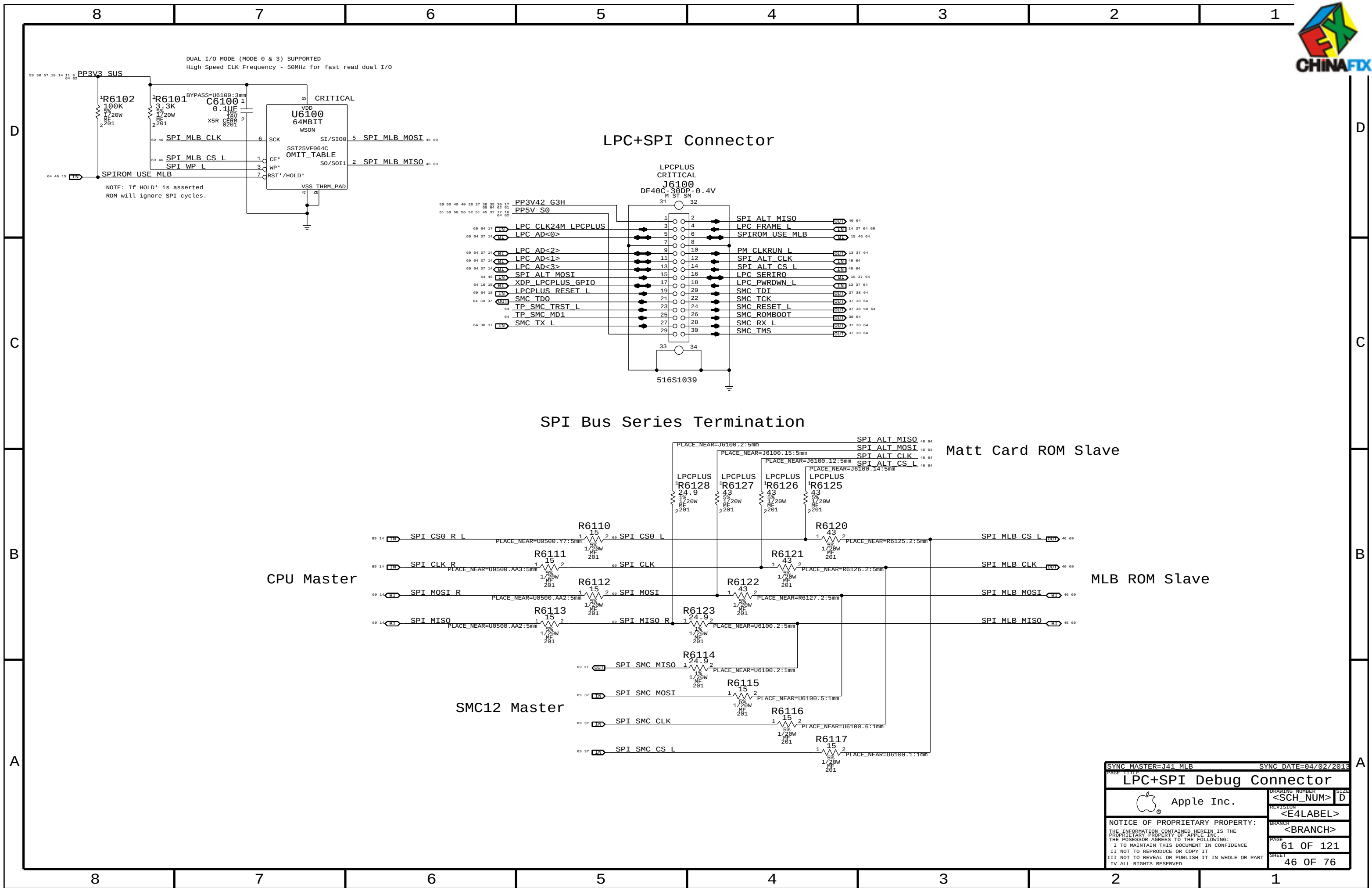


```
Write Address: 0x39
Read  Address: 0x38
```

|                                                                                                                                                                                                                                                                                     |                |                      |           |
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| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                 |                | SYNC DATE=02/06/2013 |           |
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| Thermal Sensors                                                                                                                                                                                                                                                                     |                |                      |           |
| <br>Apple Inc.                                                                                                                                                                                 | DRAWING NUMBER |                      | SIZE      |
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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------|--|
| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                                                   |  | SYNC DATE=02/06/2013                  |  |
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|  Apple Inc.                                                                                                                                                                                                                      |  | DRAWING NUMBER <b>&lt;SCH_NUM&gt;</b> |  |
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| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                                                     |  | SYNC DATE=04/02/2013 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  |                      |           |
| LPC+SPI Debug Connector                                                                                                                                                                                                                                                                                                 |  |                      |           |
|  Apple Inc.                                                                                                                                                                                                                        |  | DRAWING NUMBER       | SIZE      |
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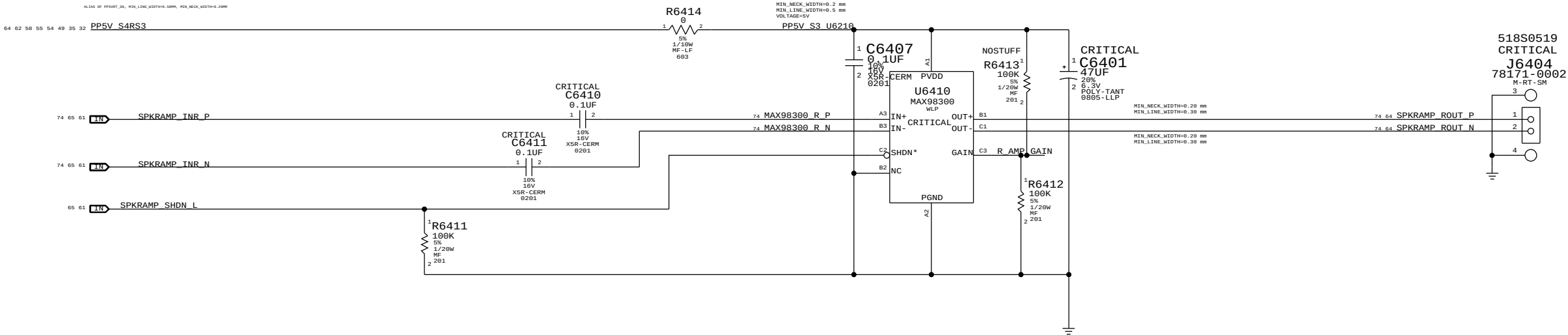


SPEAKER AMPLIFIERS

APN:353S2888

SPEAKER LOWPASS 80 HZ < FC < 132 HZ  
GAIN 6DB

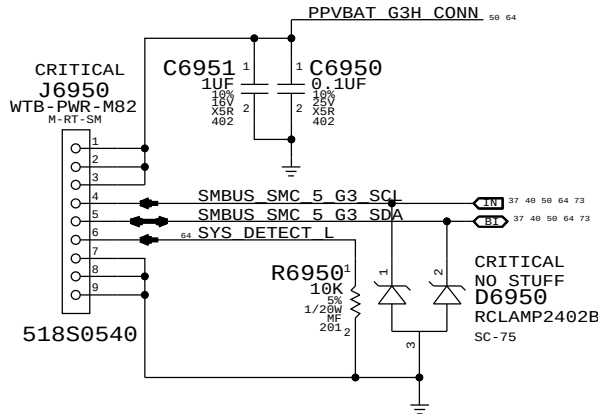
Right Speaker Connector



|                                                                                                                                                                                                                                                                                                                         |  |                      |           |
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| SYNC_MASTER=J41_MLB                                                                                                                                                                                                                                                                                                     |  | SYNC_DATE=02/06/2013 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  | Audio: Speaker Amp   |           |
|  Apple Inc.                                                                                                                                                                                                                        |  | DRAWING NUMBER       | SIZE      |
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13" SPECIFIC  
Battery Connector

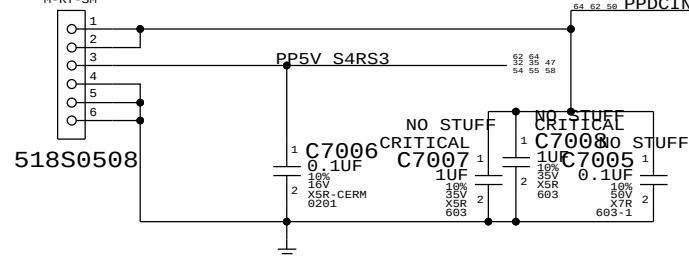


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| PAGE TITLE         |  | Battery Connector |  |
| DRAWING NUMBER     |  | SIZE              |  |
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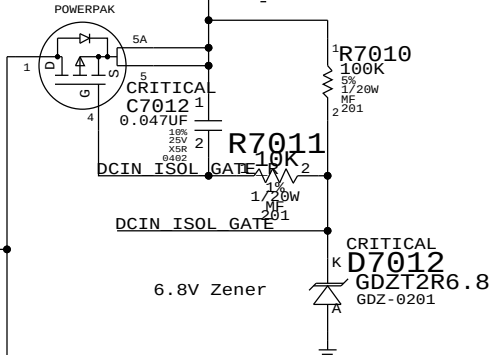
# MLB to LIO Power Cable Connector

CRITICAL  
J7000  
WTB-PWR-M82  
M-RT-SM



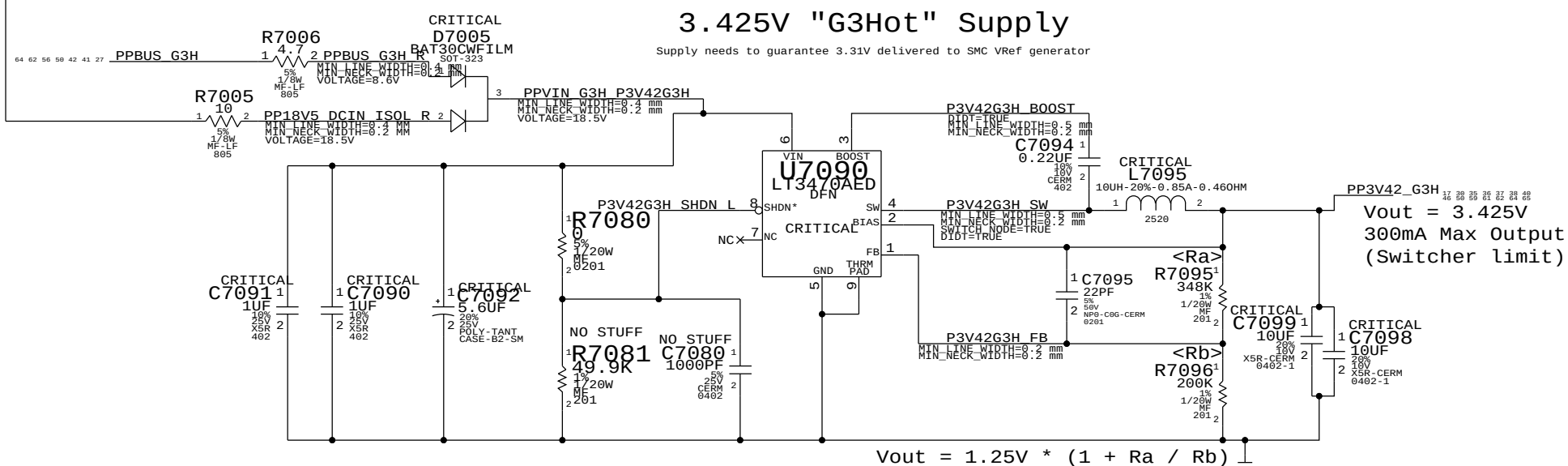
Input impedance of 68K meets  
sparkiterture requirements  
for detection of B121 (16.5V)

CRITICAL  
Q7010  
SI5419DU  
POWERPAK



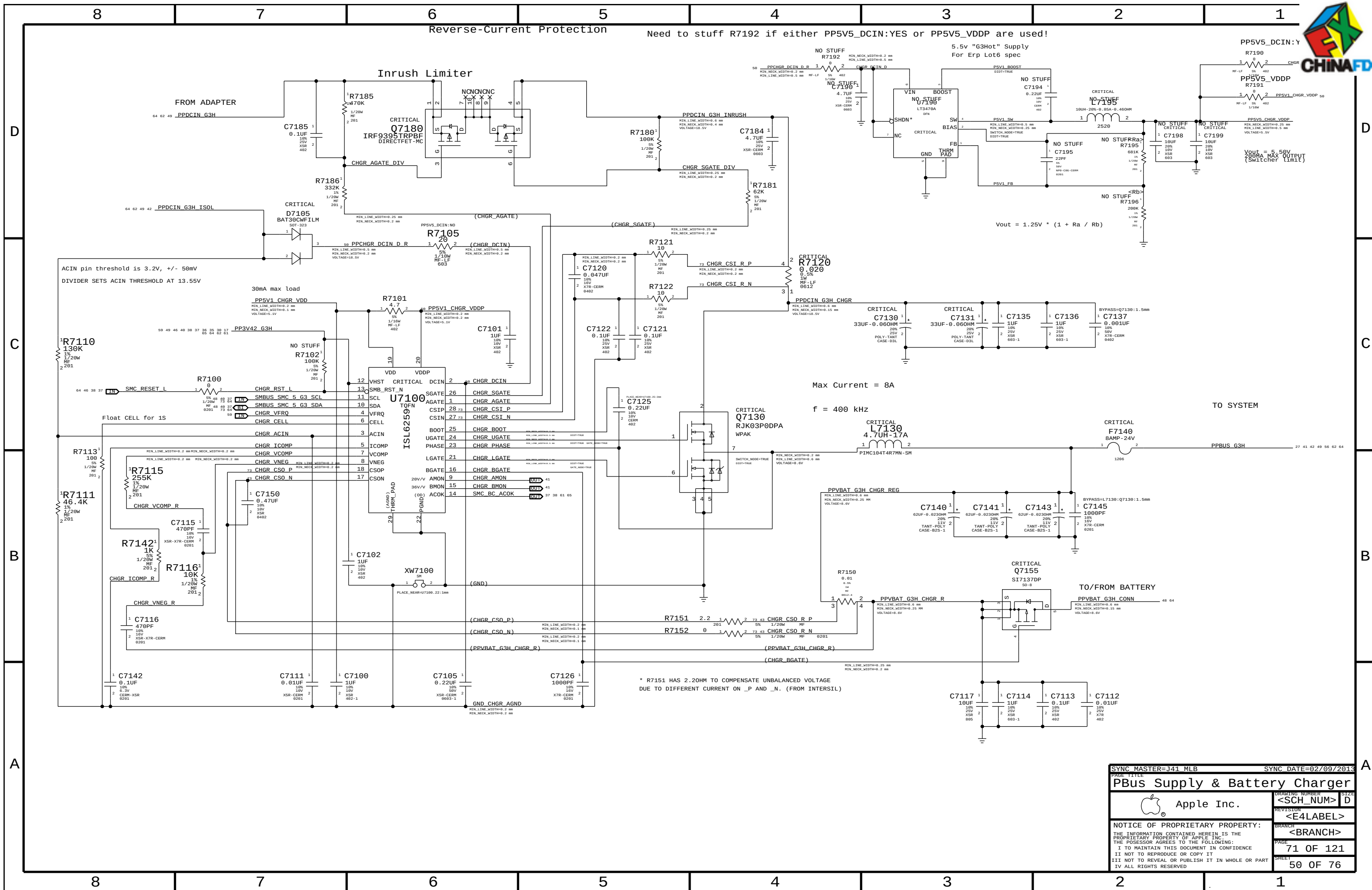
## 3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

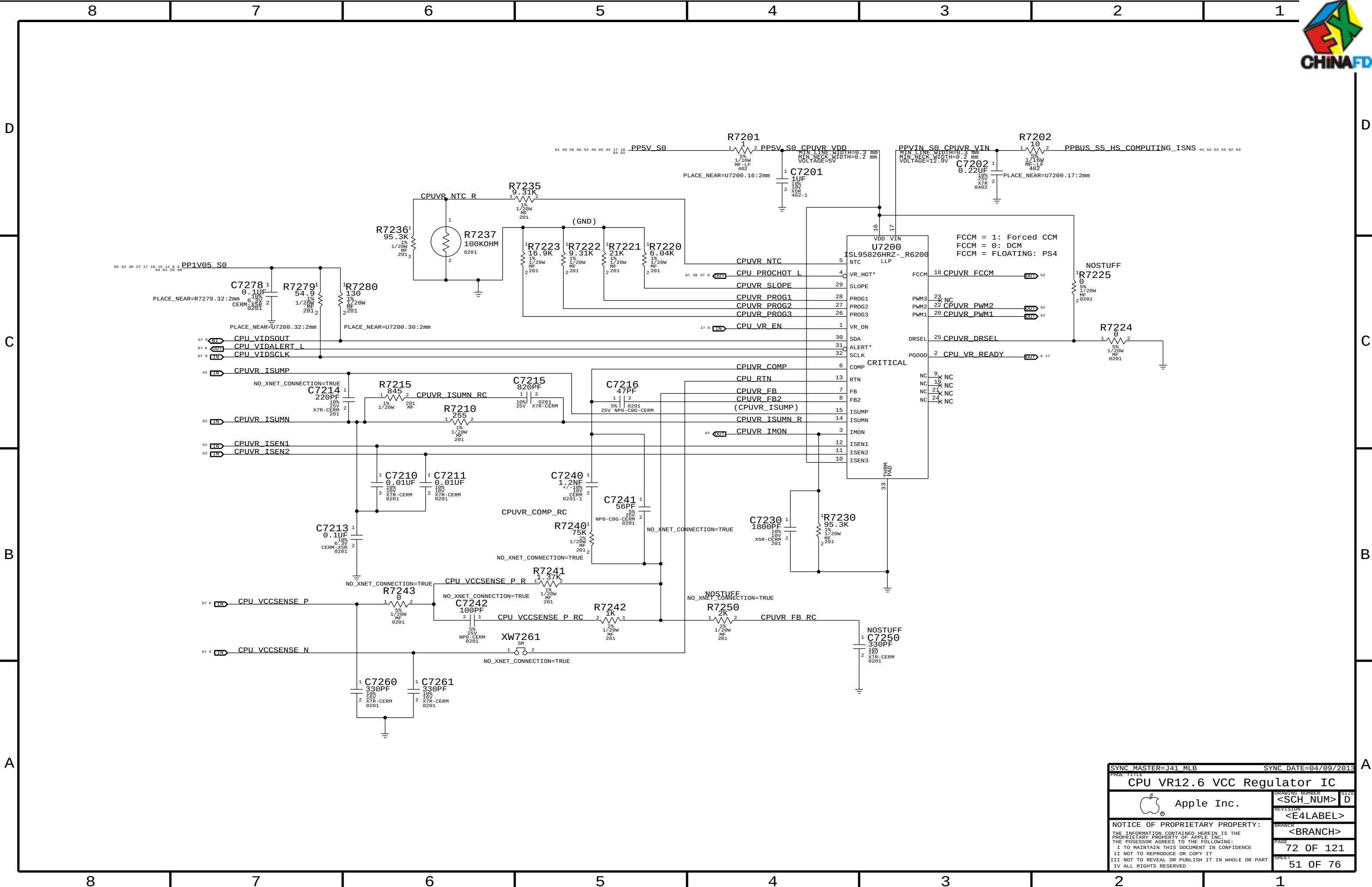


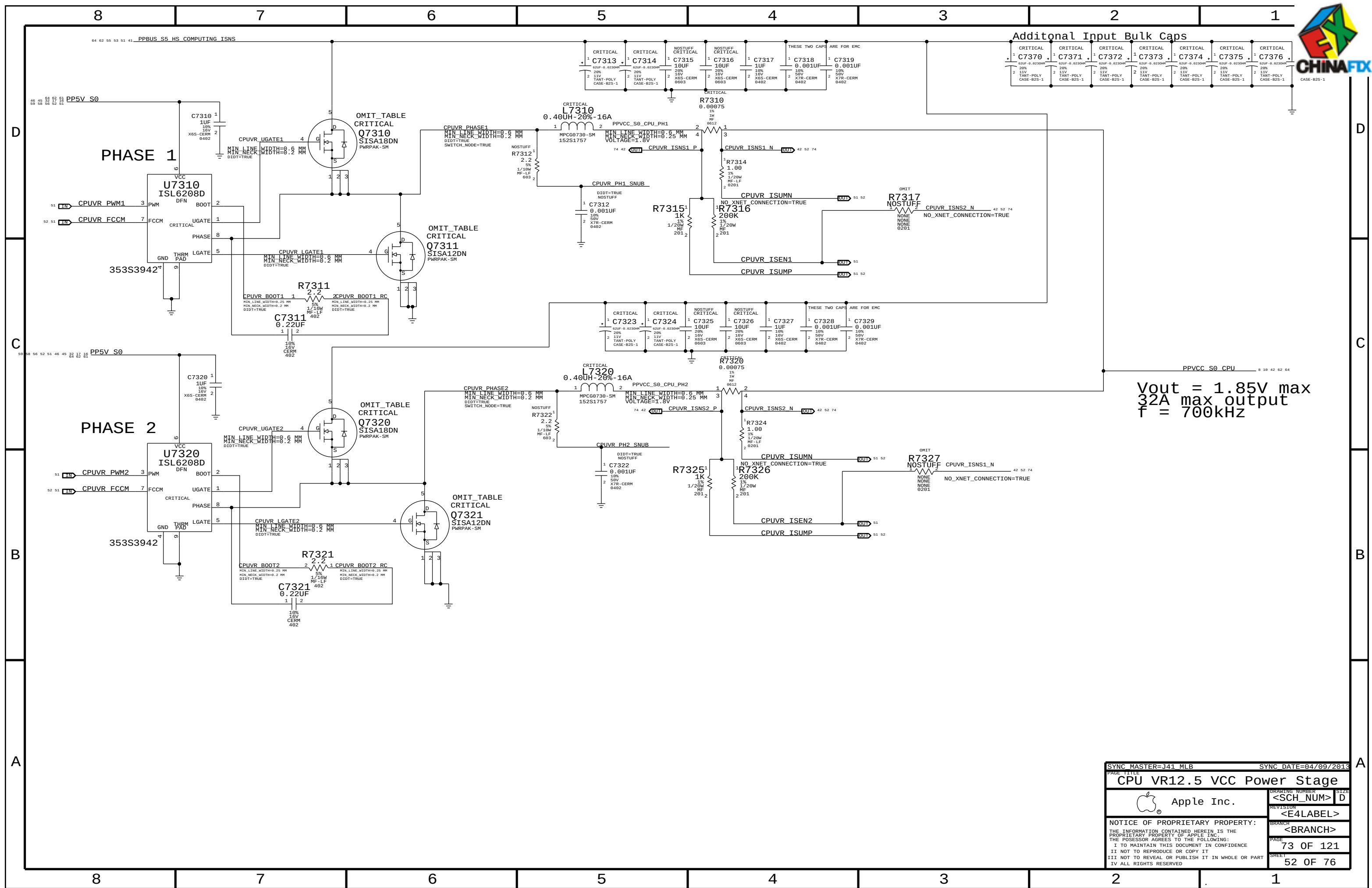
Vout = 3.425V  
300mA Max Output  
(Switcher limit)

|                                                                                                                   |  |                      |           |
|-------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=J41 MLB                                                                                               |  | SYNC DATE=02/06/2013 |           |
| PAGE TITLE                                                                                                        |  | DC-In & G3H Supply   |           |
| Apple Inc.                                                                                                        |  | DRAWING NUMBER       | SIZE      |
|                                                                                                                   |  | <SCH_NUM>            | D         |
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| SYNC MASTER=J41 MLB                                                                                               |                | SYNC DATE=04/09/2013 |           |
| PAGE TITLE                                                                                                        |                |                      |           |
| CPU VR12.5 VCC Power Stage                                                                                        |                |                      |           |
|  Apple Inc.                  | DRAWING NUMBER |                      | SIZE      |
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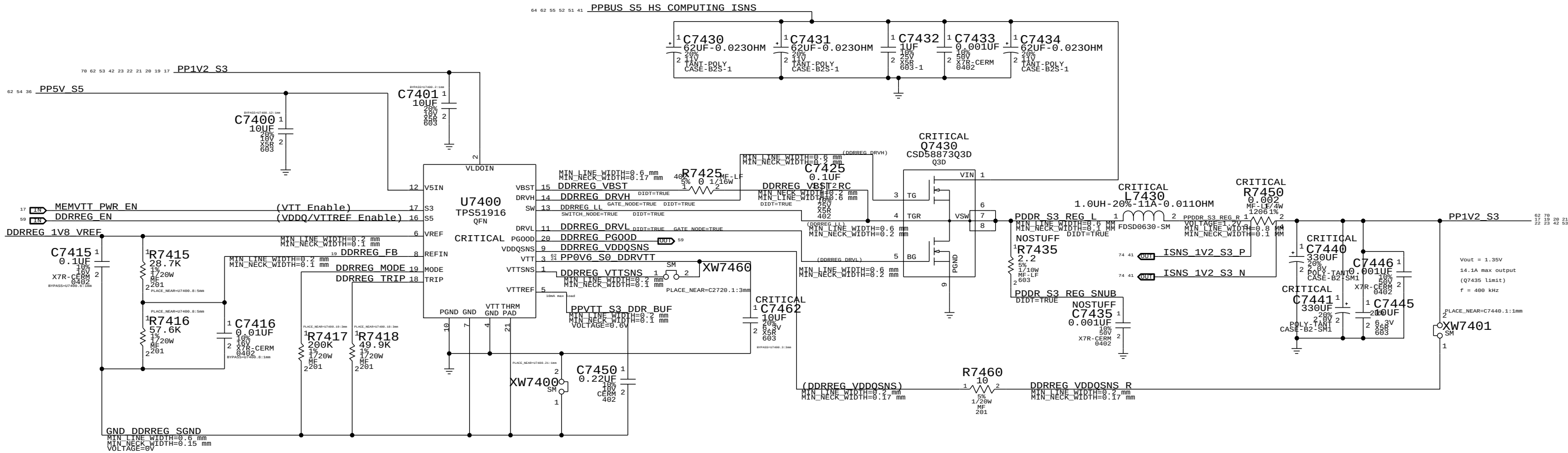
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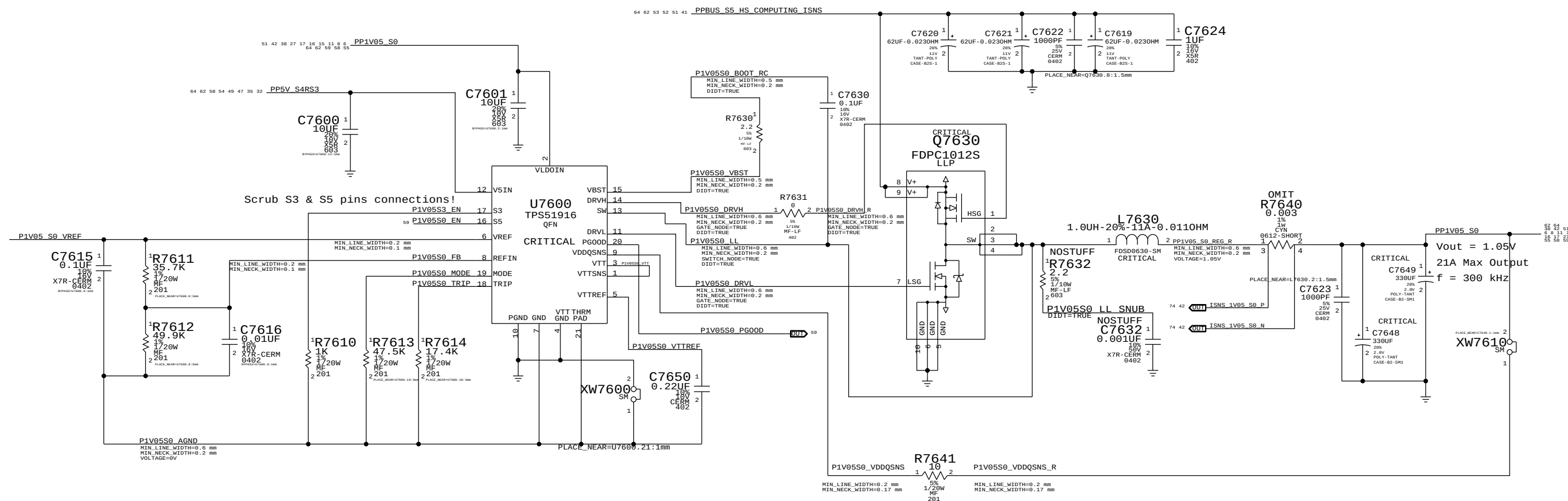


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| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                                                     |  | SYNC DATE=02/09/2013 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  |                      |           |
| LPDDR3 Supply                                                                                                                                                                                                                                                                                                           |  |                      |           |
|  Apple Inc.                                                                                                                                                                                                                        |  | DRAWING NUMBER       | SIZE      |
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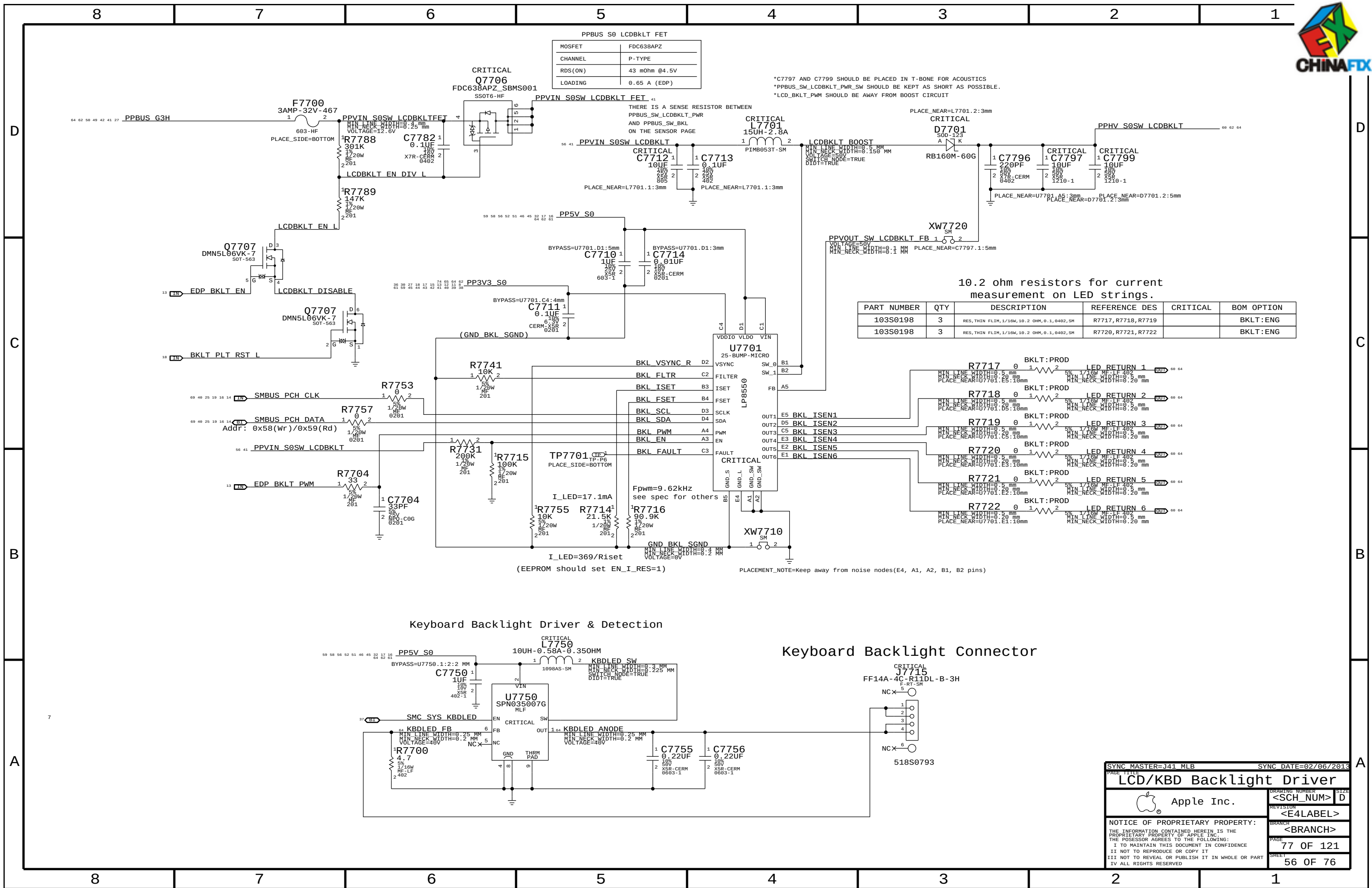


# 1.05V S0 Regulator

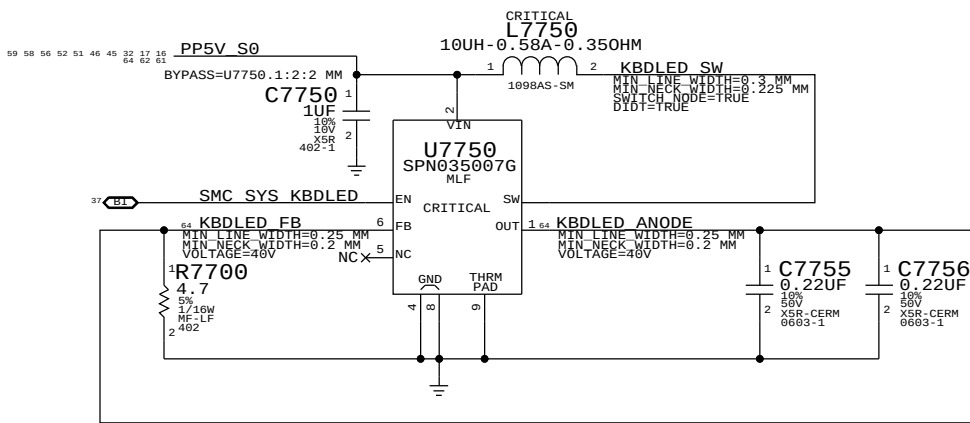


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| PAGE TITLE                                                                                                        |                |                      |           |
| 1.05V S0 Power Supply                                                                                             |                |                      |           |
|                              | DRAWING NUMBER |                      | SIZE      |
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| Apple Inc.                                                                                                        | REVISION       |                      |           |
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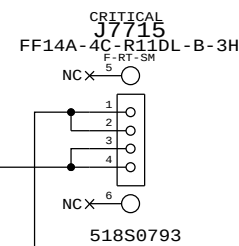




Keyboard Backlight Driver & Detection



Keyboard Backlight Connector

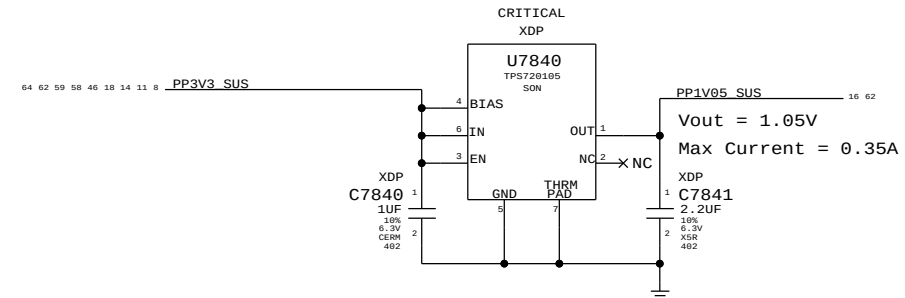


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| SYNC MASTER=J41 MLB                                                                                                                                                                                                                                                                                                     |  | SYNC DATE=02/06/2013 |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |  |                      |  |
| LCD/KBD Backlight Driver                                                                                                                                                                                                                                                                                                |  |                      |  |
|                                                                                                                                                                                                                                    |  | DRAWING NUMBER       |  |
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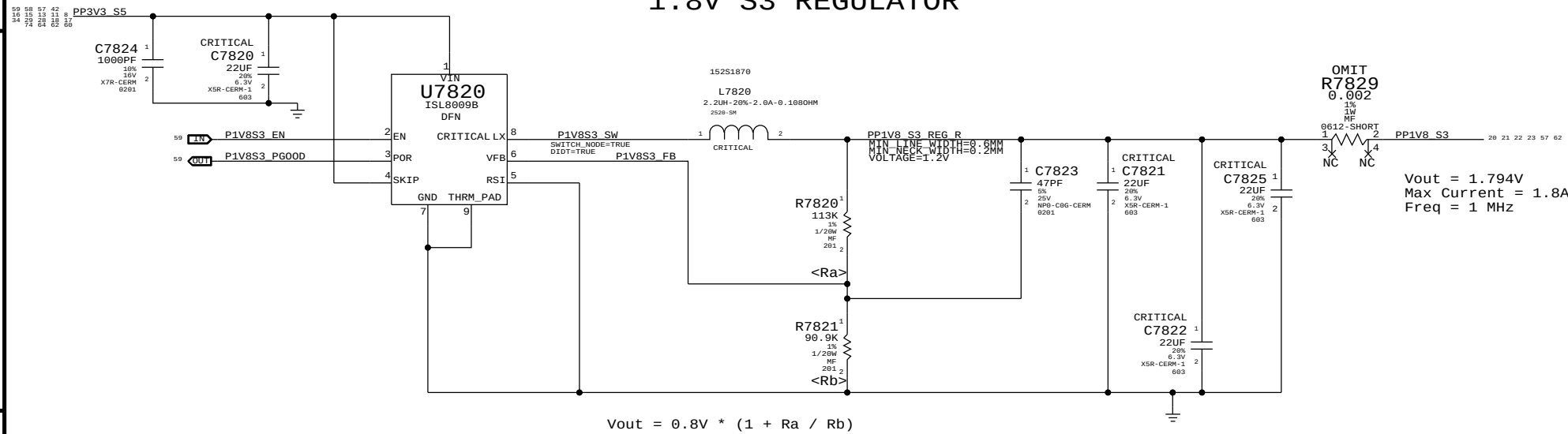


## 1.05V SUS LDO

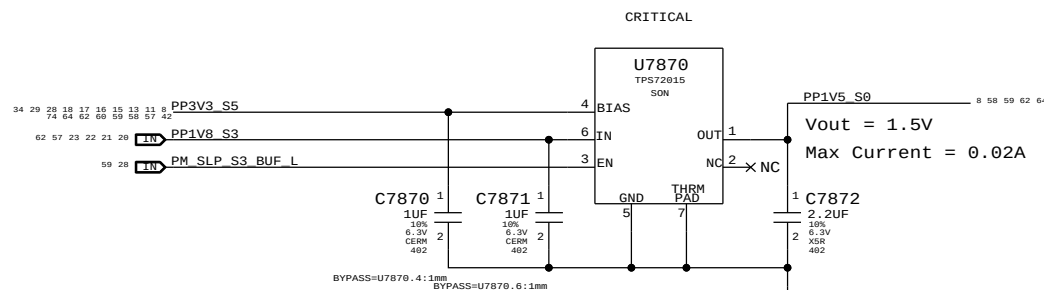
Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



## 1.8V S3 REGULATOR



## 1.5V S0 LDO



|                     |  |                      |           |
|---------------------|--|----------------------|-----------|
| SYNC MASTER=J41 MLB |  | SYNC DATE=02/06/2013 |           |
| PAGE TITLE          |  | Misc Power Supplies  |           |
| Apple Inc.          |  | DRAWING NUMBER       | SIZE      |
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Loading specs per J41/43\_PowerBudget\_Riviera\_rev0.99e

## 1.5V S0 Audio Switch

## 3.3V SUS Switch

## 1.05V PCH HSI0 Switch

## 3.3V S4 Switch

## 3.3V SSD Switch

## 3.3V S3 Switch

## 3.3V S0 Switch

## 3.3V Sensor Switch

## 5V S0 Switch

|                 |                                |
|-----------------|--------------------------------|
| Part            | TPS22924C                      |
| Type            | Load Switch                    |
| R(on)<br>@ 1.8V | 19.6 mOhm Typ<br>21.8 mOhm Max |
| Current         | 2A Max                         |

|                 |                                |
|-----------------|--------------------------------|
| Part            | TPS22924C                      |
| Type            | Load Switch                    |
| R(on)<br>@ 2.5V | 18.5 mOhm Typ<br>25.8 mOhm Max |
| Current         | 2A Max                         |

|                 |                                |
|-----------------|--------------------------------|
| Part            | TPS22924C                      |
| Type            | Load Switch                    |
| R(on)<br>@ 2.5V | 18.5 mOhm Typ<br>25.8 mOhm Max |
| Current         | 2A Max                         |

|                |                              |
|----------------|------------------------------|
| Part           | SLG5AP1453V                  |
| Type           | Load Switch                  |
| R(on)<br>@ 25C | 7.8 mOhm Typ<br>8.5 mOhm Max |
| Current        | 5.3A Max                     |

|                 |                                |
|-----------------|--------------------------------|
| Part            | TPS22924C                      |
| Type            | Load Switch                    |
| R(on)<br>@ 2.5V | 18.5 mOhm Typ<br>25.8 mOhm Max |
| Current         | 2A Max                         |

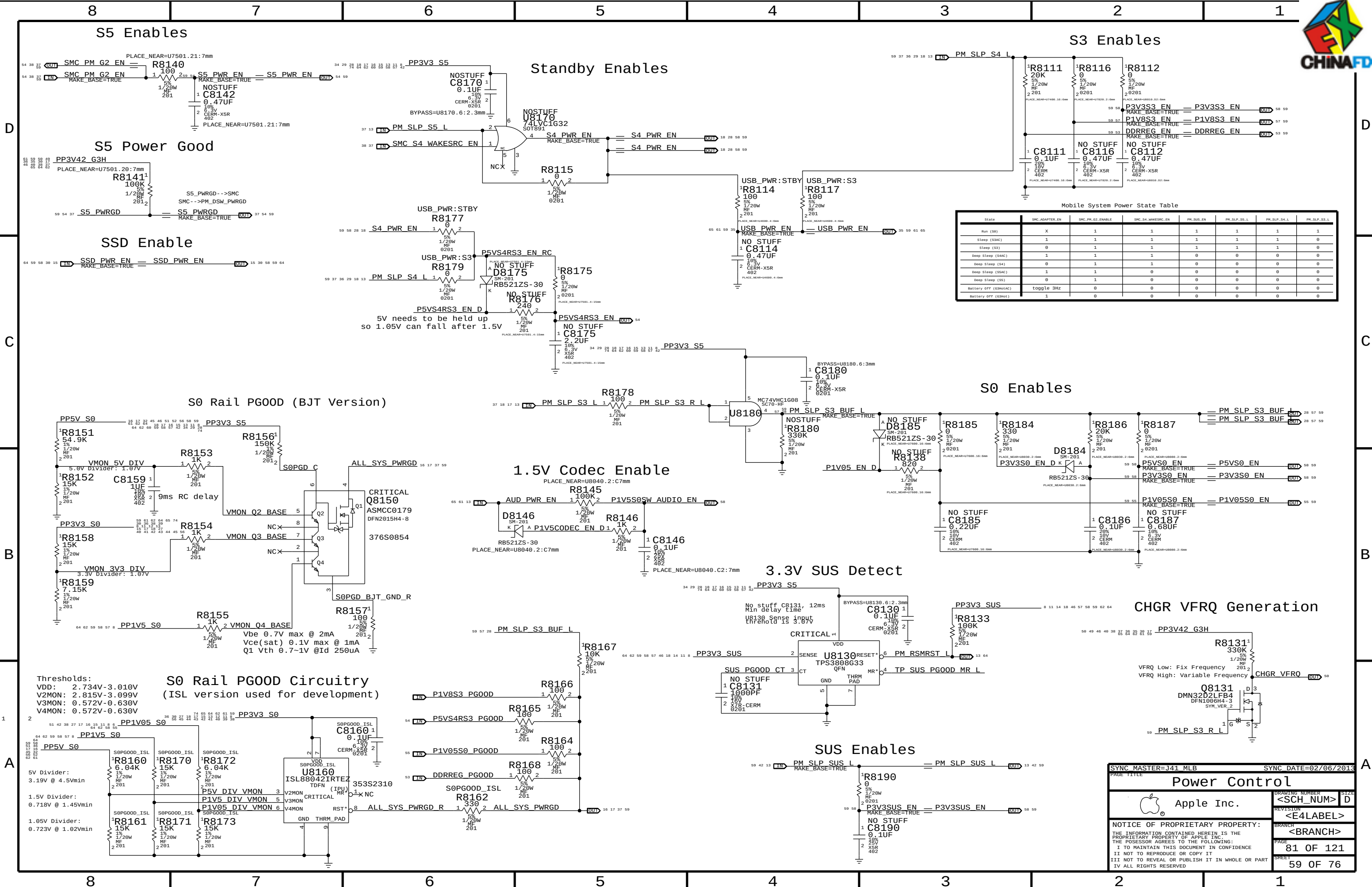
|                 |                                |
|-----------------|--------------------------------|
| Part            | TPS22924C                      |
| Type            | Load Switch                    |
| R(on)<br>@ 2.5V | 18.5 mOhm Typ<br>25.8 mOhm Max |
| Current         | 2A Max                         |

|                 |                                |
|-----------------|--------------------------------|
| Part            | TPS22924C                      |
| Type            | Load Switch                    |
| R(on)<br>@ 2.5V | 18.5 mOhm Typ<br>25.8 mOhm Max |
| Current         | 2A Max                         |

|         |                            |
|---------|----------------------------|
| Part    | SLG5AP1438V                |
| Type    | Load Switch                |
| R(on)   | 15 mOhm Typ<br>17 mOhm Max |
| Current | 2.5A                       |

|                   |                              |
|-------------------|------------------------------|
| Part              | SLG5AP1417V                  |
| Type              | Load Switch                  |
| R(on)<br>@ 4V Vgs | 9.8 mOhm Typ<br>TBD mOhm Max |
| Current           | 6A Max                       |

|                     |  |                      |           |
|---------------------|--|----------------------|-----------|
| SYNC MASTER=J41_MLB |  | SYNC DATE=02/06/2013 |           |
| PAGE TITLE          |  | Power FETs           |           |
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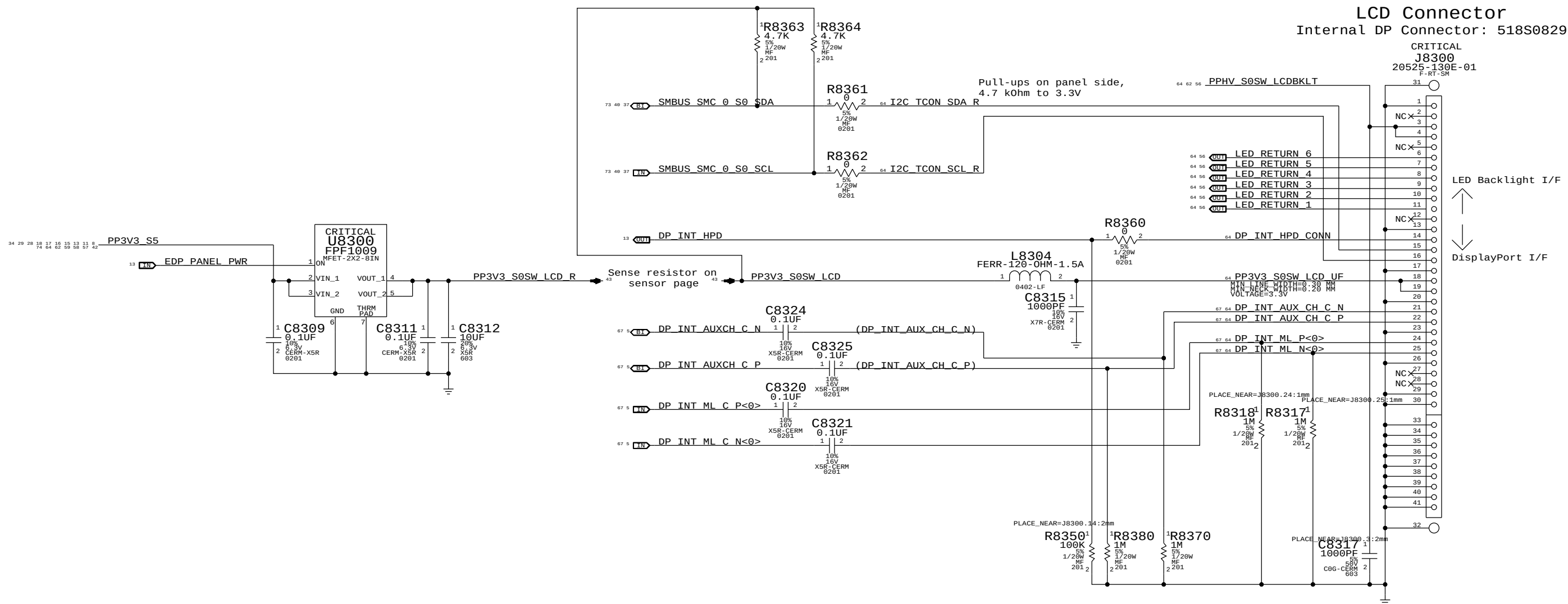
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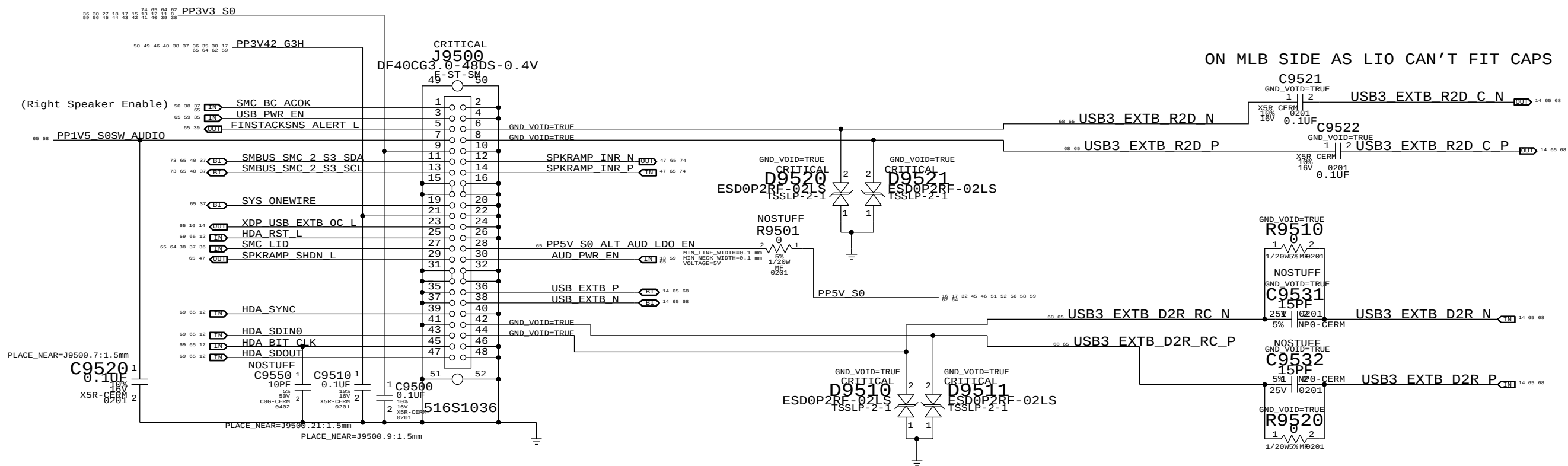
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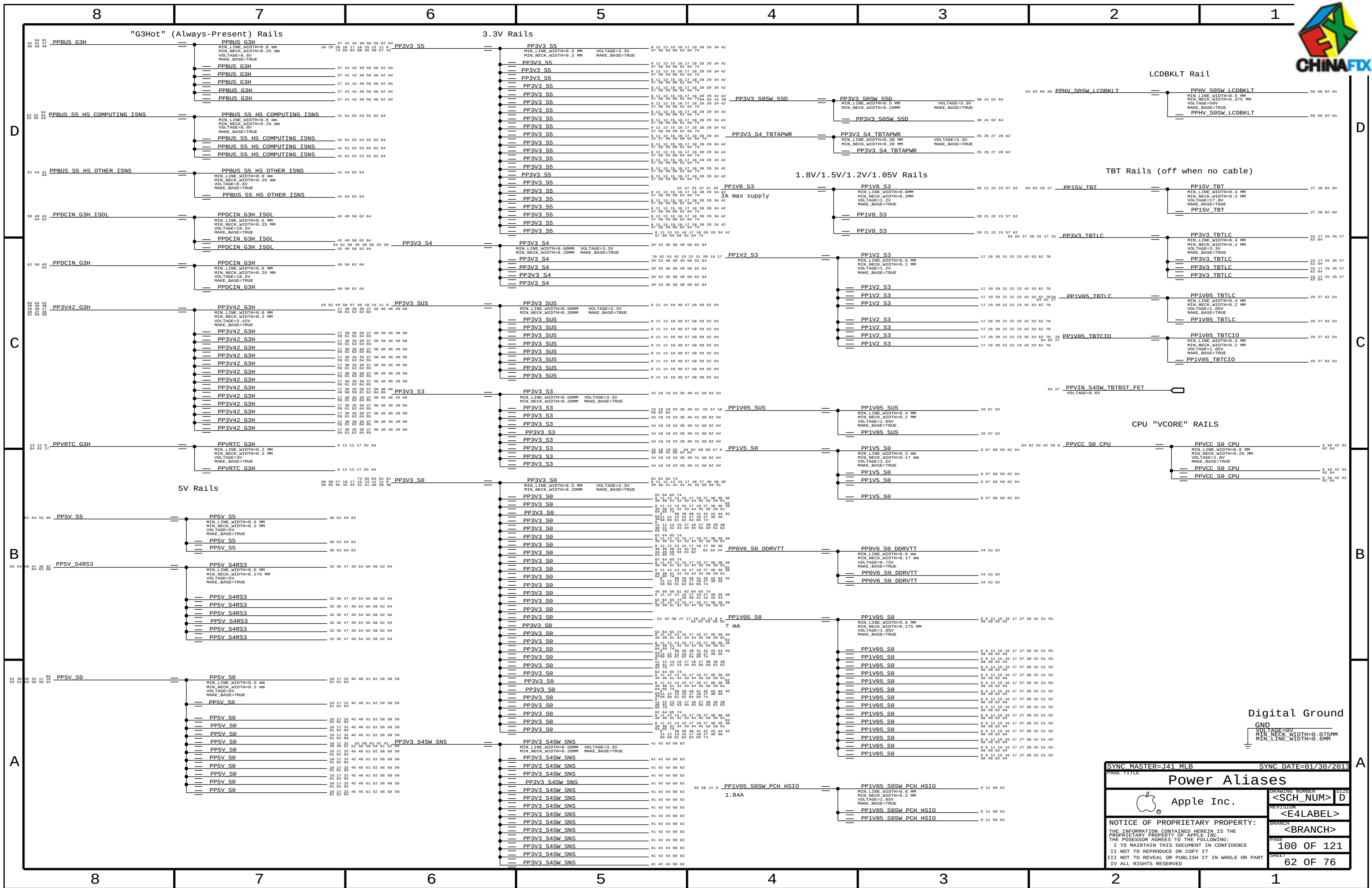
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| PAGE TITLE                                                                                                                                                                                                                                                                                                                |  |                             |                    |
| Internal DisplayPort Connector                                                                                                                                                                                                                                                                                            |  |                             |                    |
|  Apple Inc.                                                                                                                                                                                                                          |  | DRAWING NUMBER<br><SCH_NUM> | SIZE<br>D          |
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|                                                                                                                   |  |                      |           |
|-------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC_MASTER=CLEAN_343                                                                                             |  | SYNC_DATE=11/13/2012 |           |
| PART TITLE                                                                                                        |  |                      |           |
| Left I/O (LIO) Connector                                                                                          |  |                      |           |
|  Apple Inc.                  |  | DRAWING NUMBER       | SIZE      |
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SYNC MASTER=J41 MLB

SYNC DATE=01/30/2013

PAGE TITLE

Power Aliases

Apple Inc.

<SCH\_NUM>

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Memory Bit/Byte Swizzle

LPDDR3 Command/Address

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| PAGE TITLE                                                                                                        |  |                      |      |
| Signal Aliases                                                                                                    |  |                      |      |
|  Apple Inc.                  |  | DRAWING NUMBER       | SIZE |
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Functional Test Points

SD Card Aliases

J9500: LIO Connector

<

Bead Probes

|          |                    |     |            |        |
|----------|--------------------|-----|------------|--------|
| 68 61 14 | USB3_EXTB_D2R_N    | CTP | BEAD-PROBE | BPA511 |
| 68 61 14 | USB3_EXTB_D2R_P    | CTP | BEAD-PROBE | BPA510 |
| 68 65 61 | USB3_EXTB_D2R_RC_N | CTP | BEAD-PROBE | BPA520 |
| 68 65 61 | USB3_EXTB_D2R_RC_P | CTP | BEAD-PROBE | BPA521 |
| 68 61 14 | USB3_EXTB_R2D_C_N  | CTP | BEAD-PROBE | BPA513 |
| 68 61 14 | USB3_EXTB_R2D_C_P  | CTP | BEAD-PROBE | BPA512 |
| 68 65 61 | USB3_EXTB_R2D_N    | CTP | BEAD-PROBE | BPA523 |
| 68 65 61 | USB3_EXTB_R2D_P    | CTP | BEAD-PROBE | BPA522 |





J41/J43 Board-Specific Spacing & Physical Constraints

| BOARD LAYERS                                                              |  |  |  | BOARD AREAS            | BOARD UNITS (ALL OF MM) | OLE5680 |
|---------------------------------------------------------------------------|--|--|--|------------------------|-------------------------|---------|
| TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM |  |  |  | NO_TYPE, BGA, MEM_TERM | MM                      | 16.2    |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| DEFAULT           | TOP, BOTTOM | Y                     | =50_OHM_SE         | =50_OHM_SE         |                     |                      |                   |
| DEFAULT           | ISL2, ISL11 | Y                     | =45_OHM_SE         | =45_OHM_SE         |                     |                      |                   |
| DEFAULT           | ISL3, ISL10 | Y                     | =45_OHM_SE         | =45_OHM_SE         |                     |                      |                   |
| DEFAULT           | ISL4, ISL9  | Y                     | =45_OHM_SE         | =45_OHM_SE         |                     |                      |                   |
| DEFAULT           | *           | N                     | 100 MM             | 100 MM             | 10 MM               | 0 MM                 | 0 MM              |
| STANDARD          | *           | =DEFAULT              | =DEFAULT           | =DEFAULT           | =DEFAULT            | =DEFAULT             | =DEFAULT          |

Single-ended Physical Constraints

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 27P4_OHM_SE       | TOP, BOTTOM | Y                     | 0.310 MM           | 0.310 MM           |                     |                      |                   |
| 27P4_OHM_SE       | ISL2, ISL11 | Y                     | 0.182 MM           | 0.182 MM           |                     |                      |                   |
| 27P4_OHM_SE       | ISL3, ISL10 | Y                     | 0.182 MM           | 0.182 MM           |                     |                      |                   |
| 27P4_OHM_SE       | ISL4, ISL9  | Y                     | 0.182 MM           | 0.182 MM           |                     |                      |                   |
| 27P4_OHM_SE       | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 35_OHM_SE         | TOP, BOTTOM | Y                     | 0.195 MM           | 0.195 MM           |                     |                      |                   |
| 35_OHM_SE         | ISL2, ISL11 | Y                     | 0.125 MM           | 0.125 MM           |                     |                      |                   |
| 35_OHM_SE         | ISL3, ISL10 | Y                     | 0.125 MM           | 0.125 MM           |                     |                      |                   |
| 35_OHM_SE         | ISL4, ISL9  | Y                     | 0.125 MM           | 0.125 MM           |                     |                      |                   |
| 35_OHM_SE         | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 40_OHM_SE         | TOP, BOTTOM | Y                     | 0.170 MM           | 0.170 MM           |                     |                      |                   |
| 40_OHM_SE         | ISL2, ISL11 | Y                     | 0.096 MM           | 0.096 MM           |                     |                      |                   |
| 40_OHM_SE         | ISL3, ISL10 | Y                     | 0.096 MM           | 0.096 MM           |                     |                      |                   |
| 40_OHM_SE         | ISL4, ISL9  | Y                     | 0.099 MM           | 0.099 MM           |                     |                      |                   |
| 40_OHM_SE         | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 45_OHM_SE         | TOP, BOTTOM | Y                     | 0.135 MM           | 0.135 MM           |                     |                      |                   |
| 45_OHM_SE         | ISL2, ISL11 | Y                     | 0.075 MM           | 0.075 MM           |                     |                      |                   |
| 45_OHM_SE         | ISL3, ISL10 | Y                     | 0.075 MM           | 0.075 MM           |                     |                      |                   |
| 45_OHM_SE         | ISL4, ISL9  | Y                     | 0.080 MM           | 0.080 MM           |                     |                      |                   |
| 45_OHM_SE         | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 50_OHM_SE         | TOP, BOTTOM | Y                     | 0.110 MM           | 0.110 MM           |                     |                      |                   |
| 50_OHM_SE         | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 55_OHM_SE         | TOP, BOTTOM | Y                     | 0.090 MM           | 0.090 MM           |                     |                      |                   |
| 55_OHM_SE         | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

Differential Pair Physical Constraints

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 70_OHM_DIFF       | TOP, BOTTOM | Y                     | 0.165 MM           | 0.165 MM           |                     | 0.110 MM             | 0.110 MM          |
| 70_OHM_DIFF       | ISL2, ISL11 | Y                     | 0.105 MM           | 0.105 MM           |                     | 0.100 MM             | 0.100 MM          |
| 70_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.105 MM           | 0.105 MM           |                     | 0.100 MM             | 0.100 MM          |
| 70_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.110 MM           | 0.110MM            |                     | 0.095 MM             | 0.095 MM          |
| 70_OHM_DIFF       | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 80_OHM_DIFF       | TOP, BOTTOM | Y                     | 0.132 MM           | 0.132 MM           |                     | 0.130 MM             | 0.130 MM          |
| 80_OHM_DIFF       | ISL2, ISL11 | Y                     | 0.081 MM           | 0.081 MM           |                     | 0.115 MM             | 0.115 MM          |
| 80_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.081 MM           | 0.081 MM           |                     | 0.115 MM             | 0.115 MM          |
| 80_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.088 MM           | 0.088 MM           |                     | 0.110 MM             | 0.110 MM          |
| 80_OHM_DIFF       | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 90_OHM_DIFF       | TOP, BOTTOM | Y                     | 0.115 MM           | 0.115 MM           |                     | 0.200 MM             | 0.200 MM          |
| 90_OHM_DIFF       | ISL2, ISL11 | Y                     | 0.070 MM           | 0.070 MM           |                     | 0.180 MM             | 0.180 MM          |
| 90_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.070 MM           | 0.070 MM           |                     | 0.180 MM             | 0.180 MM          |
| 90_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.076 MM           | 0.076 MM           |                     | 0.180 MM             | 0.180 MM          |
| 90_OHM_DIFF       | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

Spacing Constraints

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| 1:1_SPACING      | *     | 0.100 MM             | ?      |

| SPACING_RULE_SET | LAYER       | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------------|----------------------|--------|
| 1x_DIELECTRIC    | TOP, BOTTOM | 0.071 MM             | ?      |
| 1x_DIELECTRIC    | ISL3, ISL10 | 0.053 MM             | ?      |
| 1x_DIELECTRIC    | ISL4, ISL9  | 0.050 MM             | ?      |
| 1x_DIELECTRIC    | *           | 0.090 MM             | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| DEFAULT          | *     | 0.1 MM               | ?      |
| STANDARD         | *     | =DEFAULT             | ?      |
| BGA_P075MM       | *     | 0.075 MM             | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| *                 | *                 | BGA       | BGA_P075MM       |

| NET_PHYSICAL_TYPE | AREA_TYPE | PHYSICAL_RULE_SET |
|-------------------|-----------|-------------------|
| *                 | BGA       | P070MM_BGA        |

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| P070MM_BGA        | *     |                       |                    | 0.070 MM           | 5 MM                |                      | 0.075 MM          |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 73_OHM_DIFF       | TOP, BOTTOM | Y                     | 0.165 MM           | 0.165 MM           |                     | 0.150 MM             | 0.150 MM          |
| 73_OHM_DIFF       | ISL2, ISL11 | Y                     | 0.106 MM           | 0.106 MM           |                     | 0.150 MM             | 0.150 MM          |
| 73_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.106 MM           | 0.106 MM           |                     | 0.150 MM             | 0.150 MM          |
| 73_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.110 MM           | 0.110 MM           |                     | 0.150 MM             | 0.150 MM          |
| 73_OHM_DIFF       | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 85_OHM_DIFF       | TOP, BOTTOM | Y                     | 0.120 MM           | 0.120 MM           |                     | 0.150 MM             | 0.150 MM          |
| 85_OHM_DIFF       | ISL2, ISL11 | Y                     | 0.078 MM           | 0.078 MM           |                     | 0.160 MM             | 0.160 MM          |
| 85_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.078 MM           | 0.078 MM           |                     | 0.160 MM             | 0.160 MM          |
| 85_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.082 MM           | 0.082 MM           |                     | 0.140 MM             | 0.140 MM          |
| 85_OHM_DIFF       | *           | N                     | 100 MM             | 100 MM             | =STANDARD           | =STANDARD            | =STANDARD         |

|                                                                                                                                                                                                                                                                                                                           |  |                      |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|------|
| SYNC_MASTER=CONSTRAINTS                                                                                                                                                                                                                                                                                                   |  | SYNC_DATE=10/24/2012 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                |  |                      |      |
| PCB Rule Definitions                                                                                                                                                                                                                                                                                                      |  |                      |      |
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|                                                                                                                                                                                                                                                                                                                           |  | <SCH_NUM>            | D    |
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## CPU Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| CPU_45S           | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |
| CPU_27P4S         | *     | =27P4_OHM_SE          | =27P4_OHM_SE       | =27P4_OHM_SE       | =27P4_OHM_SE        | 0.100 MM             | 0.100 MM          |

| SPACING_RULE_SET | LAYER      | LINE-TO-LINE SPACING | WEIGHT |
|------------------|------------|----------------------|--------|
| CPU_AGTL         | TOP,BOTTOM | =2x_DIELECTRIC       | ?      |
| CPU_AGTL         | *          | =STANDARD            | ?      |

Note: CPU\_8MIL and CPU\_ITP can be converted back to TABLE\_SPACING\_RULE once rdar://10308147 is resolved

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CPU_8MIL          | *                 | *         | CPU_8MIL_2ANY    |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CPU_8MIL_2ANY    | *     | 8 MIL                | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CPU_ITP           | *                 | *         | CPU_ITP_2ANY     |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CPU_ITP_2ANY     | *     | =4x_DIELECTRIC       | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CPU_COMP          | CPU_COMP          | *         | CPU_COMP_2SELF   |
| CPU_COMP          | *                 | *         | CPU_COMP_20THER  |

| SPACING_RULE_SET | LAYER      | LINE-TO-LINE SPACING | WEIGHT |
|------------------|------------|----------------------|--------|
| CPU_COMP_2SELF   | TOP,BOTTOM | =6x_DIELECTRIC       | ?      |
| CPU_COMP_20THER  | TOP,BOTTOM | =10x_DIELECTRIC      | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CPU_COMP_2SELF   | *     | =4x_DIELECTRIC       | ?      |
| CPU_COMP_20THER  | *     | =6x_DIELECTRIC       | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET    |
|-------------------|-------------------|-----------|---------------------|
| CPU_VCCSENSE      | CPU_VCCSENSE      | *         | CPU_VCCSENSE_2SELF  |
| CPU_VCCSENSE      | *                 | *         | CPU_VCCSENSE_20THER |

| SPACING_RULE_SET    | LAYER      | LINE-TO-LINE SPACING | WEIGHT |
|---------------------|------------|----------------------|--------|
| CPU_VCCSENSE_2SELF  | TOP,BOTTOM | =6x_DIELECTRIC       | ?      |
| CPU_VCCSENSE_20THER | TOP,BOTTOM | =10x_DIELECTRIC      | ?      |

| SPACING_RULE_SET    | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|---------------------|-------|----------------------|--------|
| CPU_VCCSENSE_2SELF  | *     | =4x_DIELECTRIC       | ?      |
| CPU_VCCSENSE_20THER | *     | =6x_DIELECTRIC       | ?      |

## PCI-Express Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| PCIE_80D          | *     | =80_OHM_DIFF          | =80_OHM_DIFF       | =80_OHM_DIFF       | =80_OHM_DIFF        | =80_OHM_DIFF         | =80_OHM_DIFF      |
| CLK_PCIE_80D      | *     | =80_OHM_DIFF          | =80_OHM_DIFF       | =80_OHM_DIFF       | =80_OHM_DIFF        | =80_OHM_DIFF         | =80_OHM_DIFF      |

### PCIe Clock Spacing

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CLK_PCIE          | CLK_PCIE          | *         | CLK_PCIE_2SELF   |
| CLK_PCIE          | *                 | *         | CLK_PCIE_20THER  |

| SPACING_RULE_SET | LAYER      | LINE-TO-LINE SPACING | WEIGHT |
|------------------|------------|----------------------|--------|
| CLK_PCIE_2SELF   | TOP,BOTTOM | =6x_DIELECTRIC       | ?      |
| CLK_PCIE_20THER  | TOP,BOTTOM | =10x_DIELECTRIC      | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CLK_PCIE_2SELF   | *     | =4x_DIELECTRIC       | ?      |
| CLK_PCIE_20THER  | *     | =6x_DIELECTRIC       | ?      |

### CPU PCIe Spacing

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| PCIE_CPU_TX       | PCIE_CPU_TX       | *         | PCIE_TX2TX       |
| PCIE_CPU_RX       | PCIE_CPU_RX       | *         | PCIE_RX2RX       |
| PCIE_CPU_TX       | *_CPU_TX          | *         | PCIE_TX20THERTX  |
| PCIE_CPU_RX       | *_CPU_RX          | *         | PCIE_RX20THERRX  |
| PCIE_CPU_TX       | *_CPU_RX          | *         | PCIE_TX2RX       |
| PCIE_CPU_RX       | *_CPU_TX          | *         | PCIE_RX2TX       |
| PCIE_CPU_TX       | *_TX              | *         | PCIE_20THERHS    |
| PCIE_CPU_RX       | *_TX              | *         | PCIE_20THERHS    |
| PCIE_CPU_TX       | *_RX              | *         | PCIE_20THERHS    |
| PCIE_CPU_RX       | *_RX              | *         | PCIE_20THERHS    |
| PCIE_CPU_TX       | *                 | *         | PCIE_20THER      |
| PCIE_CPU_RX       | *                 | *         | PCIE_20THER      |

| SPACING_RULE_SET | LAYER      | LINE-TO-LINE SPACING | WEIGHT |
|------------------|------------|----------------------|--------|
| PCIE_TX2TX       | TOP,BOTTOM | =5x_DIELECTRIC       | ?      |
| PCIE_RX2RX       | TOP,BOTTOM | =5x_DIELECTRIC       | ?      |
| PCIE_TX20THERTX  | TOP,BOTTOM | =5x_DIELECTRIC       | ?      |
| PCIE_RX20THERRX  | TOP,BOTTOM | =5x_DIELECTRIC       | ?      |
| PCIE_TX2RX       | TOP,BOTTOM | =7x_DIELECTRIC       | ?      |
| PCIE_RX2TX       | TOP,BOTTOM | =7x_DIELECTRIC       | ?      |
| PCIE_20THERHS    | TOP,BOTTOM | =6x_DIELECTRIC       | ?      |
| PCIE_20THER      | TOP,BOTTOM | =5x_DIELECTRIC       | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| PCIE_TX2TX       | *     | =2.5x_DIELECTRIC     | ?      |
| PCIE_RX2RX       | *     | =2.5x_DIELECTRIC     | ?      |
| PCIE_TX20THERTX  | *     | =4x_DIELECTRIC       | ?      |
| PCIE_RX20THERRX  | *     | =4x_DIELECTRIC       | ?      |
| PCIE_TX2RX       | *     | =6x_DIELECTRIC       | ?      |
| PCIE_RX2TX       | *     | =6x_DIELECTRIC       | ?      |
| PCIE_20THERHS    | *     | =4x_DIELECTRIC       | ?      |
| PCIE_20THER      | *     | =3x_DIELECTRIC       | ?      |

### PCH PCIe Spacing

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| PCIE_PCH_TX       | PCIE_PCH_TX       | *         | PCIE_TX2TX       |
| PCIE_PCH_RX       | PCIE_PCH_RX       | *         | PCIE_RX2RX       |
| PCIE_PCH_TX       | *_PCH_TX          | *         | PCIE_TX20THERTX  |
| PCIE_PCH_RX       | *_PCH_RX          | *         | PCIE_RX20THERRX  |
| PCIE_PCH_TX       | *_PCH_RX          | *         | PCIE_TX2RX       |
| PCIE_PCH_RX       | *_PCH_TX          | *         | PCIE_RX2TX       |
| PCIE_PCH_TX       | *_TX              | *         | PCIE_20THERHS    |
| PCIE_PCH_RX       | *_TX              | *         | PCIE_20THERHS    |
| PCIE_PCH_TX       | *_RX              | *         | PCIE_20THERHS    |
| PCIE_PCH_RX       | *_RX              | *         | PCIE_20THERHS    |
| PCIE_PCH_TX       | *                 | *         | PCIE_20THER      |
| PCIE_PCH_RX       | *                 | *         | PCIE_20THER      |

Note: DisplayPort tables are on Page 113

## CPU Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                           |                   | NET_TYPE     |          |                     |            |
|-------------------------------------------------------------------------------------|-------------------|--------------|----------|---------------------|------------|
|                                                                                     |                   | PHYSICAL     | SPACING  |                     |            |
|  | CPU_PECT          | CPU_45S      | CPU_COMP | CPU_PECT            | 6 38       |
|  | PM_SYNC           | CPU_45S      | CPU_AGTL | PM_SYNC             |            |
|  | PM_MEM_PWRGD      | CPU_45S      | CPU_AGTL | PM_MEM_PWRGD        |            |
|  |                   | CPU_45S      | CPU_ITP  | XDP_DBRESET_L       | 16 17      |
|  |                   | CPU_45S      | CPU_ITP  | XDP_CPU_PRDY_L      | 6 16 64    |
|  |                   | CPU_45S      | CPU_ITP  | XDP_CPU_PREQ_L      | 6 16 64    |
|  |                   | CPU_27P4S    | CPU_COMP | EDP_COMP            |            |
|  |                   | CPU_27P4S    | CPU_COMP | CPU_PEG_COMP        |            |
|  | CPU_SM_RCOMP      | CPU_27P4S    | CPU_COMP | CPU_SM_RCOMP<0>     | 6          |
|  | CPU_SM_RCOMP      | CPU_27P4S    | CPU_COMP | CPU_SM_RCOMP<1>     | 6          |
|  | CPU_SM_RCOMP      | CPU_27P4S    | CPU_COMP | CPU_SM_RCOMP<2>     | 6          |
|  |                   | CPU_45S      | CPU_ITP  | CPU_CFG<11..0>      | 6 16 64    |
|  | CPU_CATERR_L      | CPU_45S      | CPU_AGTL | CPU_CATERR_L        | 6 37       |
|  |                   | CPU_45S      | CPU_AGTL | CPU_VCCIO_SEL       |            |
|  | CPU_PROCHOT_I     | CPU_45S      | CPU_AGTL | CPU_PROCHOT_L       | 6 37 38 51 |
|  | CPU_PWRGD         | CPU_45S      | CPU_AGTL | CPU_PWRGD           | 6          |
|  | PM_THRMTRIP_I     | CPU_45S      | CPU_RMTI | PM_THRMTRIP_L       | 15 38      |
|  | DMI_CLK100M       | CLK_PCIE_80D | CLK_PCIE | DMI_CLK100M_CPU_P   |            |
|  | DMI_CLK100M       | CLK_PCIE_80D | CLK_PCIE | DMI_CLK100M_CPU_N   |            |
|  | DP_L1_REF_CLK120M | CLK_PCIE_80D | CLK_PCIE | DP_L1_REF_CLKP      |            |
|  | DP_L1_REF_CLK120M | CLK_PCIE_80D | CLK_PCIE | DP_L1_REF_CLKN      |            |
|  | ITPCPU_CLK100M    | CLK_PCIE_80D | CLK_PCIE | ITPCPU_CLK100M_P    |            |
|  | ITPCPU_CLK100M    | CLK_PCIE_80D | CLK_PCIE | ITPCPU_CLK100M_N    |            |
|  | ITPCPU_CLK100M    | CLK_PCIE_80D | CLK_PCIE | ITPXDPCPU_CLK100M_P |            |
|  | ITPCPU_CLK100M    | CLK_PCIE_80D | CLK_PCIE | ITPXDPCPU_CLK100M_N |            |
|  | ITPCPU_CLK100M    | CLK_PCIE_80D | CLK_PCIE | XDP_CPU_CLK100M_P   |            |
|  | ITPCPU_CLK100M    | CLK_PCIE_80D | CLK_PCIE | XDP_CPU_CLK100M_N   |            |
|  | XDP_TDI           | CPU_45S      | CPU_ITP  | XDP_CPU_TDI         | 6 16 64    |
|  | XDP_TDO           | CPU_45S      | CPU_ITP  | XDP_CPU_TDO         | 6 16 64    |
|  | XDP_TMS           | CPU_45S      | CPU_ITP  | XDP_CPU_TMS         | 6 16 64    |
|  | XDP_TCK           | CPU_45S      | CPU_ITP  | XDP_CPU_TCK         | 6 16 64    |
|  | XDP_TRST_L        | CPU_45S      | CPU_ITP  | XDP_CPU_PCH_TRST_L  | 6 12 16 64 |
|  | XDP_BPM_L         | CPU_45S      | CPU_ITP  | XDP_BPM_L<1..0>     | 6 16       |
|  |                   | CPU_45S      | CPU_ITP  | XDP_BPM_L<7..2>     | 6 16       |
|  |                   | CPU_45S      | CPU_ITP  | XDP_OBSDATA_B<3..0> |            |
|  |                   | CPU_45S      | CPU_ITP  | CPU_CFG<15..12>     | 6 16       |
|  | (FSR_CPURST_I)    | CPU_45S      | CPU_ITP  | XDP_CPURST_L        | 16         |

|                |                 |              |                    |      |
|----------------|-----------------|--------------|--------------------|------|
| CPU_VCCSENSE   | SENSE_1T01_P2MM | CPU_VCCSENSE | CPU_VCCSENSE_P     | 6 51 |
| CPU_VCCSENSE   | SENSE_1T01_P2MM | CPU_VCCSENSE | CPU_VCCSENSE_N     | 6 51 |
| CPU_VCCIOSENSE | SENSE_1T01_P2MM | CPU_VCCSENSE | CPU_VCCIOSENSE_P   |      |
| CPU_VCCIOSENSE | SENSE_1T01_P2MM | CPU_VCCSENSE | CPU_VCCIOSENSE_N   |      |
| CPU_AXG_SENSE  | SENSE_1T01_P2MM | CPU_VCCSENSE | CPU_AXG_SENSE_P    |      |
| CPU_AXG_SENSE  | SENSE_1T01_P2MM | CPU_VCCSENSE | CPU_AXG_SENSE_N    |      |
| CPU_VALSENSE   | CPU_27P4S       | CPU_VCCSENSE | CPU_VDDQ_SENSE_P   |      |
| CPU_VALSENSE   | CPU_27P4S       | CPU_VCCSENSE | CPU_VDDQ_SENSE_N   |      |
| CPU_VALSENSE   | CPU_27P4S       | CPU_VCCSENSE | CPU_AXG_VALSENSE_P |      |
| CPU_VALSENSE   | CPU_27P4S       | CPU_VCCSENSE | CPU_AXG_VALSENSE_N |      |
| CPU_VALSENSE   | CPU_27P4S       | CPU_VCCSENSE | CPU_VCC_VALSENSE_P |      |
| CPU_VALSENSE   | CPU_27P4S       | CPU_VCCSENSE | CPU_VCC_VALSENSE_N |      |

|                 |         |          |                |      |
|-----------------|---------|----------|----------------|------|
| CPU_SVIDALERT_L | CPU_45S | CPU_COMP | CPU_VIDALERT_L | 6 51 |
| CPU_SVIDSCLK    | CPU_45S | CPU_COMP | CPU_VIDSCLK    | 6 51 |
| CPU_SVIDSOUT    | CPU_45S | CPU_COMP | CPU_VIDSOUT    | 6 51 |

|                  |              |             |                        |          |
|------------------|--------------|-------------|------------------------|----------|
| PCIE_CPU_SSD_R2D | PCIE_80D     | PCIE_CPU_TX | PCIE_SSD_R2D_C_P<3..0> | 12 38    |
| PCIE_CPU_SSD_R2D | PCIE_80D     | PCIE_CPU_TX | PCIE_SSD_R2D_C_N<3..0> | 12 38    |
|                  | PCIE_80D     | PCIE_CPU_TX | PCIE_SSD_R2D_P<3..0>   | 36 64    |
|                  | PCIE_80D     | PCIE_CPU_TX | PCIE_SSD_R2D_N<3..0>   | 36 64    |
|                  | PCIE_80D     | PCIE_CPU_RX | PCIE_SSD_D2R_C_P<3..0> |          |
|                  | PCIE_80D     | PCIE_CPU_RX | PCIE_SSD_D2R_C_N<3..0> |          |
| PCIE_CPU_SSD_D2R | PCIE_80D     | PCIE_CPU_RX | PCIE_SSD_D2R_P<3..0>   | 12 38 64 |
| PCIE_CPU_SSD_D2R | PCIE_80D     | PCIE_CPU_RX | PCIE_SSD_D2R_N<3..0>   | 12 38 64 |
| PCIE_CLK100M_SSD | CLK_PCIE_80D | CLK_PCIE    | PCIE_CLK100M_SSD_P     | 12 38 64 |
| PCIE_CLK100M_SSD | CLK_PCIE_80D | CLK_PCIE    | PCIE_CLK100M_SSD_N     | 12 38 64 |

PCie SSD

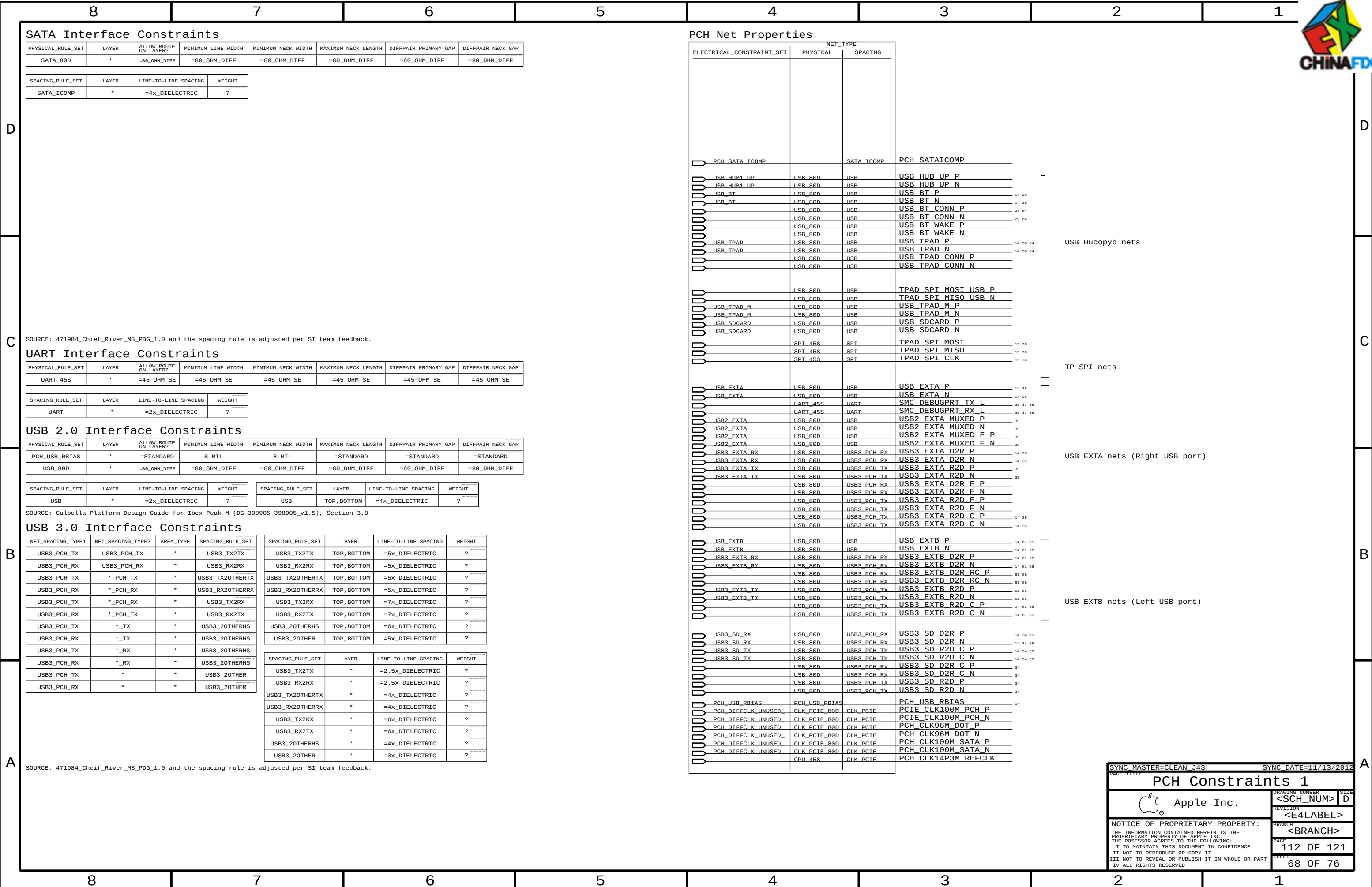
|              |        |        |                         |          |
|--------------|--------|--------|-------------------------|----------|
| DP_TBT_ML    | DP_80D | DP_TX  | DP_TBTSNK0_ML_P<3..0>   | 25       |
| DP_TBT_ML    | DP_80D | DP_TX  | DP_TBTSNK0_ML_N<3..0>   | 25       |
|              | DP_80D | DP_TX  | DP_TBTSNK0_ML_C_P<3..0> | 5 25     |
|              | DP_80D | DP_TX  | DP_TBTSNK0_ML_C_N<3..0> | 5 25     |
| DP_TBT_AUXCH | DP_80D | DP_AUX | DP_TBTSNK0_AUXCH_P      | 25       |
| DP_TBT_AUXCH | DP_80D | DP_AUX | DP_TBTSNK0_AUXCH_N      | 25       |
|              | DP_80D | DP_AUX | DP_TBTSNK0_AUXCH_C_P    | 13 25    |
|              | DP_80D | DP_AUX | DP_TBTSNK0_AUXCH_C_N    | 13 25    |
| DP_TBT_ML    | DP_80D | DP_TX  | DP_TBTSNK1_ML_P<3..0>   | 25       |
| DP_TBT_ML    | DP_80D | DP_TX  | DP_TBTSNK1_ML_N<3..0>   | 25       |
|              | DP_80D | DP_TX  | DP_TBTSNK1_ML_C_P<3..0> | 5 18 25  |
|              | DP_80D | DP_TX  | DP_TBTSNK1_ML_C_N<3..0> | 5 18 25  |
| DP_TBT_AUXCH | DP_80D | DP_AUX | DP_TBTSNK1_AUXCH_P      | 25       |
| DP_TBT_AUXCH | DP_80D | DP_AUX | DP_TBTSNK1_AUXCH_N      | 25       |
|              | DP_80D | DP_AUX | DP_TBTSNK1_AUXCH_C_P    | 13 18 25 |
|              | DP_80D | DP_AUX | DP_TBTSNK1_AUXCH_C_N    | 13 18 25 |

|           |        |       |                     |         |
|-----------|--------|-------|---------------------|---------|
| DP_INT_ML | DP_80D | DP_TX | DP_INT_ML_P<3..0>   | 60 64   |
| DP_INT_ML | DP_80D | DP_TX | DP_INT_ML_N<3..0>   | 60 64   |
|           | DP_80D | DP_TX | DP_INT_ML_C_P<3..0> | 5 60 64 |
|           | DP_80D | DP_TX | DP_INT_ML_C_N<3..0> | 5 60 64 |

|              |        |        |                   |       |
|--------------|--------|--------|-------------------|-------|
| DP_INT_AUXCH | DP_80D | DP_AUX | DP_INT_AUX_CH_C_P | 60 64 |
| DP_INT_AUXCH | DP_80D | DP_AUX | DP_INT_AUX_CH_C_N | 60 64 |
| DP_INT_AUXCH | DP_80D | DP_AUX | DP_INT_AUXCH_C_P  | 5 60  |
| DP_INT_AUXCH | DP_80D | DP_AUX | DP_INT_AUXCH_C_N  | 5 60  |
|              | DP_80D | DP_AUX | DP_INT_AUXCH_P    |       |
|              | DP_80D | DP_AUX | DP_INT_AUXCH_N    |       |

DP

|                                 |                                                                                                                                                           |                      |                |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|
| SYNC MASTER=CONSTRAINTS         |                                                                                                                                                           | SYNC DATE=09/25/2012 |                |
| PAGE TITLE                      |                                                                                                                                                           | CPU Constraints      |                |
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|                                 |                                                                                                                                                           | PAGE                 | 111 OF 121     |
|                                 |                                                                                                                                                           | SHEET                | 67 OF 76       |



USB EXTB nets (Left USB port)

|                                                                                                                                                                                                                                                                                                                      |  |                                           |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------|--|
| SYNCH MASTER=CLEAN 343                                                                                                                                                                                                                                                                                               |  | SYNCH DATE=11/13/2012                     |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                           |  |                                           |  |
| PCH Constraints 1                                                                                                                                                                                                                                                                                                    |  |                                           |  |
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|                                                                                                                                                                                                                                                                                                                      |  | PAGE<br><b>112 OF 121</b>                 |  |
|                                                                                                                                                                                                                                                                                                                      |  | SHEET<br><b>68 OF 76</b>                  |  |



## LPC Bus Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| LPC_45S           | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |
| CLK_LPC_45S       | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| LPC              | *     | =3x_DIELECTRIC       | ?      |
| CLK_LPC          | *     | =4x_DIELECTRIC       | ?      |

SOURCE: Calpella Platform Design Guide for Ibox Peak M (DG-398905-398905\_v1.5), Section 3.15

## SMBus Interface Constraints

| PHYSICAL_RULE_SET | LAYER      | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SMB_45S_R_50S     | TOP,BOTTOM | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          |                      |                   |
| SMB_45S_R_50S     | *          | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SMB              | *     | =2x_DIELECTRIC       | ?      |

## HD Audio Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| HDA_45S           | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| HDA              | *     | =2x_DIELECTRIC       | ?      |

SOURCE: Calpella Platform Design Guide for Ibox Peak M (DG-398905-398905\_v1.5), Section 3.15

## SIO Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| CLK_SLOW_45S      | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CLK_SLOW         | *     | =4x_DIELECTRIC       | ?      |

## SPI Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SPI_45S           | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SPI              | *     | =4x_DIELECTRIC       | ?      |

## XDP Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| PCH_45S           | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| PCH_ITP          | *     | =2:1_SPACING         | ?      |

## DisplayPort

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| DP_80D            | *     | =80_OHM_DIFF          | =80_OHM_DIFF       | =80_OHM_DIFF       | =80_OHM_DIFF        | =80_OHM_DIFF         | =80_OHM_DIFF      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| DP_2DP           | *     | =3x_DIELECTRIC       | ?      |
| DP_20THERHS      | *     | =4x_DIELECTRIC       | ?      |
| DP_20THER        | *     | =3x_DIELECTRIC       | ?      |
| DP_AUX           | *     | =3x_DIELECTRIC       | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| DP_TX             | DP_TX             | *         | DP_2DP           |
| DP_TX             | *_TX              | *         | DP_20THERHS      |
| DP_TX             | *_RX              | *         | DP_20THERHS      |
| DP_TX             | *                 | *         | DP_20THER        |

## System Clock Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| CLK_SLOW_45S      | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |
| CLK_25M_45S       | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CLK_SLOW         | *     | =2x_DIELECTRIC       | ?      |
| CLK_25M          | *     | =5x_DIELECTRIC       | ?      |

NOTE: 25MHz system clocks very sensitive to noise.

## PCH Net Properties

| ELECTRICAL_CONSTRAINT_SET                    |                    | NET_TYPE      |             |                         |                      |
|----------------------------------------------|--------------------|---------------|-------------|-------------------------|----------------------|
|                                              |                    | PHYSICAL      | SPACING     |                         |                      |
| <div><div></div><div></div><div></div></div> | LPC_AD             | LPC_45S       | LPC         | LPC AD<3..0>            | 14 37 46 64          |
| <div><div></div><div></div><div></div></div> | LPC_FRAME_L        | LPC_45S       | LPC         | LPC FRAME_L             | 14 37 46 64          |
| <div><div></div><div></div><div></div></div> |                    | LPC_45S       | LPC         | LPCPLUS RESET_L         | 18 46 64             |
| <div><div></div><div></div><div></div></div> | LPC_CLK33M         | CLK_LPC_45S   | CLK_LPC     | LPC CLK24M_SMC          | 17 37                |
| <div><div></div><div></div><div></div></div> |                    | CLK_LPC_45S   | CLK_LPC     | LPC CLK24M_SMC_R        | 12 17                |
| <div><div></div><div></div><div></div></div> | LPC_CLK33M         | CLK_LPC_45S   | CLK_LPC     | LPC CLK24M_LPCPLUS      | 17 46 64             |
| <div><div></div><div></div><div></div></div> |                    | CLK_LPC_45S   | CLK_LPC     | LPC CLK24M_LPCPLUS_R    | 12 17                |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> | SMBUS_PCH_CLK      | SMB_45S_R_50S | SMB         | SMBUS_PCH_CLK           | 14 16 19 25 40 56    |
| <div><div></div><div></div><div></div></div> | SMBUS_PCH_DATA     | SMB_45S_R_50S | SMB         | SMBUS_PCH_DATA          | 14 16 19 25 40 56    |
| <div><div></div><div></div><div></div></div> | SMBUS_PCH_0_CLK    | SMB_45S_R_50S | SMB         | SMB_PCH_0_CLK           | 14 40                |
| <div><div></div><div></div><div></div></div> | SMBUS_PCH_0_DATA   | SMB_45S_R_50S | SMB         | SMB_PCH_0_DATA          | 14 40                |
| <div><div></div><div></div><div></div></div> | SMBUS_SMC_1_S0_SCL | SMB_45S_R_50S | SMB         | SMBUS_SMC_1_S0_SCL      | 14 32 37 40 43 44 64 |
| <div><div></div><div></div><div></div></div> | SMBUS_SMC_1_S0_SDA | SMB_45S_R_50S | SMB         | SMBUS_SMC_1_S0_SDA      | 14 32 37 40 43 44 64 |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> | HDA_BIT_CLK        | HDA_45S       | HDA         | HDA BIT_CLK             | 12 61 65             |
| <div><div></div><div></div><div></div></div> |                    | HDA_45S       | HDA         | HDA BIT_CLK_R           | 12                   |
| <div><div></div><div></div><div></div></div> | HDA_SYNC           | HDA_45S       | HDA         | HDA SYNC                | 12 61 65             |
| <div><div></div><div></div><div></div></div> |                    | HDA_45S       | HDA         | HDA SYNC_R              | 12                   |
| <div><div></div><div></div><div></div></div> | HDA_RST_L          | HDA_45S       | HDA         | HDA_RST_R_L             | 12                   |
| <div><div></div><div></div><div></div></div> |                    | HDA_45S       | HDA         | HDA_RST_L               | 12 61 65             |
| <div><div></div><div></div><div></div></div> | HDA_SDIN0          | HDA_45S       | HDA         | HDA SDIN0               | 12 61 65             |
| <div><div></div><div></div><div></div></div> | HDA_SDOUT          | HDA_45S       | HDA         | HDA SDOUT               | 12 61 65             |
| <div><div></div><div></div><div></div></div> |                    | HDA_45S       | HDA         | HDA SDOUT_R             | 12 17                |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> | PM_SUS_CLK         | CLK_SLOW_45S  | CLK_SLOW    | PM CLK32K_SUSCLK_R      | 13 38                |
| <div><div></div><div></div><div></div></div> |                    | CLK_SLOW_45S  | CLK_SLOW    | SMC_CLK32K              | 37 38                |
| <div><div></div><div></div><div></div></div> | SPT_CLK            | SPT_45S       | SPT         | SPI_CLK_R               | 14 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_CLK                 | 46                   |
| <div><div></div><div></div><div></div></div> | SPT_MOSI           | SPT_45S       | SPT         | SPI_MOSI_R              | 14 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_MOSI                | 46                   |
| <div><div></div><div></div><div></div></div> | SPT_MISO           | SPT_45S       | SPT         | SPI_MISO                | 14 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_MISO_R              | 46                   |
| <div><div></div><div></div><div></div></div> | SPT_CS0            | SPT_45S       | SPT         | SPI_CS0_R_L             | 14 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_CS0_L               | 46                   |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_SMC_CLK             | 37 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_SMC_MOSI            | 37 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_SMC_MISO            | 37 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_SMC_CS_L            | 37 46                |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_MLB_CLK             | 46                   |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_MLB_MOSI            | 46                   |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_MLB_MISO            | 46                   |
| <div><div></div><div></div><div></div></div> |                    | SPT_45S       | SPT         | SPI_MLB_CS_L            | 46                   |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> | PCIE_AP_R2D        | PCIE_80D      | PCIE_PCH_TX | PCIE AP_R2D_P           | 29 64                |
| <div><div></div><div></div><div></div></div> | PCIE_AP_R2D        | PCIE_80D      | PCIE_PCH_TX | PCIE AP_R2D_N           | 29 64                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_TX | PCIE AP_R2D_C_P         | 14 29                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_TX | PCIE AP_R2D_C_N         | 14 29                |
| <div><div></div><div></div><div></div></div> | PCIE_AP_D2R        | PCIE_80D      | PCIE_PCH_RX | PCIE AP_D2R_P           | 14 29 64             |
| <div><div></div><div></div><div></div></div> | PCIE_AP_D2R        | PCIE_80D      | PCIE_PCH_RX | PCIE AP_D2R_N           | 14 29 64             |
| <div><div></div><div></div><div></div></div> | PCIE_CLK100M_AP    | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_AP_P       | 12 29 64             |
| <div><div></div><div></div><div></div></div> | PCIE_CLK100M_AP    | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_AP_N       | 12 29 64             |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> | PCIE_TBT_R2D       | PCIE_80D      | PCIE_PCH_TX | PCIE TBT_R2D_P<3..0>    | 25                   |
| <div><div></div><div></div><div></div></div> | PCIE_TBT_R2D       | PCIE_80D      | PCIE_PCH_TX | PCIE TBT_R2D_N<3..0>    | 25                   |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_TX | PCIE TBT_R2D_C_P<3..0>  | 14 25                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_TX | PCIE TBT_R2D_C_N<3..0>  | 14 25                |
| <div><div></div><div></div><div></div></div> | PCIE_TBT_D2R       | PCIE_80D      | PCIE_PCH_RX | PCIE TBT_D2R_P<3..0>    | 14 25                |
| <div><div></div><div></div><div></div></div> | PCIE_TBT_D2R       | PCIE_80D      | PCIE_PCH_RX | PCIE TBT_D2R_N<3..0>    | 14 25                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_RX | PCIE TBT_D2R_C_P<3..0>  | 25                   |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_RX | PCIE TBT_D2R_C_N<3..0>  | 25                   |
| <div><div></div><div></div><div></div></div> | PCIE_CLK100M_TBT   | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_TBT_P      | 12 25                |
| <div><div></div><div></div><div></div></div> | PCIE_CLK100M_TBT   | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_TBT_N      | 12 25                |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> |                    | CLK_PCIE_80D  | CLK_PCIE    | PEG_CLK100M_P           |                      |
| <div><div></div><div></div><div></div></div> |                    | CLK_PCIE_80D  | CLK_PCIE    | PEG_CLK100M_N           |                      |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> | XDP_TDI            | PCH_45S       | PCH_ITP     | XDP_PCH_TDI             | 12 16 64             |
| <div><div></div><div></div><div></div></div> | XDP_TDO            | PCH_45S       | PCH_ITP     | XDP_PCH_TDO             | 12 16 64             |
| <div><div></div><div></div><div></div></div> | XDP_TMS            | PCH_45S       | PCH_ITP     | XDP_PCH_TMS             | 12 16 64             |
| <div><div></div><div></div><div></div></div> | XDP_TCK            | PCH_45S       | PCH_ITP     | XDP_PCH_TCK             | 12 16 64             |
|                                              |                    |               |             |                         |                      |
| <div><div></div><div></div><div></div></div> | PCIE_CAM           | PCIE_80D      | PCIE_PCH_TX | PCIE CAMERA_R2D_P       | 31 32                |
| <div><div></div><div></div><div></div></div> | PCIE_CAM           | PCIE_80D      | PCIE_PCH_TX | PCIE CAMERA_R2D_N       | 31 32                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_TX | PCIE CAMERA_R2D_C_P     | 14 32                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_TX | PCIE CAMERA_R2D_C_N     | 14 32                |
| <div><div></div><div></div><div></div></div> | PCIE_CAM           | PCIE_80D      | PCIE_PCH_RX | PCIE CAMERA_D2R_P       | 14 32                |
| <div><div></div><div></div><div></div></div> | PCIE_CAM           | PCIE_80D      | PCIE_PCH_RX | PCIE CAMERA_D2R_N       | 14 32                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_RX | PCIE CAMERA_D2R_C_P     | 31 32                |
| <div><div></div><div></div><div></div></div> |                    | PCIE_80D      | PCIE_PCH_RX | PCIE CAMERA_D2R_C_N     | 31 32                |
| <div><div></div><div></div><div></div></div> | PCIE_CLK100M_CAM   | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_CAMERA_P   | 12 32                |
| <div><div></div><div></div><div></div></div> | PCIE_CLK100M_CAM   | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_CAMERA_N   | 12 32                |
| <div><div></div><div></div><div></div></div> |                    | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_CAMERA_C_P | 31 32                |
| <div><div></div><div></div><div></div></div> |                    | CLK_PCIE_80D  | CLK_PCIE    | PCIE_CLK100M_CAMERA_C_N | 31 32                |

## Clock Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                           |                    | NET_TYPE     |          |                      |
|-------------------------------------------------------------------------------------|--------------------|--------------|----------|----------------------|
|                                                                                     |                    | PHYSICAL     | SPACING  |                      |
|  | SYSCLK_CLK32K_RTC  | CLK_SLOW_45S | CLK_SLOW | SYSCLK_CLK32K_RTCX1  |
|  | SYSCLK_CLK25M_SB   | CLK_25M_45S  | CLK_25M  | SYSCLK_CLK25M_CAMERA |
|  |                    | CLK_25M_45S  | CLK_25M  | CLK25M_CAM_CLKP      |
|  |                    | CLK_25M_45S  | CLK_25M  | CLK25M_CAM_XTALP_R   |
|  |                    | CLK_25M_45S  | CLK_25M  | CLK25M_CAM_XTALP     |
|  |                    | CLK_25M_45S  | CLK_25M  | CLK25M_CAM_XTALN     |
|  |                    | CLK_25M_45S  | CLK_25M  | CLK25M_CAM_CLKN      |
|  | SYSCLK_CLK25M_TBT  | CLK_25M_45S  | CLK_25M  | SYSCLK_CLK25M_TBT    |
|  |                    | CLK_25M_45S  | CLK_25M  | SYSCLK_CLK25M_TBT_R  |
|  | SYSCLK_CLK25M_XTAL | CLK_25M_45S  | CLK_25M  | SYSCLK_CLK25M_X1     |
|  |                    | CLK_25M_45S  | CLK_25M  | SYSCLK_CLK25M_X2     |
|  |                    | CLK_25M_45S  | CLK_25M  | SYSCLK_CLK25M_X2_R   |
|  |                    | CLK_25M_45S  | CLK_25M  | SDCLK_CLK25M_X2      |
|  |                    | CLK_25M_45S  | CLK_25M  | SDCLK_CLK25M_X2_R    |
|  |                    | CLK_25M_45S  | CLK_25M  | SDSCLK_CLK25M_X1     |

|                                                                                                                   |  |                      |            |
|-------------------------------------------------------------------------------------------------------------------|--|----------------------|------------|
| SYNC_MASTER=J41 MLB                                                                                               |  | SYNC_DATE=12/14/2012 |            |
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| PCH Constraints 2                                                                                                 |  |                      |            |
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## Memory Bus Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| MEM_40S           | *     | =40_OHM_SE            | =40_OHM_SE         | =40_OHM_SE         | =40_OHM_SE          | =40_OHM_SE           | =40_OHM_SE        |
| MEM_50S           | *     | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =50_OHM_SE           | =50_OHM_SE        |
| MEM_70D           | *     | =70_OHM_DIFF          | =70_OHM_DIFF       | =70_OHM_DIFF       | =70_OHM_DIFF        | =70_OHM_DIFF         | =70_OHM_DIFF      |
| MEM_73D           | *     | =73_OHM_DIFF          | =73_OHM_DIFF       | =73_OHM_DIFF       | =73_OHM_DIFF        | =73_OHM_DIFF         | =73_OHM_DIFF      |

## Spacing Rule Sets

| SPACING_RULE_SET  | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|-------------------|-------|----------------------|--------|
| MEM_DATA2SELF     | *     | =2X_DIELECTRIC       | ?      |
| MEM_DATA2OTHERMEM | *     | =8X_DIELECTRIC       | ?      |
| MEM_DQS2OWNDATA   | *     | =3X_DIELECTRIC       | ?      |
| MEM_CMD2CMD       | *     | =3X_DIELECTRIC       | ?      |
| MEM_CMD2CTRL      | *     | =3X_DIELECTRIC       | ?      |
| MEM_CTRL2CTRL     | *     | =3X_DIELECTRIC       | ?      |
| MEM_CLK2CLK       | *     | =6X_DIELECTRIC       | ?      |
| MEM_20THERMEM     | *     | =4X_DIELECTRIC       | ?      |
| MEM_2PWR          | *     | =2X_DIELECTRIC       | 10000  |
| MEM_2GND          | *     | =2X_DIELECTRIC       | 10000  |
| MEM_20THER        | *     | =6X_DIELECTRIC       | ?      |

## Memory to Power Spacing

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_PWR           | MEM_*             | *         | MEM_2PWR         |
| MEM_PWR           | *                 | *         | DEFAULT          |

| NET_PHYSICAL_TYPE | AREA_TYPE | PHYSICAL_RULE_SET |
|-------------------|-----------|-------------------|
| MEM_70D           | MEM_TERM  | MEM_73D           |
| MEM_40S           | MEM_TERM  | MEM_50S           |

## Memory to GND Spacing

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| GND               | MEM_*             | *         | MEM_2GND         |

## Memory Bus Spacing Group Assignments

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_A_DQS_0       | MEM_A_DATA_0      | *         | MEM_DQS2OWNDATA  |
| MEM_A_DQS_1       | MEM_A_DATA_1      | *         | MEM_DQS2OWNDATA  |
| MEM_A_DQS_2       | MEM_A_DATA_2      | *         | MEM_DQS2OWNDATA  |
| MEM_A_DQS_3       | MEM_A_DATA_3      | *         | MEM_DQS2OWNDATA  |
| MEM_A_DQS_4       | MEM_A_DATA_4      | *         | MEM_DQS2OWNDATA  |
| MEM_A_DQS_5       | MEM_A_DATA_5      | *         | MEM_DQS2OWNDATA  |
| MEM_A_DQS_6       | MEM_A_DATA_6      | *         | MEM_DQS2OWNDATA  |
| MEM_A_DQS_7       | MEM_A_DATA_7      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_0       | MEM_B_DATA_0      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_1       | MEM_B_DATA_1      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_2       | MEM_B_DATA_2      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_3       | MEM_B_DATA_3      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_4       | MEM_B_DATA_4      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_5       | MEM_B_DATA_5      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_6       | MEM_B_DATA_6      | *         | MEM_DQS2OWNDATA  |
| MEM_B_DQS_7       | MEM_B_DATA_7      | *         | MEM_DQS2OWNDATA  |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_*_DATA_*      | =SAME             | *         | MEM_DATA2SELF    |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET  |
|-------------------|-------------------|-----------|-------------------|
| MEM_*_DATA_*      | MEM_*             | *         | MEM_DATA2OTHERMEM |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CMD           | MEM_CMD           | *         | MEM_CMD2CMD      |
| MEM_CMD           | MEM_CTRL          | *         | MEM_CMD2CTRL     |
| MEM_CTRL          | MEM_CTRL          | *         | MEM_CTRL2CTRL    |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CLK           | MEM_CLK           | *         | MEM_CLK2CLK      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_*             | MEM_*             | *         | MEM_20THERMEM    |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_A_DQS_0       | *                 | *         | MEM_20THER       |
| MEM_A_DQS_1       | *                 | *         | MEM_20THER       |
| MEM_A_DQS_2       | *                 | *         | MEM_20THER       |
| MEM_A_DQS_3       | *                 | *         | MEM_20THER       |
| MEM_A_DQS_4       | *                 | *         | MEM_20THER       |
| MEM_A_DQS_5       | *                 | *         | MEM_20THER       |
| MEM_A_DQS_6       | *                 | *         | MEM_20THER       |
| MEM_A_DQS_7       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_0       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_1       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_2       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_3       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_4       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_5       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_6       | *                 | *         | MEM_20THER       |
| MEM_B_DQS_7       | *                 | *         | MEM_20THER       |

|              |   |   |            |
|--------------|---|---|------------|
| MEM_A_DATA_0 | * | * | MEM_20THER |
| MEM_A_DATA_1 | * | * | MEM_20THER |
| MEM_A_DATA_2 | * | * | MEM_20THER |
| MEM_A_DATA_3 | * | * | MEM_20THER |
| MEM_A_DATA_4 | * | * | MEM_20THER |
| MEM_A_DATA_5 | * | * | MEM_20THER |
| MEM_A_DATA_6 | * | * | MEM_20THER |
| MEM_A_DATA_7 | * | * | MEM_20THER |
| MEM_B_DATA_0 | * | * | MEM_20THER |
| MEM_B_DATA_1 | * | * | MEM_20THER |
| MEM_B_DATA_2 | * | * | MEM_20THER |
| MEM_B_DATA_3 | * | * | MEM_20THER |
| MEM_B_DATA_4 | * | * | MEM_20THER |
| MEM_B_DATA_5 | * | * | MEM_20THER |
| MEM_B_DATA_6 | * | * | MEM_20THER |
| MEM_B_DATA_7 | * | * | MEM_20THER |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_A_DATA_0      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_1      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_2      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_3      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_4      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_5      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_6      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_7      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_0      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_1      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_2      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_3      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_4      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_5      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_6      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_7      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_CMD           | *                 | *         | MEM_20THER       |
| MEM_CTRL          | *                 | *         | MEM_20THER       |
| MEM_CLK           | *                 | *         | MEM_20THER       |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_A_DATA_0      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_1      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_2      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_3      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_4      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_5      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_6      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_A_DATA_7      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_0      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_1      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_2      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_3      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_4      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_5      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_6      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |
| MEM_B_DATA_7      | MEM_*_DATA_*      | *         | MEM_20THERMEM    |

## Memory Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET_TYPE |              |                       |                      |
|---------------------------|----------|--------------|-----------------------|----------------------|
|                           | PHYSICAL | SPACING      |                       |                      |
| MEM_A_CLK0                | MEM_70D  | MEM_CLK      | MEM A CLK P<0>        | 7 21 24              |
| MEM_A_CLK0                | MEM_70D  | MEM_CLK      | MEM A CLK N<0>        | 7 21 24              |
| MEM_A_CLK1                | MEM_70D  | MEM_CLK      | MEM A CLK P<1>        | 7 21 24              |
| MEM_A_CLK1                | MEM_70D  | MEM_CLK      | MEM A CLK N<1>        | 7 21 24              |
| MEM_A_CTRL                | MEM_40S  | MEM_CTRL     | MEM A CS L<1..0>      | 7 20 21 24           |
| MEM_A_CTRL                | MEM_40S  | MEM_CTRL     | MEM A ODT<0>          | 7 20 21 24 63        |
| MEM_A_CKE0                | MEM_40S  | MEM_CMD      | MEM A CKE<1..0>       | 7 20 24              |
| MEM_A_CKE1                | MEM_40S  | MEM_CMD      | MEM A CKE<3..2>       | 7 21 24              |
| MEM_A_CMD0                | MEM_40S  | MEM_CMD      | MEM A CAA<9..0>       | 7 20 24 63           |
| MEM_A_CMD1                | MEM_40S  | MEM_CMD      | MEM A CAB<9..0>       | 7 21 24 63           |
| MEM_A_DQ_BYTE0            | MEM_40S  | MEM_A_DATA_0 | MEM A DQ<7..0>        | 7 63                 |
| MEM_A_DQ_BYTE1            | MEM_40S  | MEM_A_DATA_1 | MEM A DQ<15..8>       | 7 63                 |
| MEM_A_DQ_BYTE2            | MEM_40S  | MEM_A_DATA_2 | MEM A DQ<23..16>      | 7 63                 |
| MEM_A_DQ_BYTE3            | MEM_40S  | MEM_A_DATA_3 | MEM A DQ<31..24>      | 7 63                 |
| MEM_A_DQ_BYTE4            | MEM_40S  | MEM_A_DATA_4 | MEM A DQ<39..32>      | 7 21 63              |
| MEM_A_DQ_BYTE5            | MEM_40S  | MEM_A_DATA_5 | MEM A DQ<47..40>      | 7 63                 |
| MEM_A_DQ_BYTE6            | MEM_40S  | MEM_A_DATA_6 | MEM A DQ<55..48>      | 7 63                 |
| MEM_A_DQ_BYTE7            | MEM_40S  | MEM_A_DATA_7 | MEM A DQ<63..56>      | 7 63                 |
| MEM_A_DQS0                | MEM_70D  | MEM_A_DQS_0  | MEM A DQS P<0>        | 7 63                 |
| MEM_A_DQS0                | MEM_70D  | MEM_A_DQS_0  | MEM A DQS N<0>        | 7 63                 |
| MEM_A_DQS1                | MEM_70D  | MEM_A_DQS_1  | MEM A DQS P<1>        | 7 63                 |
| MEM_A_DQS1                | MEM_70D  | MEM_A_DQS_1  | MEM A DQS N<1>        | 7 63                 |
| MEM_A_DQS2                | MEM_70D  | MEM_A_DQS_2  | MEM A DQS P<2>        | 7 63                 |
| MEM_A_DQS2                | MEM_70D  | MEM_A_DQS_2  | MEM A DQS N<2>        | 7 63                 |
| MEM_A_DQS3                | MEM_70D  | MEM_A_DQS_3  | MEM A DQS P<3>        | 7 63                 |
| MEM_A_DQS3                | MEM_70D  | MEM_A_DQS_3  | MEM A DQS N<3>        | 7 63                 |
| MEM_A_DQS4                | MEM_70D  | MEM_A_DQS_4  | MEM A DQS P<4>        | 7 63                 |
| MEM_A_DQS4                | MEM_70D  | MEM_A_DQS_4  | MEM A DQS N<4>        | 7 63                 |
| MEM_A_DQS5                | MEM_70D  | MEM_A_DQS_5  | MEM A DQS P<5>        | 7 63                 |
| MEM_A_DQS5                | MEM_70D  | MEM_A_DQS_5  | MEM A DQS N<5>        | 7 63                 |
| MEM_A_DQS6                | MEM_70D  | MEM_A_DQS_6  | MEM A DQS P<6>        | 7 21 63              |
| MEM_A_DQS6                | MEM_70D  | MEM_A_DQS_6  | MEM A DQS N<6>        | 7 21 63              |
| MEM_A_DQS7                | MEM_70D  | MEM_A_DQS_7  | MEM A DQS P<7>        | 7 63                 |
| MEM_A_DQS7                | MEM_70D  | MEM_A_DQS_7  | MEM A DQS N<7>        | 7 63                 |
| MEM_B_CLK0                | MEM_70D  | MEM_CLK      | MEM B CLK P<0>        | 7 22 24              |
| MEM_B_CLK0                | MEM_70D  | MEM_CLK      | MEM B CLK N<0>        | 7 22 24              |
| MEM_B_CLK1                | MEM_70D  | MEM_CLK      | MEM B CLK P<1>        | 7 23 24              |
| MEM_B_CLK1                | MEM_70D  | MEM_CLK      | MEM B CLK N<1>        | 7 23 24              |
| MEM_B_CTRL                | MEM_40S  | MEM_CTRL     | MEM B CS L<1..0>      | 7 22 23 24           |
| MEM_B_CTRL                | MEM_40S  | MEM_CTRL     | MEM B ODT<0>          | 7 22 23 24 63        |
| MEM_B_CKE0                | MEM_40S  | MEM_CMD      | MEM B CKE<1..0>       | 7 22 24              |
| MEM_B_CKE1                | MEM_40S  | MEM_CMD      | MEM B CKE<3..2>       | 7 23 24              |
| MEM_B_CMD0                | MEM_40S  | MEM_CMD      | MEM B CAA<9..0>       | 7 22 24 63           |
| MEM_B_CMD1                | MEM_40S  | MEM_CMD      | MEM B CAB<9..0>       | 7 23 24 63           |
| MEM_B_DQ_BYTE0            | MEM_40S  | MEM_B_DATA_0 | MEM B DQ<7..0>        | 7 63                 |
| MEM_B_DQ_BYTE1            | MEM_40S  | MEM_B_DATA_1 | MEM B DQ<15..8>       | 7 63                 |
| MEM_B_DQ_BYTE2            | MEM_40S  | MEM_B_DATA_2 | MEM B DQ<23..16>      | 7 63                 |
| MEM_B_DQ_BYTE3            | MEM_40S  | MEM_B_DATA_3 | MEM B DQ<31..24>      | 7 63                 |
| MEM_B_DQ_BYTE4            | MEM_40S  | MEM_B_DATA_4 | MEM B DQ<39..32>      | 7 23 63              |
| MEM_B_DQ_BYTE5            | MEM_40S  | MEM_B_DATA_5 | MEM B DQ<47..40>      | 7 63                 |
| MEM_B_DQ_BYTE6            | MEM_40S  | MEM_B_DATA_6 | MEM B DQ<55..48>      | 7 63                 |
| MEM_B_DQ_BYTE7            | MEM_40S  | MEM_B_DATA_7 | MEM B DQ<63..56>      | 7 63                 |
| MEM_B_DQS0                | MEM_70D  | MEM_B_DQS_0  | MEM B DQS P<0>        | 7 63                 |
| MEM_B_DQS0                | MEM_70D  | MEM_B_DQS_0  | MEM B DQS N<0>        | 7 63                 |
| MEM_B_DQS1                | MEM_70D  | MEM_B_DQS_1  | MEM B DQS P<1>        | 7 63                 |
| MEM_B_DQS1                | MEM_70D  | MEM_B_DQS_1  | MEM B DQS N<1>        | 7 63                 |
| MEM_B_DQS2                | MEM_70D  | MEM_B_DQS_2  | MEM B DQS P<2>        | 7 63                 |
| MEM_B_DQS2                | MEM_70D  | MEM_B_DQS_2  | MEM B DQS N<2>        | 7 63                 |
| MEM_B_DQS3                | MEM_70D  | MEM_B_DQS_3  | MEM B DQS P<3>        | 7 63                 |
| MEM_B_DQS3                | MEM_70D  | MEM_B_DQS_3  | MEM B DQS N<3>        | 7 63                 |
| MEM_B_DQS4                | MEM_70D  | MEM_B_DQS_4  | MEM B DQS P<4>        | 7 63                 |
| MEM_B_DQS4                | MEM_70D  | MEM_B_DQS_4  | MEM B DQS N<4>        | 7 63                 |
| MEM_B_DQS5                | MEM_70D  | MEM_B_DQS_5  | MEM B DQS P<5>        | 7 63                 |
| MEM_B_DQS5                | MEM_70D  | MEM_B_DQS_5  | MEM B DQS N<5>        | 7 63                 |
| MEM_B_DQS6                | MEM_70D  | MEM_B_DQS_6  | MEM B DQS P<6>        | 7 23 63              |
| MEM_B_DQS6                | MEM_70D  | MEM_B_DQS_6  | MEM B DQS N<6>        | 7 23 63              |
| MEM_B_DQS7                | MEM_70D  | MEM_B_DQS_7  | MEM B DQS P<7>        | 7 63                 |
| MEM_B_DQS7                | MEM_70D  | MEM_B_DQS_7  | MEM B DQS N<7>        | 7 63                 |
|                           |          | MEM_PWR      | PP1V2 S3              | 17 19 20 21 22 23 42 |
|                           |          | MEM_PWR      | PP0V6 S3 MEM VREFCA A | 18 19 20 21          |
|                           |          | MEM_PWR      | PP0V6 S3 MEM VREFDQ A | 18 19 20 21          |
|                           |          | MEM_PWR      | PP0V6 S3 MEM VREFCA B | 18 19 22 23          |
|                           |          | MEM_PWR      | PP0V6 S3 MEM VREFDQ B | 18 19 22 23          |

|                                                                                                                   |  |                      |            |
|-------------------------------------------------------------------------------------------------------------------|--|----------------------|------------|
| SYNC_MASTER=CONSTRAINTS                                                                                           |  | SYNC_DATE=09/25/2012 |            |
| PAGE TITLE                                                                                                        |  |                      |            |
| Memory Constraints                                                                                                |  |                      |            |
|  Apple Inc.                  |  | DRAWING NUMBER       | SIZE       |
|                                                                                                                   |  | <SCH_NUM>            | D          |
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## DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

### Thunderbolt SPI Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| TBT_SPI_45S       | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| TBT_SPI          | *     | =2x_DIELECTRIC       | ?      |

### Thunderbolt/DP Connector Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| TBDTP_80D         | *     | =80_OHM_DIFF          | =80_OHM_DIFF       | =80_OHM_DIFF       | =80_OHM_DIFF        | =80_OHM_DIFF         | =80_OHM_DIFF      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| TBDTP_TX          | TBDTP_TX          | *         | TBDTP_TX2TX      |
| TBDTP_RX          | TBDTP_RX          | *         | TBDTP_RX2RX      |
| TBDTP_TX          | TBDTP_RX          | *         | TBDTP_TX2RX      |
| TBDTP_RX          | TBDTP_TX          | *         | TBDTP_TX2RX      |
| TBDTP_TX          | *_TX              | *         | TBDTP_20THERHS   |
| TBDTP_RX          | *_TX              | *         | TBDTP_20THERHS   |
| TBDTP_TX          | *_RX              | *         | TBDTP_20THERHS   |
| TBDTP_RX          | *_RX              | *         | TBDTP_20THERHS   |
| TBDTP_TX          | *                 | *         | TBDTP_20THER     |
| TBDTP_RX          | *                 | *         | TBDTP_20THER     |

| SPACING_RULE_SET | LAYER      | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|------------|----------------------|--------|
| TBDTP_TX2TX      | TOP,BOTTOM | =6x_DIELECTRIC       | ?      |
| TBDTP_RX2RX      | TOP,BOTTOM | =6x_DIELECTRIC       | ?      |
| TBDTP_TX2RX      | TOP,BOTTOM | =10x_DIELECTRIC      | ?      |
| TBDTP_20THERHS   | TOP,BOTTOM | =10x_DIELECTRIC      | ?      |
| TBDTP_20THER     | TOP,BOTTOM | =6x_DIELECTRIC       | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| TBDTP_TX2TX      | *     | =4x_DIELECTRIC       | ?      |
| TBDTP_RX2RX      | *     | =4x_DIELECTRIC       | ?      |
| TBDTP_TX2RX      | *     | =6x_DIELECTRIC       | ?      |
| TBDTP_20THERHS   | *     | =6x_DIELECTRIC       | ?      |
| TBDTP_20THER     | *     | =4x_DIELECTRIC       | ?      |

## Thunderbolt/DP Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET_TYPE  |          |                              |
|---------------------------|-----------|----------|------------------------------|
|                           | PHYSICAL  | SPACING  |                              |
| TBT_A_R2D                 | TBDTP_80D | TBDTP_TX | TBT A R2D C P<1..0> 25 28    |
| TBT_A_R2D                 | TBDTP_80D | TBDTP_TX | TBT A R2D C N<1..0> 25 28    |
|                           | TBDTP_80D | TBDTP_TX | TBT A R2D P<1..0> 28         |
|                           | TBDTP_80D | TBDTP_TX | TBT A R2D N<1..0> 28         |
| DP_TBTPA_ML1              | DP_80D    | DP_TX    | DP TBTPA ML C P<1> 25 28     |
| DP_TBTPA_ML1              | DP_80D    | DP_TX    | DP TBTPA ML C N<1> 25 28     |
| DP_TBTPA_ML3              | DP_80D    | DP_TX    | DP TBTPA ML C P<3> 25 28     |
| DP_TBTPA_ML3              | DP_80D    | DP_TX    | DP TBTPA ML C N<3> 25 28     |
|                           | DP_80D    | DP_TX    | DP TBTPA ML P<3..1:2> 28     |
|                           | DP_80D    | DP_TX    | DP TBTPA ML N<3..1:2> 28     |
|                           | DP_80D    | DP_TX    | DP A LSX ML P<1> 28          |
|                           | DP_80D    | DP_TX    | DP A LSX ML N<1> 28          |
|                           | TBDTP_80D | TBDTP_RX | TBT A D2R C P<1..0> 28       |
|                           | TBDTP_80D | TBDTP_RX | TBT A D2R C N<1..0> 28       |
| TBT_A_D2R1                | TBDTP_80D | TBDTP_RX | TBT A D2R P<1> 25 28         |
| TBT_A_D2R1                | TBDTP_80D | TBDTP_RX | TBT A D2R N<1> 25 28         |
| TBT_A_D2R0                | TBDTP_80D | TBDTP_RX | TBT A D2R P<0> 25 28         |
| TBT_A_D2R0                | TBDTP_80D | TBDTP_RX | TBT A D2R N<0> 25 28         |
| TBT_A_AUXCH               | DP_80D    | DP_AUX   | DP TBTPA AUXCH C P 25 28     |
| TBT_A_AUXCH               | DP_80D    | DP_AUX   | DP TBTPA AUXCH C N 25 28     |
|                           | DP_80D    | DP_AUX   | DP TBTPA AUXCH P 28          |
|                           | DP_80D    | DP_AUX   | DP TBTPA AUXCH N 28          |
|                           | DP_80D    | DP_AUX   | DP A AUXCH DDC P 28          |
|                           | DP_80D    | DP_AUX   | DP A AUXCH DDC N 28          |
|                           | TBDTP_80D | TBDTP_RX | TBT A D2R1 AUXDDC P 28       |
|                           | TBDTP_80D | TBDTP_RX | TBT A D2R1 AUXDDC N 28       |
| TBT_B_R2D                 | TBDTP_80D | TBDTP_TX | TBT B R2D C P<1..0> 64       |
| TBT_B_R2D                 | TBDTP_80D | TBDTP_TX | TBT B R2D C N<1..0> 64       |
|                           | TBDTP_80D | TBDTP_TX | TBT B R2D P<1..0> 64         |
|                           | TBDTP_80D | TBDTP_TX | TBT B R2D N<1..0> 64         |
| DP_TBTPB_ML               | DP_80D    | DP_TX    | NC DP TBTPB ML CP<3..1:2> 64 |
| DP_TBTPB_ML               | DP_80D    | DP_TX    | NC DP TBTPB ML CN<3..1:2> 64 |
|                           | DP_80D    | DP_TX    | DP TBTPB ML P<3..1:2> 64     |
|                           | DP_80D    | DP_TX    | DP TBTPB ML N<3..1:2> 64     |
|                           | DP_80D    | DP_TX    | DP B LSX ML P<1> 64          |
|                           | DP_80D    | DP_TX    | DP B LSX ML N<1> 64          |
|                           | TBDTP_80D | TBDTP_RX | TBT B D2R C P<1..0> 64       |
|                           | TBDTP_80D | TBDTP_RX | TBT B D2R C N<1..0> 64       |
| TBT_B_D2R                 | TBDTP_80D | TBDTP_RX | TBT B D2R P<1..0> 64         |
| TBT_B_D2R                 | TBDTP_80D | TBDTP_RX | TBT B D2R N<1..0> 64         |
| TBT_B_AUXCH               | DP_80D    | DP_AUX   | NC DP TBTPB AUXCH CP 25 64   |
| TBT_B_AUXCH               | DP_80D    | DP_AUX   | NC DP TBTPB AUXCH CN 25 64   |
|                           | DP_80D    | DP_AUX   | DP TBTPB AUXCH P 64          |
|                           | DP_80D    | DP_AUX   | DP TBTPB AUXCH N 64          |
|                           | DP_80D    | DP_AUX   | DP B AUXCH DDC P 64          |
|                           | DP_80D    | DP_AUX   | DP B AUXCH DDC N 64          |
|                           | TBDTP_80D | TBDTP_RX | TBT B D2R1 AUXDDC P 64       |
|                           | TBDTP_80D | TBDTP_RX | TBT B D2R1 AUXDDC N 64       |

Only used on dual-port hosts.

## Thunderbolt IC Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET_TYPE    |         |                           |
|---------------------------|-------------|---------|---------------------------|
|                           | PHYSICAL    | SPACING |                           |
|                           | DP_80D      | DP_TX   | DP TBTSRC ML C P<3..0> 25 |
|                           | DP_80D      | DP_TX   | DP TBTSRC ML C N<3..0> 25 |
|                           | DP_80D      | DP_AUX  | DP TBTSRC AUXCH C P 25    |
|                           | DP_80D      | DP_AUX  | DP TBTSRC AUXCH C N 25    |
| TBT_SPI_CLK               | TBT_SPI_45S | TBT_SPI | TBT SPI CLK 25            |
| TBT_SPI_MOSI              | TBT_SPI_45S | TBT_SPI | TBT SPI MOSI 25           |
| TBT_SPI_MISO              | TBT_SPI_45S | TBT_SPI | TBT SPI MISO 25           |
| TBT_SPI_CS_L              | TBT_SPI_45S | TBT_SPI | TBT SPI CS L 25           |

Only used on hosts supporting Thunderbolt video-in

|                                                                                                                   |  |                      |            |
|-------------------------------------------------------------------------------------------------------------------|--|----------------------|------------|
| SYNC_MASTER=CONSTRAINTS                                                                                           |  | SYNC_DATE=09/25/2012 |            |
| PAGE TITLE                                                                                                        |  |                      |            |
| Thunderbolt Constraints                                                                                           |  |                      |            |
|  Apple Inc.                  |  | DRAWING NUMBER       | SIZE       |
|                                                                                                                   |  | <SCH_NUM>            | D          |
|                                                                                                                   |  | REVISION             |            |
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Camera Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET_TYPE   |              |                        |       |
|---------------------------|------------|--------------|------------------------|-------|
|                           | PHYSICAL   | SPACING      |                        |       |
| S2_MEM_CLK                | S2_MFM_85D | S2_MFM_CLK   | MEM_CAM_CLK_P          | 31 32 |
| S2_MEM_CLK                | S2_MEM_85D | S2_MEM_CLK   | MEM_CAM_CLK_N          | 31 32 |
| S2_MEM_CTRL               | S2_MFM_45S | S2_MEM_CTRL  | MEM_CAM_CKE            | 31 32 |
| S2_MEM_CTRL               | S2_MFM_45S | S2_MFM_CTRL  | MEM_CAM_CS_L           | 31 32 |
| S2_MEM_CMD                | S2_MFM_45S | S2_MEM_CTRL  | MEM_CAM_ODT            | 32    |
| S2_MEM_CMD                | S2_MFM_45S | S2_MEM_CTRL  | MEM_CAM_CAS_L          | 31 32 |
| S2_MEM_CMD                | S2_MFM_45S | S2_MEM_CTRL  | MEM_CAM_RAS_L          | 31 32 |
| S2_MEM_CMD                | S2_MFM_45S | S2_MEM_CMD   | MEM_CAM_WE_L           | 31 32 |
| S2_MEM_CMD                | S2_MFM_45S | S2_MEM_CMD   | MEM_CAM_BA<0>          | 31 32 |
| S2_MEM_CMD                | S2_MFM_45S | S2_MEM_CMD   | MEM_CAM_BA<1>          | 31 32 |
| S2_MEM_CMD                | S2_MFM_45S | S2_MEM_CMD   | MEM_CAM_BA<2>          | 31 32 |
| S2_MEM_DQS0               | S2_MFM_85D | S2_MEM_DQS0  | MEM_CAM_DQS_P<0>       | 31 32 |
| S2_MEM_DQS0               | S2_MFM_85D | S2_MEM_DQS0  | MEM_CAM_DQS_N<0>       | 31 32 |
| S2_MEM_DQS1               | S2_MFM_85D | S2_MEM_DQS1  | MEM_CAM_DQS_P<1>       | 31 32 |
| S2_MEM_DQS1               | S2_MFM_85D | S2_MEM_DQS1  | MEM_CAM_DQS_N<1>       | 31 32 |
| S2_MEM_DATA_0             | S2_MFM_45S | S2_MFM_DATA0 | MEM_CAM_DM<0>          | 31 32 |
| S2_MEM_DATA_1             | S2_MFM_45S | S2_MFM_DATA1 | MEM_CAM_DM<1>          | 31 32 |
| S2_MEM_A                  | S2_MFM_45S | S2_MEM_CMD   | MEM_CAM_A<14..0>       | 31 32 |
| S2_MEM_DATA_0             | S2_MFM_45S | S2_MFM_DATA0 | MEM_CAM_DQ<7..0>       | 31 32 |
| S2_MEM_DATA_1             | S2_MFM_45S | S2_MEM_DATA1 | MEM_CAM_DQ<15..8>      | 31 32 |
|                           |            |              |                        |       |
| MIPI_DATA_S2              | MIPT_85D   | MIPI_DATA    | MIPI_DATA_P            | 31 32 |
| MIPI_DATA_S2              | MIPT_85D   | MIPI_DATA    | MIPI_DATA_N            | 31 32 |
| MIPI_DATA                 | MIPT_85D   | MIPI_DATA    | MIPI_DATA_CONN_P       | 32 64 |
| MIPI_DATA                 | MIPT_85D   | MIPI_DATA    | MIPI_DATA_CONN_N       | 32 64 |
|                           |            |              |                        |       |
| MIPI_CLK_S2               | MIPT_85D   | CLK_MIPI     | MIPI_CLK_P             | 31 32 |
| MIPI_CLK_S2               | MIPT_85D   | CLK_MIPI     | MIPI_CLK_N             | 31 32 |
| MIPI_CLK                  | MIPT_85D   | CLK_MIPI     | MIPI_CLK_CONN_P        | 32 64 |
| MIPI_CLK                  | MIPT_85D   | CLK_MIPI     | MIPI_CLK_CONN_N        | 32 64 |
|                           |            |              |                        |       |
| S2_MEM_PWR                |            | S2_MEM_PWR   | PP1V35_CAM             | 31 32 |
| S2_MEM_PWR                |            | S2_MEM_PWR   | PP0V675_CAM_VREF       | 31 32 |
| S2_MEM_PWR                |            | S2_MEM_PWR   | PP0V675_MEM_CAM_VREFCA | 32    |
| S2_MEM_PWR                |            | S2_MEM_PWR   | PP0V675_MEM_CAM_VREFDQ | 32    |

MIPI Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| MIPI_85D          | *     | =85_OHM_DIFF          | =85_OHM_DIFF       | =85_OHM_DIFF       | =85_OHM_DIFF        | =85_OHM_DIFF         | =85_OHM_DIFF      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| MIPI_20THER      | *     | =4X_DIELECTRIC       | ?      |
| MIPI_2CLK        | *     | =6X_DIELECTRIC       | ?      |
| MIPICKL_20THER   | *     | =70X_DIELECTRIC      | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MIPI_DATA         | *                 | *         | MIPI_20THER      |
| MIPI_DATA         | CLK_MIPI          | *         | MIPI_2CLK        |
| CLK_MIPI          | *                 | *         | MIPICKL_20THER   |

Memory Bus Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| S2_MEM_45S        | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |
| S2_MEM_85D        | *     | =85_OHM_DIFF          | =85_OHM_DIFF       | =85_OHM_DIFF       | =85_OHM_DIFF        | =85_OHM_DIFF         | =85_OHM_DIFF      |

Spacing Rule Sets

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| S2_DATA2SELF     | *     | =2X_DIELECTRIC       | ?      |
| S2_DQS20WNDATA   | *     | =2X_DIELECTRIC       | ?      |
| S2_CMD2CMD       | *     | =2X_DIELECTRIC       | ?      |
| S2_CMD2CTRL      | *     | =2X_DIELECTRIC       | ?      |
| S2_CTRL2CTRL     | *     | =2X_DIELECTRIC       | ?      |
| S2_20THERMEM     | *     | =4X_DIELECTRIC       | ?      |
| S2MEM_2PWR       | *     | =2X_DIELECTRIC       | ?      |
| S2MEM_2GND       | *     | =2X_DIELECTRIC       | ?      |
| S2MEM_20THER     | *     | =6X_DIELECTRIC       | ?      |

| SPACING_RULE_SET | LAYER      | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|------------|----------------------|--------|
| S2_DATA2SELF     | TOP,BOTTOM | =4X_DIELECTRIC       | ?      |
| S2_DQS20WNDATA   | TOP,BOTTOM | =4X_DIELECTRIC       | ?      |
| S2_CMD2CMD       | TOP,BOTTOM | =4X_DIELECTRIC       | ?      |
| S2_CMD2CTRL      | TOP,BOTTOM | =4X_DIELECTRIC       | ?      |
| S2_CTRL2CTRL     | TOP,BOTTOM | =4X_DIELECTRIC       | ?      |
| S2_20THERMEM     | TOP,BOTTOM | =6X_DIELECTRIC       | ?      |
| S2MEM_2PWR       | TOP,BOTTOM | =4X_DIELECTRIC       | ?      |
| S2MEM_2GND       | TOP,BOTTOM | =4X_DIELECTRIC       | ?      |
| S2MEM_20THER     | TOP,BOTTOM | =10X_DIELECTRIC      | ?      |

Memory Bus Spacing Group Assignments

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| S2_MEM_DATA*      | *                 | *         | S2MEM_20THER     |
| S2_MEM_DQS*       | *                 | *         | S2MEM_20THER     |
| S2_MEM_CMD        | *                 | *         | S2MEM_20THER     |
| S2_MEM_CTRL       | *                 | *         | S2MEM_20THER     |
| S2_MEM_CLK        | *                 | *         | S2MEM_20THER     |
| S2_MEM_DATA*      | =SAME             | *         | S2_DATA2SELF     |
| S2_MEM_CMD        | S2_MEM_CMD        | *         | S2_CMD2CMD       |
| S2_MEM_CMD        | S2_MEM_CTRL       | *         | S2_CMD2CTRL      |
| S2_MEM_CTRL       | S2_MEM_CTRL       | *         | S2_CTRL2CTRL     |
| S2_MEM_*          | S2_MEM_*          | *         | S2_20THERMEM     |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| S2_MEM_DQS1       | S2_MEM_DATA1      | *         | S2_DQS20WNDATA   |
| S2_MEM_DQS0       | S2_MEM_DATA0      | *         | S2_DQS20WNDATA   |

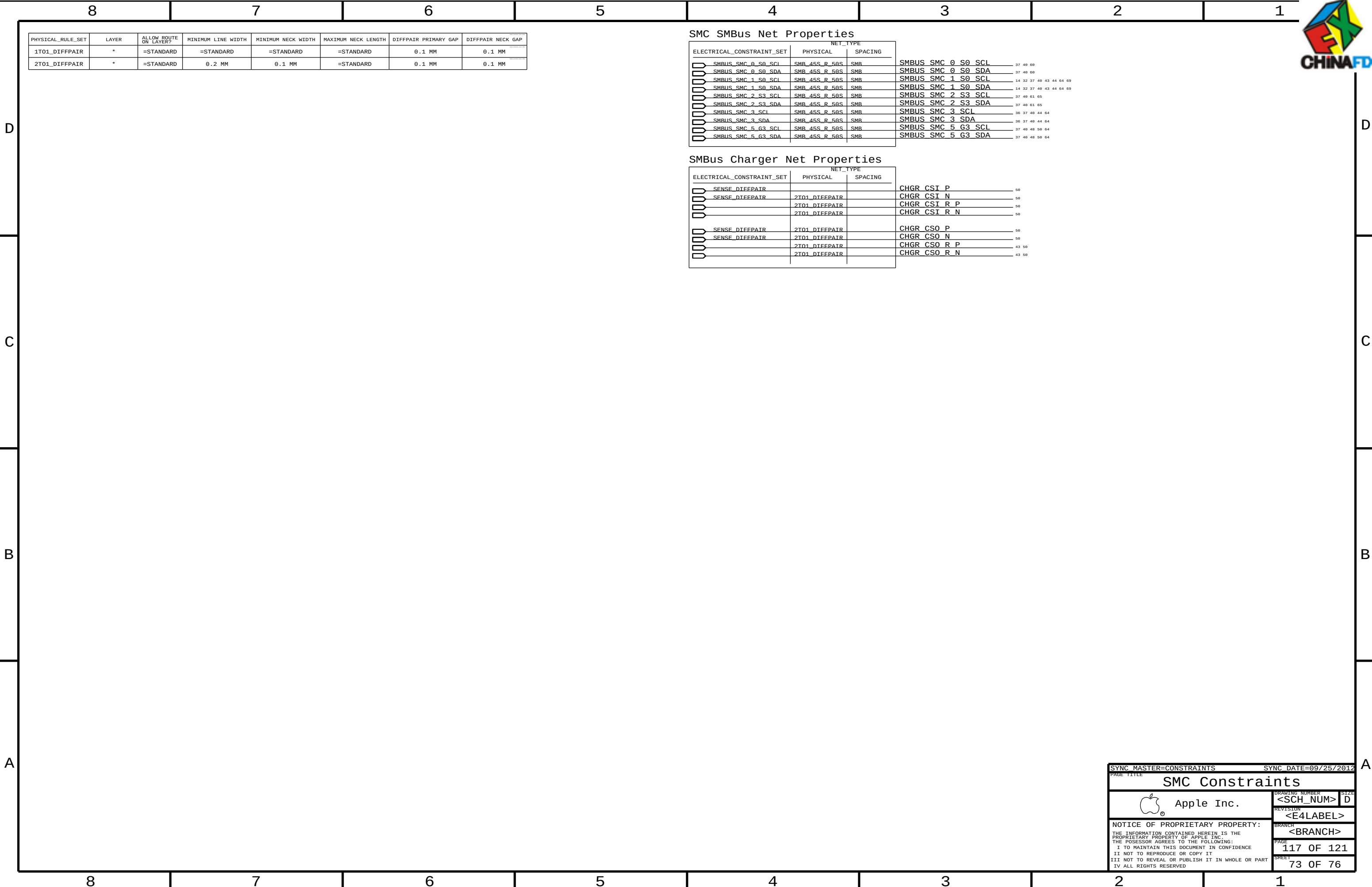
Memory to Power Spacing

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| S2_MEM_PWR        | S2_MEM_*          | *         | S2MEM_2PWR       |
| S2_MEM_PWR        | *                 | *         | DEFAULT          |

Memory to GND Spacing

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| GND               | S2_MEM_*          | *         | S2MEM_2GND       |

|                                                                                                                   |  |                             |           |
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| Camera Constraints                                                                                                |  |                             |           |
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|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|---|

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SENSE_1T01_45S    | *     | =1T01_DIFFPAIR        | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =1T01_DIFFPAIR       | =1T01_DIFFPAIR    |
| SENSE_1T01_P2MM   | *     | =1T01_DIFFPAIR        | 0.200 MM           | 0.100 MM           | =1T01_DIFFPAIR      | =1T01_DIFFPAIR       | =1T01_DIFFPAIR    |
| THERM_1T01_45S    | *     | =1T01_DIFFPAIR        | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =1T01_DIFFPAIR       | =1T01_DIFFPAIR    |
| SPKR_DIFFPAIR     | *     | =1T01_DIFFPAIR        | 0.300 MM           | 0.100 MM           | =1T01_DIFFPAIR      | =1T01_DIFFPAIR       | =1T01_DIFFPAIR    |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SENSE            | *     | =2:1_SPACING         | ?      |
| THERM            | *     | =2:1_SPACING         | ?      |
| AUDIO            | *     | =2:1_SPACING         | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND              | *     | =STANDARD            | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE_SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND_P2MM         | *     | 0.20 MM              | 10000  |
| PWR_P2MM         | *     | 0.20 MM              | 10000  |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CPU_COMP          | GND               | *         | GND_P2MM         |
| CPU_VCCSENSE      | GND               | *         | GND_P2MM         |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| GND               | CLK_PCIE          | *         | GND_P2MM         |
| GND               | PCIE*             | *         | GND_P2MM         |
| GND               | SATA*             | *         | GND_P2MM         |
| GND               | USB*              | *         | GND_P2MM         |
| GND               | LVDS*             | *         | GND_P2MM         |
| SB_POWER          | CLK_PCIE          | *         | PWR_P2MM         |
| SB_POWER          | SATA*             | *         | PWR_P2MM         |
| SB_POWER          | SATA*             | *         | PWR_P2MM         |

J11/J13 Specific Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET_TYPE        |         |                                                                                  |
|---------------------------|-----------------|---------|----------------------------------------------------------------------------------|
|                           | PHYSICAL        | SPACING |                                                                                  |
| SENSE_DIFFPAIR            | THERM_1T01_45S  | THERM   | INLET THMSNS D1 P 44                                                             |
| SENSE_DIFFPAIR            | THERM_1T01_45S  | THERM   | INLET THMSNS D1 N 44                                                             |
| SENSE_DIFFPAIR            | THERM_1T01_45S  | THERM   | TBTTHMSNS D2 R P 44                                                              |
| SENSE_DIFFPAIR            | THERM_1T01_45S  | THERM   | TBTTHMSNS D2 R N 44                                                              |
| SENSE_DIFFPAIR            | THERM_1T01_45S  | THERM   | TBTTHMSNS D2 P 44                                                                |
| SENSE_DIFFPAIR            | THERM_1T01_45S  | THERM   | TBTTHMSNS D2 N 44                                                                |
| SENSE_DIEFPATR            | THERM_1T01_45S  | THERM   | TBT MLBBOT THMSNS P 44                                                           |
| SENSE_DIEFPATR            | THERM_1T01_45S  | THERM   | TBT MLBBOT THMSNS N 44                                                           |
| SENSE_DIEFPATR            | THERM_1T01_45S  | THERM   | MLBBOT THMSNS D3 P 44                                                            |
| SENSE_DIEFPATR            | THERM_1T01_45S  | THERM   | MLBBOT THMSNS D3 N 44                                                            |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | TBDTHMSNS D2 P 44                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | TBDTHMSNS D2 N 44                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | CPU THMSNS D2 P 44                                                               |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | CPU THMSNS D2 N 44                                                               |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | CPUVCCIOS0 CS N 44                                                               |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | CPUVCCIOS0 CS P 44                                                               |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | CPUVR ISNS1 P 42 52                                                              |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | CPUVR ISNS1 N 42 52                                                              |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | CPUVR ISNS2 P 42 52                                                              |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | CPUVR ISNS2 N 42 52                                                              |
| SENSE_1T01_P2MM           | SENSE_1T01_P2MM | SENSE   | CPUVR ISNS1 P R 42 43                                                            |
| SENSE_1T01_P2MM           | SENSE_1T01_P2MM | SENSE   | CPUVR ISNS1 N R 42 43                                                            |
| SENSE_1T01_45S            | SENSE_1T01_45S  | SENSE   | CPUVR ISUM R P 42                                                                |
| SENSE_1T01_45S            | SENSE_1T01_45S  | SENSE   | CPUVR ISUM R N 42                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | ISNS CPUDDR P 42                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | ISNS CPUDDR N 42                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS P3V3S5 N 42                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS P3V3S5 P 42                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS 3V3 S0 P 41                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS 3V3 S0 N 41                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS CAMERA P 41                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS CAMERA N 41                                                                 |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS P3V3 S0 N 41                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS P3V3 S0 P 41                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | ISNS 1V05 S0 P 42 56                                                             |
| SENSE_DIEFPATR            | SENSE_1T01_P2MM | SENSE   | ISNS 1V05 S0 N 42 56                                                             |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS BMON GAIN P 41                                                              |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS BMON GAIN N 41                                                              |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS HS COMPUTING N 41 43                                                        |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS HS COMPUTING P 41 43                                                        |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS HS OTHER N 41                                                               |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS HS OTHER P 41                                                               |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS 1V2 S3 N 41 53                                                              |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS 1V2 S3 P 41 53                                                              |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS AIRPORT N 41                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS AIRPORT P 41                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS SSD N 41                                                                    |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS SSD P 41                                                                    |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS LCDBKLT N 41                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS LCDBKLT P 41                                                                |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS PANEL N 43                                                                  |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS PANEL P 43                                                                  |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS HS GAIN N 43 44                                                             |
| SENSE_DIEFPATR            | SENSE_1T01_45S  | SENSE   | ISNS HS GAIN P 43 44                                                             |
| AUD_DIFF                  | 1T01_DIEFPATR   | AUDIO   | SPKRAMP INR P 47 61 65                                                           |
| AUD_DIFF                  | 1T01_DIEFPATR   | AUDIO   | SPKRAMP INR N 47 61 65                                                           |
| SPKR_OUT                  | 1T01_DIEFPATR   | AUDIO   | MAX98300 R P 47                                                                  |
| SPKR_OUT                  | 1T01_DIEFPATR   | AUDIO   | MAX98300 R N 47                                                                  |
| SPKR_OUT                  | SPKR_DIEFPATR   | AUDIO   | SPKRAMP ROUT P 47 64                                                             |
| SPKR_OUT                  | SPKR_DIEFPATR   | AUDIO   | SPKRAMP ROUT N 47 64                                                             |
|                           | SB_POWER        |         | PP3V3 S5 8 11 13 15 16 17 18 20 29 34 42                                         |
|                           | SB_POWER        |         | PP3V3 S0 57 58 59 60 62 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 |
|                           | GND             |         | GND                                                                              |

|                                                                                                                   |  |                              |            |
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| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM_LINE_WIDTH | MINIMUM_NECK_WIDTH | MAXIMUM_NECK_LENGTH | DIFFPAIR_PRIMARY_GAP | DIFFPAIR_NECK_GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SD_45SE           | *     | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          |                      |                   |

SD Card Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET_T       |         | PE                      |
|---------------------------|-------------|---------|-------------------------|
|                           | PHYSICAL    | SPACING |                         |
| H614 SDDATA               | SD_45SE     |         | SDCONN_DATA<0..3> 33 34 |
| H614 SDCLK                | SD_45SE     |         | SDCONN_CLK 33 34        |
| H614                      | SD_45SE     |         | SDCONN_WP 33 34         |
| H614                      | SD_45SE     |         | SDCONN_CMD 33 34        |
| H614                      | SD_45SE     |         | SDCONN_DETECT_L 33 34   |
| H614                      | SD_45SE     | SPT     | SD_SPI_CLK 34           |
| H614                      | SD_45SE     | SPT     | SD_SPI_CS_L 34          |
| H614                      | SD_45SE     | SPT     | SD_SPI_MQSI 34          |
| H614                      | SD_45SE     | SPT     | SD_SPI_MISO 34          |
| H614                      | CLK_25M_45S |         | SDCLK_CLK_25M_X1 34 69  |
| H614                      | CLK_25M_45S |         | SDCLK_CLK25M_X2_R       |

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<RDAR://COMPONENT/508934> J43 HW EE SCHEMATIC | PROTO 0  
<RDAR://COMPONENT/508937> J43 HW EE SCHEMATIC | PROTO 1  
<RDAR://COMPONENT/508941> J43 HW EE SCHEMATIC | EVT  
<RDAR://COMPONENT/508945> J43 HW EE SCHEMATIC | DVT

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afp://kismet.apple.com/Kismet-Projects/J41-J43

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<rdar://component/497589> MobileMac HW | Architecture

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